

Investment and Finance

Introduction and Fundamental Concepts

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- Cand.scient.oecon from Aarhus University, 2024.
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- **Lectures:** 2x2 lectures per week. Please tell me if you have suggestions for improvement. For instance, if it going too slow, too fast, comments to the stucture, etc.
- **Exercises:** Uploaded every week. You should try to solve them in advance of the tutorials.
- **Textbook:** Investment Science, David Luenberger, Oxford University Press, 2009. We will follow the language of the book in terms of the technical terminology and presentation. In addition, we might decide to upload supplementary articles along the way. In some cases, material will be presented on the slides, which is not covered in Luenberger (2009).
- **The exam:** Four hour written exam. You can bring computer, books, pocket calculator etc. However, internet is not allowed.

The course will use elements from the following areas/courses

- Linear algebra (Lineær algebra)
- Basic probability theory (Introduktion til sandsynlighedsteori og statistik)
- Math (Matematisk analyse 1 og 2)
- Basic accounting (Principper i matematik-økonomi)

Information will be provided on a “continuous” basis on Brightspace.

- Lectures:
 - First Half
 - Tuesday 10-12, Aud. D4
 - Friday 8-10, Aud. D4
 - Second Half
 - Tuesday 10-12, Aud. D4
 - Wednesday 10-12, Aud. D1
- Tutorials:
 - MØ1: Thursday 14-16, Koll G3, Instructor Anders Borg Bilberg
 - MØ2: Monday 12-14, Koll G3, Instructor Peter Anders Skriver

Course outline for the first half

- Introduction and basic concepts
- Fixed income securities and interest rate term structures
- Portfolio theory (Markowitz' framework)
- The Capital Asset Pricing Model (CAPM)
- Pricing models

Out line of this week's lectures

- Introduction to fundamental assumptions
- Investment problems
- Interest on investments
- Computing present values
- The internal rate of return
- Comparison of evaluation criteria
- Fundamental Concepts
 - Financial markets
 - Arbitrage
 - Complete markets

Throughout the course you will encounter the following fundamental assumptions again and again:

- No arbitrage opportunities in the market.
- Investors are risk averse.
- Future price values are uncertain and will be modeled as stochastic variables/processes.

No arbitrage opportunities in the market

An arbitrage is defined as a situation where money can be earned (now or in the future) at no cost and with no risk involved. (We will define the concept more formally later in this slide set.)

Example: Say you know bank A where you can deposit money at an annual rate of 12% and another bank B where you can borrow money at 10%. You can arbitrage this by borrowing 100 USD at bank B and deposit the money at bank A. After one year you can collect $112 - 110 = 2$ USD. You can also run the scheme at much higher levels!

- Arbitrage opportunities should not occur in financial markets and if they do, they should disappear rapidly. Also known as “no free lunch”.

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- Arbitrage opportunities should not occur in financial markets and if they do, they should disappear rapidly. Also known as “no free lunch”.
- Law of one price: Identical future cash flows should cost the same today. The law of one price basically tells us that an investor should not care whether \$1 dollar comes from one place or another (perfectly reasonable from an economic perspective). Note: in fact this does not always hold. E.g. “Green bonds” (not covered in this course).

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- We will see how no-arbitrage is a very powerful assumption for pricing purposes.

Risk averse investors

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- A very popular framework to specify risk and return is the so-called mean-variance approach.

Uncertainty of future prices

Asset prices react to the arrival of new information flowing to the market. Hence, price paths should be modeled as dynamic and stochastic.

- **One-period model:** One will often use stochastic variables to model the future prices.
- **Multi-period model:** Modelled as a sequence of stochastic variables (discrete stochastic process) or binomial trees.
- **Continuous time model:** Stochastic processes in continuous time.

Typical investment problems

- **Pricing by replication:** Given a stream of certain or uncertain cash flows, it is often of interest to price this today. If we are able to replicate the cash flow stream using other assets traded, we know by no-arbitrage that the price of the cash flow stream should equal the price of the replicating portfolio. More on that later.

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- **Hedging:** By hedging we refer to the process of reducing financial risks with the use of financial assets (typically financial derivatives).
Examples: 1. the case of buying insurance on your car. 2. the case of a baker who wants to hedge against unfavorable movements in the price of wheat. This can be achieved via financial derivatives such as forwards and futures contracts.

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- **Portfolio selection:** An investor with capital needs to figure out how to allocate the capital to financial assets in a way that maximizes her utility, e.g. maximizes return while minimizing risk.

Time value of money

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- n denotes the number of years the money is invested.
- m denotes the compounding periods per year.

Compounding conventions

Different conventions exist for computing the interest on a given principal investment A over time.

- Simple compounding:

$$A \left(1 + \frac{r}{m} mn \right) = A(1 + rn) \quad (1)$$

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- Continuous compounding:

$$A \lim_{m \rightarrow \infty} \left(1 + \frac{r}{m} \right)^{nm} = Ae^{rn} \quad (3)$$

Since $\exp x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n} \right)^n$.

The impact of interest compounding

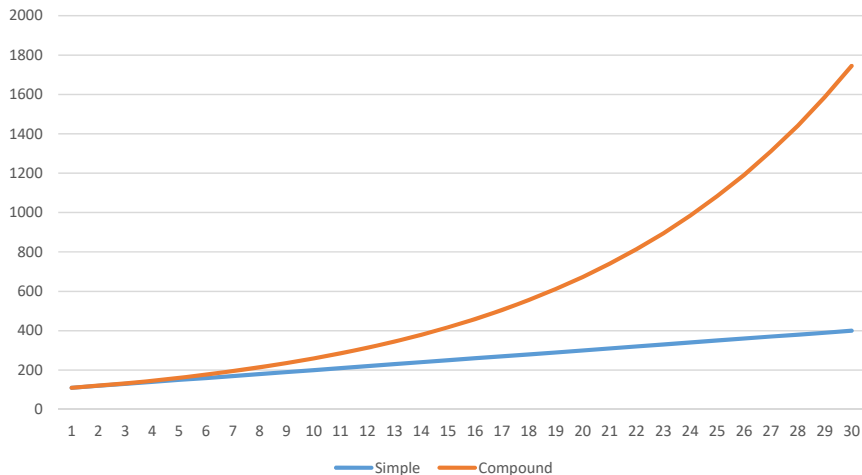


Figure: $A = 100$, $r = 10\%$, $m = 1$

The effective rate

For convenience, it can be nice to convert a given rate r under some compounding scheme into an equivalent rate r' defined under yearly compounding. We denote the r' as the effective rate. For instance, with m compounding periods connected to the rate r we get

$$1 + r' = \left(1 + \frac{r}{m}\right)^m \Rightarrow r' = \left(1 + \frac{r}{m}\right)^m - 1. \quad (4)$$

Increasing the compounding frequency for a fixed r increases r' .

Present value

Instead of computing the future value of an investment today, we can reverse the concepts and find the present value of future payments.

Consider the two different options:

- 1 Receive 110 USD in one year.
- 2 Receive 100 USD today and invest in bank at 10% for one year.

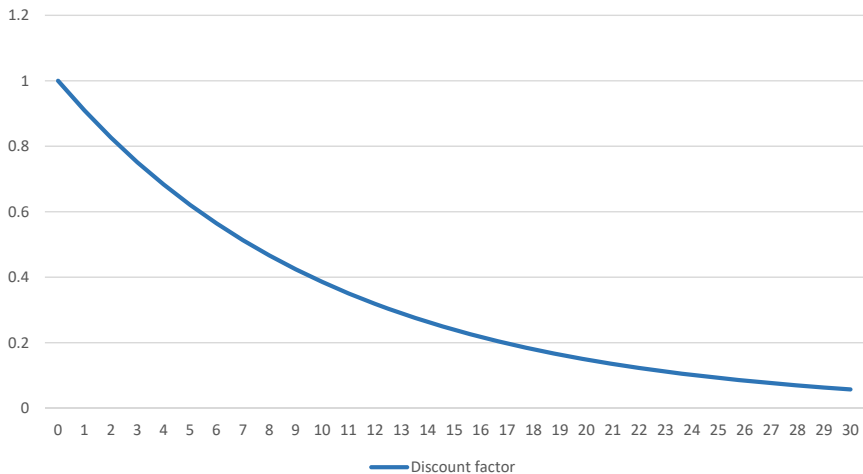
Both options result in the same stream of cash flows. Namely, 0 USD at time 0 and 110 at time 1. The forward value of both CF streams at time 1 must clearly be 110, while the present value must be 100. In order to go from the forward value to the present value you must multiply by the discount factor $d_1 = \frac{100}{110} = \frac{1}{1+0.1}$.

In general, we have that

$$d_n = \frac{1}{\left(1 + \frac{r}{m}\right)^{mn}}. \quad (5)$$

- We call d_n for the discount factor for the time point n .
- With the discount curve at hand, we can compute present values for payments falling at different time points. This makes it easier to compare different investment alternatives.

Discount curve d_t with $r = 10\%$ and annual compounding.



Valuing streams of cash flows

We can write a stream of cash flows associated with an investment in vector format as

$$x = (x_0, x_1, \dots, x_n), \quad (6)$$

where x_i is fixed payment at time i already known at time 0.

We can compute the net present value (NPV) by discounting all the cash flows back to 0

$$NPV(x) = \sum_{i=0}^n x_i d_i = x_0 + \sum_{i=1}^n \frac{x_i}{(1+r)^i}, \quad (7)$$

where the last equality holds for annual compounding (or the rate r is the period rate).

We say that two cash flow streams x and y are equivalent if

$$NPV(x) = NPV(y). \quad (8)$$

Similarly, we can write down the time- n forward value of a cash flow stream as

$$FV = \sum_{i=0}^n x_i (1 + r)^{n-i}. \quad (9)$$

Internal rate of return (IRR)

Let $x = (x_0, x_1, \dots, x_n)$, be a CF stream. Then the IRR is defined as the rate that solves

$$0 = \sum_{i=0}^n \frac{x_i}{(1 + IRR)^i}. \quad (10)$$

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- In practice, one finds the IRR using some numerical solver.

We have discussed two approaches that can be used to evaluate if a given investment alternative x is profitable

- 1 If $NPV(x) > 0$
- 2 If $IRR(x) > r$.

Tree cutting example on blackboard

Discussion of the evaluation criteria, IRR

Advantages:

- ① The IRR does not depend on the term structure.
- ② For investments with different time horizons and the possibility to be repeated, comparison of the alternatives using the IRR will give you the correct evaluation.

Problems:

- ① IRR is a relative number. It does not tell you much in isolation. You need to be able to compare with another investment opportunity.
- ② It does not tell you anything about the horizon of the investment.
- ③ It is not well-defined for arbitrary payment streams.

Discussion of the evaluation criteria, NPV

Advantages:

- 1 Always yield to correct answers if computed correctly. In particular, for projects that can be repeated, you should compute NPV of the projects in question up to a point where the horizons of the repeated projects are of the same length.

Problems:

- 1 You need to know the term structure in order to compute discount factors.

Tree cutting example continued

We will encounter and build upon the following three concepts throughout the course:

- Structure of a financial market
- Arbitrage
- Complete markets

Structure of a financial market

We define a financial market with n assets and with T periods as follows:

- ① $P \in \mathbb{R}^n$ denotes the prices of the assets

$$P = [P_1, P_2, \dots, P_n]^\top, \quad (11)$$

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- ② and $B \in \mathbb{R}^n \times \mathbb{R}^T$ is the matrix containing the assets future payments

$$B = \begin{bmatrix} B_{11} & B_{12} & \cdots & B_{1T} \\ B_{21} & B_{22} & \cdots & B_{2T} \\ \vdots & \vdots & \ddots & \vdots \\ B_{n1} & B_{n2} & \cdots & B_{nT} \end{bmatrix}. \quad (12)$$

If $B_{ij} > 0$ ($B_{ij} < 0$) you receive (pay) the amount at time j .

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If $B_{ij} > 0$ ($B_{ij} < 0$) you receive (pay) the amount at time j .

- ③ You pay the price P_i today at time 0 in exchange for the future payment stream $B_{i\cdot} = (B_{i1}, \dots, B_{iT})$.

Portfolios

A portfolio is denoted by $\theta \in \mathbb{R}^n$ and the i th element θ_i denotes the position in asset i .

- The price of the portfolio equals

$$P^{pf} = \theta^\top P. \quad (13)$$

- The portfolio has the cash flow stream

$$B^{pf} = \theta^\top B. \quad (14)$$

- If $\theta_i > 0$ we buy the asset and we go long in i .
- If $\theta_i < 0$ we sell the asset and we go short in i .
- When you go short you borrow the asset from someone else and sell it. In the future you deliver the asset back to owner.

Example on Blackboard.

Arbitrage (No such thing as a free lunch)

We define an arbitrage if it is possible to construct a portfolio that has one of the two features:

- ① Type 1: $P^{pf} = 0$ and $B_j^{pf} \geq 0$ and $B_j^{pf} > 0$ for at least one j . We get a positive payment in the future at zero price today.
- ② Type 2: $P^{pf} < 0$ and $B_j^{pf} \geq 0$ for all j . We receive money today without having to pay anything in the future.

We call the market arbitrage free if it is not possible to construct a portfolio that is an arbitrage.

A financial market should not permit arbitrage and the no-arbitrage assumption is almost always assumed when new theory is developed.

Market imperfections

An arbitrage opportunity might be difficult to exploit because of various market imperfections, such as trading cost, lack of liquidity, wide bid-ask spread, short-selling restrictions etc.

Furthermore, even without these market imperfections the opportunity would most likely disappear very fast, since the opportunity would be attractive to all the arbitrageurs out there exploiting it. This will level out prices.

First fundamental Theorem of asset pricing

Theorem

The following two statements are equivalent:

- *The market defined by P and B is arbitrage free.*
- *There exists a $d \in \mathbb{R}_+^T$ that solves*

$$P = Bd. \quad (15)$$

Remarks:

- d_i is the discount factor for the time point i .
- There can be many different vectors d that solves the pricing relation in equation (15).

Example continued

Complete financial markets

A financial market is complete if all possible stream of payments B^{pf} can be constructed by forming a portfolio of the existing assets in the market. Mathematically:

$$\forall B^{pf} \in \mathbb{R}^T, \exists \theta \in \mathbb{R}^n \text{ so } \theta^\top B = B^{pf}. \quad (16)$$

Remarks:

- In order for a market to be complete, the matrix B should have $\text{rank}(B) = T$. Hence, there should be as many linearly independent rows as columns.
- If the market is arbitrage free and complete, the vector d solving equation (15) is unique.

Example continued

Conclusion

In this slide set we introduced the fundamental concepts that the course builds upon. Moreover, we discussed the time value of money and introduced an interest rate to measure this.

In the coming lectures, we will discuss the market where the time value of money is typically extracted from. Namely, the bond market. We will see how one can extract a full term structure of interest rates from the bond market.

However, first we will introduce the market and present the methodology behind risk management of bond portfolios.

Investment and Finance

Fixed Income Securities

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Out line of this week's lectures

- Motivation
- Bonds
- Yield to maturity
- Measures of price sensitivity to changes in YTM
 - Modified duration
 - Macaulay
 - Convexity
- Sensitivity measures for portfolios
- Immunization

The overall market for interest rates is far more complex than the fixed rate bank account introduced in the introductory slide set. In practice the fixed income markets is comprised of many different securities such as: savings deposits, money market loans (LIBOR, CIBOR, EURIBOR), government bonds, corporate bonds, mortgage backed bonds etc.

In this lecture we will focus on government bonds where it can be assumed that the credit risk component is negligible (as opposed to corporate bonds).

- A bond is a financial security that promises to pay a fixed stream of payments in the future.
- Issued by governments, corporations, (homeowners).
- A bond is characterized by:
 - Face value, F .
 - Coupon rate, c . Coupon in nominal value is denoted C .
 - Maturity date, T .
 - Number of payments per year, m , (for government bonds, normally 2 in the US, Italy, Japan, UK and 1 in Denmark, France, Germany.)

Draw timeline of payments on blackboard.

Zero coupon bonds (strips)

Zero coupon bonds are bonds with no payments before the maturity of the bond.

ZCBs are convenient for two reasons:

- 1 They make it easy to construct any cash flow.
- 2 Greater sensitivities to changes in interest rates than coupon bonds (more on this later).

Yield to maturity

Yield to maturity is the single sensible rate that, when used to discount a bond's cash flow, produces the bond's market price.

Yield is not a good measure of realized return to maturity. Only if the yield to maturity of a bond does not change over the lifetime of the bond, is the realized return on that bond equal to its yield.

Definition

Take a coupon bond with m coupons a year, coupon rate c , maturity T , face value F and price $P(T)$. Then the yield λ on that bond solves the equation

$$P(T) = \sum_{k=1}^{mT} \frac{\frac{c}{m}F}{\left(1 + \frac{\lambda}{m}\right)^k} + \frac{F}{\left(1 + \frac{\lambda}{m}\right)^{mT}}. \quad (1)$$

Since the yield corresponding to a given bond is unique, it is often used as an alternative way of quoting a bond's price. Note how the YTM equals the IRR by definition.

Take a perpetuity that pays out the same amount $A = a/m$, m times per year (note: a denotes the annual payment) for all eternity. Using a straightforward analog to the price in (1) we have

$$P = \sum_{k=1}^{\infty} \frac{A}{\left(1 + \frac{\lambda}{m}\right)^k} = \frac{A}{\lambda/m} = \frac{a}{\lambda}. \quad (2)$$

Recall from Mathematical Analysis

$$\sum_{k=1}^{\infty} Ar^k = \frac{A \cdot r}{1 - r}$$

An annuity is similar to a perpetuity but with a finite maturity.
An application of the pricing formula for perpetuities yields the *annuity formula* (Blackboard)

$$P_A(T) = \frac{a}{\lambda} \left[1 - \frac{1}{(1 + \lambda/m)^{mT}} \right], \quad (3)$$

where we again denote the annual payment by a and m is the number of payments per year.

An alternative pricing formula for coupon bonds

- Using the annuity formula (3) one finds that equation (1) can be rewritten as

$$P(T) = \frac{Fc}{\lambda} \left[1 - \frac{1}{(1 + \lambda/m)^{mT}} \right] + \frac{F}{(1 + \lambda/m)^{mT}}. \quad (4)$$

- Notice:
 - that for $c = \lambda$ the price reduces to $P(T) = F$, i.e. trading at par.
 - that for $c > \lambda$ we have $P(T) > F$.
 - that for $c < \lambda$ we have $P(T) < F$.

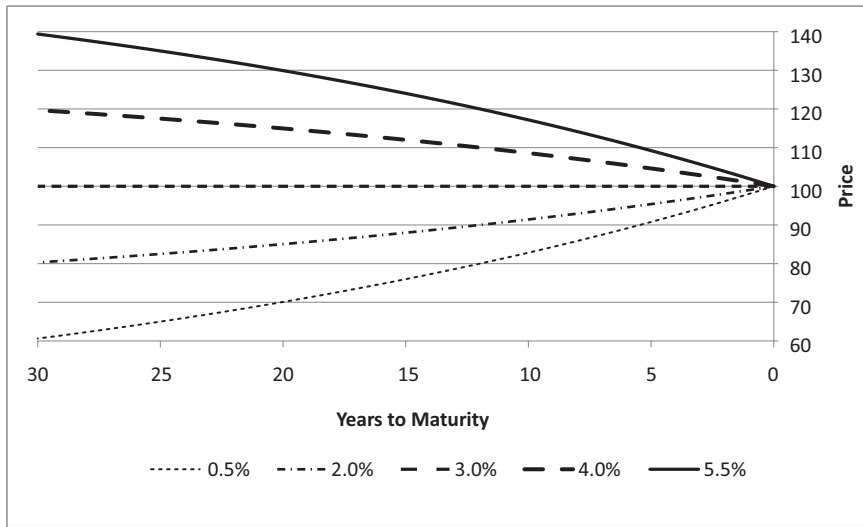


Figure: Prices of bonds with varying coupons as a function of time to maturity and with yield at 3%. The graph depicts what is known as pull-to-par.

Yield to maturity and realized return

- Sometimes ytm is described as a given bond's return if it is held to maturity. Take for example the pricing equation of a bond with coupon rate of $4\frac{1}{2}\%$ with $T = 1.5$, $m = 2$ and $P = 105.86$

$$\sum_{i=1}^3 \frac{2.25}{\left(1 + \frac{\lambda}{2}\right)^i} + \frac{100}{\left(1 + \frac{\lambda}{2}\right)^3} = 105.86 .$$

- Multiplying both sides by $\left(1 + \frac{\lambda}{2}\right)^3$ yields

$$\sum_{i=1}^3 2.25 \left(1 + \frac{\lambda}{2}\right)^{3-i} + 100 = 105.86 \left(1 + \frac{\lambda}{2}\right)^3 .$$

If we assume that every coupon payment can be reinvested at the ytm, then the sum of the reinvested coupon payments equals the price invested for 1.5 years at the ytm.

- Generally, there is no reason why the coupon payments can be reinvested at the ytm! Hence, the description of ytm as the return if held to maturity is flawed.

Accrued interest

Until now we have assumed that cash flows occur in even intervals. Obviously trading also takes place on dates outside the coupon payment dates. In that case one has to take accrued interest into account.

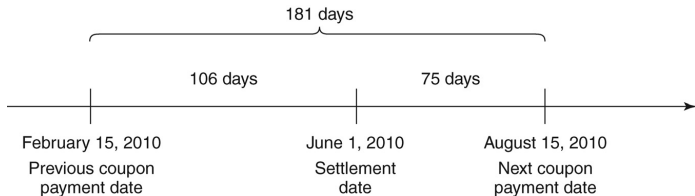
- Take a coupon bond with maturity T_n paying out at dates six months apart and that the last payment date was T_1 the next is T_2 . Assume that investor B wishes to buy the bond from investor A at time t , $T_1 < t < T_2$. How do they settle on a price?
- Since some of the coupon $\frac{cF}{m}$ paid at time T_2 belongs to investor A (accrued interest), she has to be compensated when selling the bond.
- The market convention is to count the number of calendar days between the payment dates $T_2 - T_1$ and the number of calendar days between the last payment date and today $t - T_1$, and then determine the accrued interest as (*actual/actual*)

$$AI = \frac{t - T_1}{T_2 - T_1} \frac{cF}{m}.$$

- The **full price** P^{full} of the bond is then equal to the **flat price** (quoted price) P^{flat} added the accrued interest AI

$$P^{full} = PV(\text{future cash flows}) = P^{flat} + AI. \quad (5)$$

- Example: Take a $3\frac{5}{8}\%$ coupon bond on June 1, 2010, which matures on August 15, 2019. The coupon dates are February 15 and August 15.



- The accrued interest on the face value F to be paid on June 1, 2010 is

$$AI = \frac{106}{181} \frac{3\frac{5}{8}\%}{2} F = 1.0615\% F.$$

- The quoted price (flat price) is 102.81. The full price is then

$$P^{full} = 102.81 + 1.0615 = 103.87.$$

One could ask: why quote the flat price of a coupon bond?

The flat price has the convenient feature of not falling after a coupon payment.

- This can be realized from the following. Let $P^{b,flat}$ and $P^{a,flat}$ denote the flat price before and after the coupon payment. In the instant just before the coupon payment we have the relation

$$P^{b,flat} + \frac{cF}{m} = P^{b,full} = \frac{cF}{m} + PV(\text{cash flows after coupon})$$

$\Downarrow$

$$P^{b,flat} = P^{b,full} - \frac{cF}{m} = PV(\text{cash flows after coupon}).$$

- On the other hand, right after the coupon payment

$$P^{a,flat} + 0 = P^{a,full} = PV(\text{cash flows after coupon}).$$

- From which we see that

$$P^{b,flat} = P^{a,flat}$$

$$P^{b,full} = P^{a,full} + \frac{cF}{m} > P^{a,full}.$$



Figure: Full and flat prices for the $3\frac{5}{8}\%$ s of August 15, 2019, over time with a constant discount function.

It is nice to think of changes in prices reflecting changes in market conditions. For this reason it is clear that quoting the flat price will give you a more transparent measure of (changes in) market conditions.

Day-count conventions

- “actual/actual”: Used when computing accrued interests for government bonds.
- “actual/360”: Commonly used in money markets.
- “30/360”: Corporate bonds and the fixed leg in interest rate swaps.

Whatever convention is used does not really matter in practice. The flat price will adjust such that equation (5) always holds.

Measures of price sensitivity

The dominant risk factor affecting the valuation of most fixed income securities is interest rate risk, i.e. the change in the price as interest rates change.

Measures of price sensitivity to changes in interest rates are among other things important for the following reasons:

- 1 Traders wishing to hedge a bond with another bond or a portfolio of bonds need to know how the bonds respond to changes in interest rates.
- 2 If you as an investor have a particular view on future interest rates, it is of interest to know how bond prices react if you are to be right (or wrong).
- 3 Risk managers need to know how volatile their portfolios are.
- 4 Important for asset liability management. To minimize risk, the sensitivity of the asset values to changes in interest rates should be equal to the sensitivity of the liabilities to changes in rates.

Price sensitivity measures based on parallel yield shifts

Of course when you talk about the price sensitivity to changes in interest rates, you have to specify what rates. In this lecture we will specify price sensitivity measure based on parallel shifts in the yield. Although, theoretically inconsistent and unrealistic these measures are still important to understand for the following reasons.

- ① First, easy to compute and easy to understand.
- ② Widely used in the financial industry.
- ③ A lot of intuition can be gained from understanding these measures.

Modified Duration

The modified duration measures the relative change in the value of a security for a change in the yield of a bond.

$$D_{Mod} = -\frac{1}{P} \frac{\Delta P}{\Delta \lambda}. \quad (6)$$

Or in the limit

$$D_{Mod} = -\frac{1}{P} \frac{dP(\lambda)}{d\lambda}. \quad (7)$$

If we multiply both sides of (7) with $-d\lambda$ we get

$$\frac{dP(\lambda)}{P} = -D_{Mod} d\lambda. \quad (8)$$

Hence, the percentage change in the price equals the negative of the duration times the change in the yield.

Remember that the price of a bond with maturity T , m coupons per year and face value $F = 100$ is

$$P(\lambda) = \sum_{k=1}^{mT} \frac{100c/m}{(1 + \lambda/m)^k} + \frac{100}{(1 + \lambda/m)^{mT}} \quad (9)$$

or in more compact form

$$P(\lambda) = \frac{100c}{\lambda} \left(1 - \frac{1}{(1 + \lambda/m)^{mT}} \right) + \frac{100}{(1 + \lambda/m)^{mT}}. \quad (10)$$

We can get expressions for the duration by taking the derivative of either (9) or (10) with respect to λ .

For example if we differentiate (9) with respect to λ we get

$$\begin{aligned}\frac{dP}{d\lambda} &= \sum_{k=1}^{mT} -\frac{k}{m} \frac{100c/m}{(1 + \lambda/m)^{k+1}} - T \frac{100}{(1 + \lambda/m)^{mT+1}} \\ &= -\frac{1}{(1 + \lambda/m)} \left(\sum_{k=1}^{mT} \frac{k}{m} \frac{100c/m}{(1 + \lambda/m)^k} + T \frac{100}{(1 + \lambda/m)^{mT}} \right). \quad (11)\end{aligned}$$

Alternatively, we could of course have differentiated (10) (*Blackboard*)

$$\frac{dP}{d\lambda} = -\frac{100c}{\lambda^2} \left(1 - \frac{1}{(1 + \lambda/m)^{mT}} \right) - T \left(1 - \frac{c}{\lambda} \right) \frac{100}{(1 + \lambda/m)^{mT+1}}. \quad (12)$$

With the first order derivative of the price w.r.t. to yield available, we can now compute the modified duration in closed form using either of the derivatives (11) and (12)

$$D_{Mod} = -\frac{1}{P} \frac{dP(\lambda)}{d\lambda} \quad (13)$$

$$= \frac{1}{P} \frac{1}{(1 + \lambda/m)} \left(\sum_{k=1}^{mT} \frac{k}{m} \frac{100c/m}{(1 + \lambda/m)^k} + T \frac{100}{(1 + \lambda/m)^{mT}} \right) \quad (14)$$

$$= \frac{1}{P} \left(\frac{100c}{\lambda^2} \left(1 - \frac{1}{(1 + \lambda/m)^{mT}} \right) + T \left(1 - \frac{c}{\lambda} \right) \frac{100}{(1 + \lambda/m)^{mT+1}} \right). \quad (15)$$

Macaulay Duration

A related quantity to modified duration is the Macaulay duration defined as

$$\begin{aligned} D_{Mac} &= (1 + \lambda/m) D_{Mod} \\ &= \frac{1}{P} \left(\sum_{k=1}^{mT} \frac{k}{m} \frac{100c/m}{(1 + \lambda/m)^k} + T \frac{100}{(1 + \lambda/m)^{mT}} \right). \end{aligned} \quad (16)$$

And the compact version

$$D_{Mac} = \frac{(1 + \frac{\lambda}{m})}{P} \left(\frac{100c}{\lambda^2} \left(1 - \frac{1}{(1 + \frac{\lambda}{m})^{mT}} \right) + T \left(1 - \frac{c}{\lambda} \right) \frac{100}{(1 + \frac{\lambda}{m})^{mT+1}} \right) \quad (17)$$

Interpretation of the Macaulay duration

Until now it has been quite unclear why the duration has the name duration. In order to realize why, we rewrite the Macaulay duration as

$$\begin{aligned} D_{Mac} &= \left(\sum_{k=1}^{mT} \frac{k}{m} \frac{1}{P} \frac{100c/m}{(1 + \lambda/m)^k} + T \frac{1}{P} \frac{100}{(1 + \lambda/m)^{mT}} \right) \\ &= \left(\sum_{k=1}^{mT} \frac{k}{m} \frac{PV(100c/m, k/m; \lambda)}{P} + T \frac{PV(100, T; \lambda)}{P} \right). \end{aligned}$$

Notice how each payment date is weighted by the fraction of that particular payments value relative to the total value of all the payments. Hence, the Macaulay duration can be interpreted as the “average maturity” of the bond. Therefore the terminology duration!

Duration for zero coupon bonds

Recall that a zero coupon bond has the price

$$P = \frac{100}{(1 + \lambda/m)^{mT}} \quad (18)$$

which gives

$$\frac{dP}{dy} = -T \frac{100}{(1 + \lambda/m)^{mT+1}}.$$

Then we easily see that

$$D_{Mod} = \frac{T}{(1 + \lambda/m)} \quad (19)$$

$$D_{Mac} = T. \quad (20)$$

Hence, the Macaulay duration is just the maturity.

Duration for par bonds

Recall that for a par coupon bond $y = c$. Hence, using (12) we get

$$\frac{dP}{dy} = -\frac{100}{\lambda} \left(1 - \frac{1}{(1 + \lambda/m)^{mT}} \right).$$

Then we see that (remember that $P = 100$)

$$D_{Mod} = \frac{1}{\lambda} \left(1 - \frac{1}{(1 + \lambda/m)^{mT}} \right) \quad (21)$$

$$D_{Mac} = \frac{(1 + \lambda/m)}{\lambda} \left(1 - \frac{1}{(1 + \lambda/m)^{mT}} \right). \quad (22)$$

Duration for perpetuities

Recall that a perpetuity has the price

$$P = \frac{a}{\lambda}$$

where a is the annual payment. This gives

$$\frac{dP}{d\lambda} = -\frac{a}{\lambda^2}.$$

Then we easily see that

$$D_{Mod} = -\frac{1}{P} \left(-\frac{a}{\lambda^2} \right) = \frac{1}{\lambda}$$
$$D_{Mac} = \frac{1 + \lambda/m}{\lambda}.$$

Hence, a finite duration.

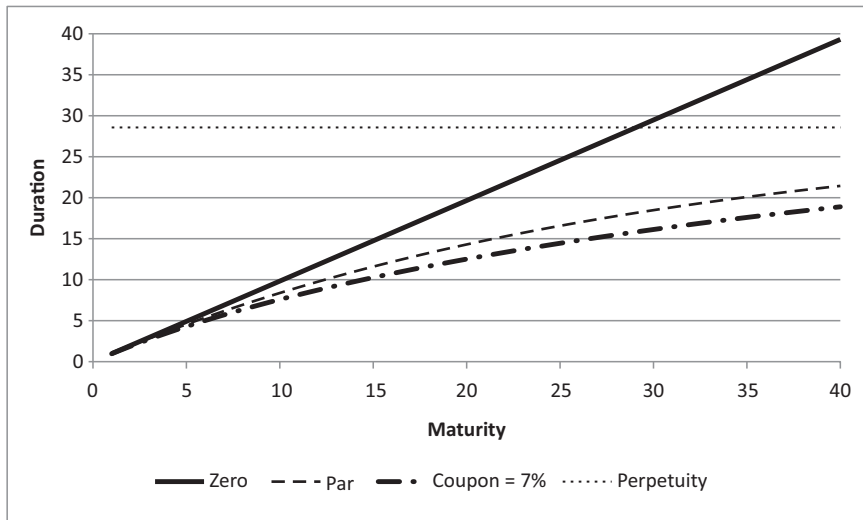


Figure: Duration Across Bonds Yielding 3.50%.

The slope of the price-yield curve change when yields change. Hence, so does the duration. We can capture the sensitivity of the slope to changes in yield via the second derivative of the price function $\frac{d^2 P(\lambda)}{d\lambda^2}$. Convexity is defined via the second derivative as

$$C = \frac{1}{P} \frac{d^2 P(\lambda)}{d\lambda^2}. \quad (23)$$

- If $\frac{d^2 P(\lambda)}{d\lambda^2} > 0$ and hence $C > 0$ for an asset, then the price function is convex, which is the case for standard bonds.
- Examples of assets with negative convexity are callable bonds and mortgage backed securities (maybe more on this later).

If we differentiate the first derivative (11) again, we get the second derivative

$$\begin{aligned}\frac{d^2 P}{d\lambda^2} &= \sum_{k=1}^{mT} \frac{k}{m} \frac{k+1}{m} \frac{100c/m}{(1+\lambda/m)^{k+2}} + T \frac{mT+1}{m} \frac{100}{(1+\lambda/m)^{mT+2}} \\ &= \frac{1}{(1+\frac{\lambda}{m})^2} \left(\sum_{k=1}^{mT} \frac{k}{m} \frac{k+1}{m} \frac{100c/m}{(1+\frac{\lambda}{m})^k} + T \left(T + \frac{1}{m} \right) \frac{100}{(1+\frac{\lambda}{m})^{mT}} \right).\end{aligned}\quad (24)$$

Which gives us the convexity

$$C = \frac{1}{P(1+\lambda/m)^2} \left(\sum_{k=1}^{mT} \frac{k}{m} \frac{k+1}{m} \frac{100c/m}{(1+\lambda/m)^k} + T \left(T + \frac{1}{m} \right) \frac{100}{(1+\lambda/m)^{mT}} \right) \quad (25)$$

- The convexity of a zero coupon bond equals

$$C^{ZCB} = \frac{T \left(T + \frac{1}{m} \right)}{\left(1 + \frac{\lambda}{m} \right)^2}. \quad (26)$$

- From (26) it is clear that zero coupon bonds have increasing convexity with increasing maturity.
- This results in a more curved (convex) price rate curve for longer maturities.
- Since coupon bonds are portfolio of zero coupon bonds this effect is normally also present in their price rate curve.

Estimating price changes via duration and convexity

To illustrate how the duration and convexity measures can be related to price changes, we first use a *second order Taylor approximation* to approximate the price after a small change in λ

$$P(\lambda + \Delta\lambda) \approx P(\lambda) + \frac{dP(\lambda)}{d\lambda} \Delta\lambda + \frac{1}{2} \frac{d^2P(\lambda)}{d\lambda^2} (\Delta\lambda)^2 \quad (27)$$

$\Downarrow$

$$\frac{\Delta P}{P} \approx \frac{1}{P} \frac{dP(\lambda)}{d\lambda} \Delta\lambda + \frac{1}{2} \frac{1}{P} \frac{d^2P(\lambda)}{d\lambda^2} (\Delta\lambda)^2 \quad (28)$$

$\Downarrow$

$$\frac{\Delta P}{P} \approx -D \cdot \Delta\lambda + \frac{1}{2} C \cdot (\Delta\lambda)^2. \quad (29)$$

Likewise we could have used a first order approximation to find that

$$\frac{\Delta P}{P} \approx -D \cdot \Delta\lambda. \quad (30)$$

The second order approximation works well for even quite convex price functions, while care must be taken if only the first order approximation is used.

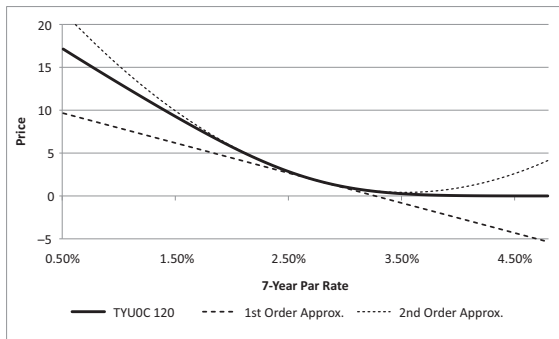


Figure: Price-rate curve for a given bond and its first- and second-order approximations.

Notice how being long convexity is essentially being long volatility. The more rates move, the better. This can also be seen from equation (29) since $(\Delta\lambda)^2$ is always positive.

Price sensitivity of portfolios

Until now we have considered three different measures of price sensitivities for single assets. Investors owning a portfolio of assets are interested in the sensitivity of the value of the total portfolio to changes in rates.

- The value of the portfolio P^{PF} is equal to the sum of the individual components value P^i

$$P^{PF} = \sum P^i. \quad (31)$$

- The modified duration of the portfolio

$$D^{PF} = -\frac{1}{P^{PF}} \frac{dP^{PF}}{d\lambda} = \sum -\frac{P^i}{P^i} \frac{1}{P^{PF}} \frac{dP^i}{d\lambda} = \sum \frac{P^i}{P^{PF}} D^i. \quad (32)$$

- Finally for the convexity

$$C^{PF} = \frac{1}{P^{PF}} \frac{d^2 P^{PF}}{d\lambda^2} = \sum \frac{P^i}{P^i} \frac{1}{P^{PF}} \frac{d^2 P^i}{d\lambda^2} = \sum \frac{P^i}{P^{PF}} C^i. \quad (33)$$

Immunization with duration

Consider the following financial market

$$P = [100, 100, 98]^T$$
$$B = \begin{pmatrix} 10 & 10 & 110 \\ 40 & 40 & 40 \\ 8 & 108 & 0 \end{pmatrix}$$

Say a new zero coupon bond with face value 100 is introduced in the market with maturity $T = 5$ years and price $P = 64.52$. An investor owning the ZCB wishes to hedge/immunize the bond position against fluctuations in the yield using the existing bonds trading in the market by matching the price and the duration.

Example on blackboard

The investor wishes to improve on the immunization by taking into account the convexity.

Example continued on blackboard

Conclusion

In this slide set we introduced the bond market and studied how various sensitivity measures can be computed. In addition, we covered a risk management application where an investor matches the risk measures of a constructed portfolio to a given investment.

All the risk measures introduced were based on parallel shifts in the yields of the bonds, and we only had to compute the individual yield to maturities of the bonds in question to use the theory.

In the next week we will present a more consistent theory of interest rates and present various types of interest rates. We will also introduce the concept of an interest rate term structure.

Bonus: General Calculation of Duration and Convexity

Given a cashflow of $(x_{t_1}, x_{t_2}, \dots, x_{t_n})$, and m specifying the number of yearly payments.

- The value of the cashflow is:

$$P = \sum_{k=1}^n \frac{x_{t_k}}{\left(1 + \frac{\lambda}{m}\right)^{t_k}} \quad (34)$$

- The modified duration of the cashflow

$$D_{Mod} = -\frac{1}{P} \frac{\Delta P}{\Delta \lambda} = \frac{1}{P} \frac{1}{\left(1 + \lambda/m\right)} \sum_{k=1}^n \frac{t_k}{m} \frac{x_{t_k}}{\left(1 + \lambda/m\right)^{t_k}} \quad (35)$$

- Finally for the convexity

$$C^{PF} = \frac{1}{P \left(1 + \lambda/m\right)^2} \sum_{k=1}^n \frac{t_k}{m} \frac{t_k + 1}{m} \frac{x_{t_k}}{\left(1 + \lambda/m\right)^{t_k}} \quad (36)$$

Investment and Finance

The Interest Rate Term Structure

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Out line of this week's lectures

- Various interest rates
 - Spot rates
 - Forward rates
 - Par rates
- Floating rate bonds
- Bootstrapping of spot rates
- Curve fitting
 - Linear interpolation
 - Nelson-Siegel
- Fisher-Weil duration

Various interest rates

- In the lectures last week, we defined the yield to maturity as the rate that when used for discounting a bond's cash flows gives the price of the bond. Hence, for each bond you have a different yield. As such it is not an appropriate measure of the *time value of money*.

Various interest rates

- In the lectures last week, we defined the yield to maturity as the rate that when used for discounting a bond's cash flows gives the price of the bond. Hence, for each bond you have a different yield. As such it is not an appropriate measure of the *time value of money*.
- In contrast, in the first week of the course, we saw how to measure the time value of money via the term structure of discount factors. It is not straightforward to interpret a discount curve, and it is more intuitive to use interest rates, such as *spot rates and forward rates*. The spot rate term structure highlights the fact that the annual rate attached to money being borrowed depends on the length of the loan.

Spot rates aka ZCB rates (Discrete compounding)

The annualized interest earned on a spot loan with maturity t years is denoted the spot rate s_t .

Definition

Take a spot loan/ZCB of value P which promises to pay F at maturity t years. Then the spot rate, based on a compounding frequency of m times per year, s_t is defined as the solution to

$$P = \frac{F}{\left(1 + \frac{s_t}{m}\right)^{mt}}. \quad (1)$$

Hence

$$s_t = m \left(\left(\frac{F}{P} \right)^{\frac{1}{mt}} - 1 \right). \quad (2)$$

Often we will assume yearly compounding $m = 1$.

Spot rates aka ZCB rates (Continuous compounding)

Definition

Take a spot loan/ZCB of value P which promises to pay F at maturity t years. Then the spot rate, based on continuous compounding, s_t is defined as the solution to

$$P = Fe^{-s_t t}. \quad (3)$$

Hence

$$s_t = -\frac{1}{t} \ln \left(\frac{P}{F} \right) = \frac{1}{t} \ln \left(\frac{F}{P} \right). \quad (4)$$

If we compare equations (1) and (3) with the present value formula (7) from the first slide set, we observe the following relation between spot rates and discount factors.

- Discrete compounding:

$$d_t = \frac{1}{\left(1 + \frac{s_t}{m}\right)^{mt}}. \quad (5)$$

And

$$s_t = m \left(\left(\frac{1}{d_t} \right)^{\frac{1}{mt}} - 1 \right). \quad (6)$$

d_t is the value of \$1 discounted for t years at the rate s_t (discrete or continuous).

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Pricing of coupon bonds

From the definition of the spot rates, we also see that the pricing equation (7) from the first slide set applied to a coupon bond paying out m times per year can be written as:

- Discrete compounding:

$$P = \sum_{i=1}^n \frac{\frac{c}{m}F}{\left(1 + \frac{s_{t_i}}{m}\right)^i} + \frac{F}{\left(1 + \frac{s_{t_n}}{m}\right)^n}, \quad (9)$$

with $t_i = \frac{i}{m}$.

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- Continuous compounding:

$$P = \sum_{i=1}^n \frac{c}{m} F e^{-s_{t_i} t_i} + F e^{-s_{t_n} t_n}, \quad (10)$$

where $t_i = \frac{i}{m}$.

Definition

A forward rate f_{t_1, t_2} is a rate for money borrowed between two future dates t_1 and t_2 but with a rate agreed upon today.

Consider two different investment alternatives ($m = 1$):

- 1) Invest 1 USD for two years at the rate s_2 . After two years this returns $(1 + s_2)^2$.

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- 1) Invest 1 USD for two years at the rate s_2 . After two years this returns $(1 + s_2)^2$.
- 2) Invest 1 USD for one year at the rate s_1 followed by a new loan running from year 1 to year 2 at a rate $f_{1,2}$ agreed upon today. After two years this returns $(1 + s_1)(1 + f_{1,2})$.

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 - 2) Invest 1 USD for one year at the rate s_1 followed by a new loan running from year 1 to year 2 at a rate $f_{1,2}$ agreed upon today. After two years this returns $(1 + s_1)(1 + f_{1,2})$.
- By no arbitrage, the two investment alternatives should result in the same amount. Hence,

$$(1 + s_2)^2 = (1 + s_1)(1 + f_{1,2}).$$

Rearranging

$$f_{1,2} = \frac{(1 + s_2)^2}{1 + s_1} - 1. \quad (11)$$

- Discrete compounding: For simplicity assume that the compounding frequency equals $m = 1$. More generally we have for $j > i$ the relation

$$(1 + s_j)^j = (1 + s_i)^i (1 + f_{i,j})^{j-i}. \quad (12)$$

Rearranging (12) and using (5) to write in terms of discount factors

$$f_{i,j} = \left(\frac{(1 + s_j)^j}{(1 + s_i)^i} \right)^{\frac{1}{j-i}} - 1 = \left(\frac{d(i)}{d(j)} \right)^{\frac{1}{j-i}} - 1. \quad (13)$$

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$$e^{s_{t_2} t_2} = e^{s_{t_1} t_1} e^{f_{t_1, t_2} (t_2 - t_1)}, \quad (14)$$

which rearranged gives

$$f_{t_1, t_2} = \frac{1}{t_2 - t_1} (s_{t_2} t_2 - s_{t_1} t_1). \quad (15)$$

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- Example on blackboard

Extra slide with $m > 1$

- When $m > 1$ formulas (12) take the form

$$\left(1 + \frac{s_{t_j}}{m}\right)^j = \left(1 + \frac{s_{t_i}}{m}\right)^i \left(1 + \frac{f_{t_i, t_j}}{m}\right)^{j-i}, \quad (16)$$

and (13)

$$f_{t_i, t_j} = m \left(\left(\frac{\left(1 + \frac{s_{t_j}}{m}\right)^j}{\left(1 + \frac{s_{t_i}}{m}\right)^i} \right)^{\frac{1}{j-i}} - 1 \right) = m \left(\left(\frac{d(t_i)}{d(t_j)} \right)^{\frac{1}{j-i}} - 1 \right), \quad (17)$$

where $t_j = \frac{j}{m}$ for all $j = 1, 2, \dots$

Definition

The par rate is defined as the (unique) coupon rate $c(T)$ on a coupon bond with maturity T , that makes the present value of the bond equal to par. Hence, it is the coupon that solves

$$1 = \frac{c(T)}{m} \sum_{i=1}^{mT} d\left(\frac{i}{m}\right) + d(T). \quad (18)$$

Let

$$A(T) = \sum_{i=1}^{mT} d\left(\frac{i}{m}\right) \quad (19)$$

denote the **annuity factor**, i.e. the present value of receiving \$ 1 at every payment date until time T . Isolating $c(T)$ in (18) we find that the par rate equals

$$c(T) = \frac{m(1 - d(T))}{A(T)}. \quad (20)$$

It should be noted here that par rates change as market conditions change, i.e. as the discount factors change.

Prices in terms of par rates

We can use the annuity factor to quote coupon bond prices in a convenient way ($F = 1$)

$$P = \frac{c}{m}A(T) + d(T). \quad (21)$$

Deducting (18) from (21) we get *blackboard*

$$P = 1 + \frac{c - c(T)}{m}A(T). \quad (22)$$

Notice, for $c > c(T)$ the price of the coupon bond is greater than par, $P > 1$, and vice versa.

Floating rate bonds

Theorem

Consider a floating rate bond that has a floating coupon c_i equal to the prevailing spot rate for the period that has just passed $s_{i-1,i}$. Then the post coupon price of the floating rate bond is equal to par at payment dates.

Proof on blackboard

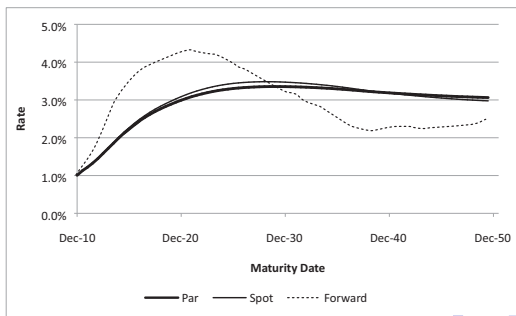
The shape of the term structure of rates

Recall equation (12) with $i = j - 1$

$$(1 + s_j)^j = (1 + s_{j-1})^{j-1} (1 + f_{j-1,j}).$$

This relation allows us to discuss the shape of the term structure.

- If $f_{j-1,j} > s_{j-1}$ this implies that $s_j > s_{j-1}$ and the spot rate curve is increasing.
- It is decreasing if $f_{j-1,j} < s_{j-1}$ implying that $s_j < s_{j-1}$.



The shape of the term structure of rates

Luenberger (2009) mentions three potential explanations for the shape of the term structure:

- 1 Expectations: If the market believes the rates to increase in the future this is reflected in an increasing spot rate term structure.

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- ② Liquidity preferences/risk premia: Longer maturity bonds are more risky. If you need to sell them prematurely you have higher risk in comparison to short maturity bonds. Hence, you require a bigger rate to invest in these.
- ③ Market segmentation: Different investors invest in bonds with different maturities. Hence, prices can reflect this.

Extracting discount courses from market prices

In general we can divide the approaches to determine the discount curve into two categories

- 1 Bootstrapping
- 2 Curve fitting

Bootstrapping

Take a market of n bonds with payments at n different dates

$$\mathbf{P} = \begin{pmatrix} P_1 \\ P_2 \\ \vdots \\ P_n \end{pmatrix} = \begin{pmatrix} B_{11} & 0 & \cdots & 0 \\ B_{21} & B_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ B_{n1} & B_{n2} & \cdots & B_{nn} \end{pmatrix} \begin{pmatrix} d_1 \\ d_2 \\ \vdots \\ d_n \end{pmatrix} = \mathbf{B}\mathbf{d} \quad (23)$$

Bootstrapping procedure on Blackboard.

It is rarely the case that the market has the nice structure as in previous slide. In general, we need n bonds that are "sufficiently different" to solve for the discount factors

$$\mathbf{P} = \begin{pmatrix} P_1 \\ P_2 \\ \vdots \\ P_n \end{pmatrix} = \begin{pmatrix} B_{11} & B_{12} & \cdots & B_{1n} \\ B_{21} & B_{22} & \cdots & B_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ B_{n1} & B_{n2} & \cdots & B_{nn} \end{pmatrix} \begin{pmatrix} d_1 \\ d_2 \\ \vdots \\ d_n \end{pmatrix} = \mathbf{B}\mathbf{d} \quad (24)$$

- We need to matrix $\mathbf{B}$ to have full rank. This is the case when the n bonds have linearly independent cash flow streams.
- We have

$$\mathbf{d} = \mathbf{B}^{-1}\mathbf{P}. \quad (25)$$

Pros:

- The computation of the discount factors $d_i, i = 1, \dots, n$ is exact and all bond prices are priced with no error.
- Simple to compute discount factors.

Cons:

- We only get information about the discount factors at time points $i = 1, \dots, n$. Often one would use some interpolation between payment dates to get the full curve.
- Noise in prices is incorporated into discount factors.
- Linear interpolation results in some undesired shapes of the spot and forward rate curves (non-smoothness).

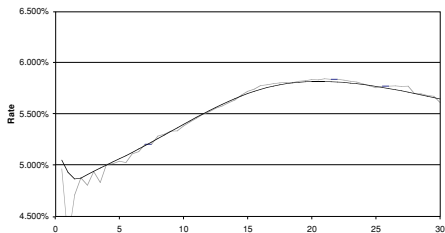


FIGURE 4.5 Spot Curves from Raw C-STRIPS Data and from a Curve Fitting Procedure

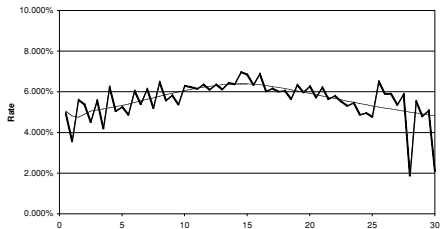


FIGURE 4.6 Forward Curves from Raw C-STRIPS Data and from a Curve Fitting Procedure

Curve fitting is a technique to extract a discount function for all maturities from the set of traded bond prices.

Curve fitting is useful for a number reason:

- 1 Equipped with the discount function, any fixed income cash flow can be valued.
- 2 Since we don't have market prices of bonds for all maturities, curve fitting can be used to fill in the missing prices.

Example: Say you extracted from the market the discount factors d_1 , d_2 and d_3 . How would you price a bond with coupon payments at time points 0.5, 1.5 and 2.5?

Problems with linear interpolation

- We argued that one could interpolate on the discount factors or the spot rates directly. Here the discount factors for the available maturities are needed. Finding the discount factors is easier said than done. For example let $\mathbf{d}$ denote the vector of discount factors, $\mathbf{B}$ the matrix of all payoffs of the bonds and finally let $\mathbf{P}$ be the vector of prices. Then the following holds

$$\underset{N \times 1}{\mathbf{P}} = \underset{N \times n}{\mathbf{B}} \underset{n \times 1}{\mathbf{d}}, \quad (26)$$

where n denotes the number of all future payment dates and N the number of bonds considered.

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- Even if $n = N$ the linear interpolation results in some undesired shapes of the spot and forward rate curves (non-smoothness).

Alternatives to linear interpolation

- Rather than finding a full discount/spot rate curve from interpolating between the available data points, it is often better to use some parametric model for the spot rate curve.

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 - Select the set of bonds used to estimate the curve, with prices $\mathbf{P}^{market}$.
 - Select a **smooth functional form** for the discount factors $d_t(\beta)$ or the spot rates $s_t(\beta)$, where the β is a vector of parameters determining the shape of the curve.

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 - Generate the theoretical prices for the bonds by using the parametric form of the discount functions or the spot rates.
 - Estimate the vector of parameters β^* as the parameters that minimizes the “distance” between market prices and theoretical prices according to some criteria. E.g. the sum of squared errors:

$$SSE = \sum_{i=1}^N \left(P_i^{Market} - P_i^{Model} \right)^2.$$

The Nelson-Siegel model for the yield curve is given by:

$$s_t = \beta_0 + \beta_1 \frac{1 - e^{-\lambda t}}{\lambda t} + \beta_2 \left(\frac{1 - e^{-\lambda t}}{\lambda t} - e^{-\lambda t} \right)$$

Where:

- β_0 is the level factor,
- β_1 is the slope factor,
- β_2 is the curvature factor,
- λ is the decay factor that controls the exponential decay.

Problems with curve-fitting methods

- In general there are fewer bonds in the long end of the curve.
- Illiquidity both in the long and the short end.
- In general the interest rate term structure evolve over time, when market conditions change. The curve-fitting methods discussed in the slide set are static. They only provide the user with a snapshot of the market.
- You will learn more about dynamic models later in the course.

In the lectures on duration measures, we focused on measures based on parallel yield shifts. A more consistent approach is to consider the price sensitivity to parallel shifts in the spot rate curve. The resulting risk measure is termed Fisher-Weil duration.

We will consider the case of continuous compounding. Consider the pricing equation for a given cash flow sequence $\mathbf{x} = (x_{t_0}, x_{t_1}, \dots, x_{t_n})$

$$P = \sum_{i=0}^n x_{t_i} e^{-s_{t_i} t_i}. \quad (27)$$

We introduce a parallel shift δ (not to be confused with the yield of last slide set) to the spot rate curve

$$P(\lambda) = \sum_{i=0}^n x_{t_i} e^{-(s_{t_i} + \delta) t_i}. \quad (28)$$

The price sensitivity to a parallel shift in the spot rate curve is then equal to

$$\frac{dP(\delta)}{d\delta} \Big|_{\delta=0} = - \sum_{i=0}^n t_i x_{t_i} e^{-s_{t_i} t_i}. \quad (29)$$

Definition

The Fisher-Weil duration of the cash flow stream $\mathbf{x}$ is defined as

$$D_{FW} = - \frac{1}{P(0)} \frac{dP(\delta)}{d\delta} \Big|_{\delta=0} = \frac{1}{P(0)} \sum_{i=0}^n t_i x_{t_i} e^{-s_{t_i} t_i}. \quad (30)$$

Similarly, we can define a first-order sensitivity measure when spot rates are compounded discretely. Here we choose to present the definition without going into details.

Definition

With compounding m times per year, the Quasi-Modified duration equals

$$D_Q = -\frac{1}{P(0)} \frac{dP(\delta)}{d\delta} \Big|_{\delta=0} = \frac{1}{P(0)} \sum_{k=1}^n \frac{k}{m} x_k \left(1 + \frac{s_{k/m}}{m}\right)^{-(k+1)}. \quad (31)$$

Example on Blackboard

Conclusion

So far in the course we have focused on fixed income securities markets and we have had attention on securities with fixed future payments. Financial markets comprise many other types of assets. In particular, for many assets only the initial price is known while the future cash flows are uncertain.

We will add stochasticity to the setup in the weeks to come. Next week, our focus will be on the single-period portfolio allocation framework introduced by Markowitz. → exciting and groundbreaking stuff!

Investment and Finance

Portfolio Theory

Jakob Bloch

Aarhus University
School of Business and Social Sciences
Department of Economics and Business Economics

Out line of this week's lectures

- Motivation
- Assumptions and notation
- Short selling
- Intro about portfolio diversification
- Plotting portfolios in a $(\sigma, \bar{r})$ -diagram.
- The Markowitz problem
 - Without a risk-free asset
 - The minimum variance portfolio
 - The two-fund separation theorem
 - Including a risk-free asset
 - The one fund theorem
 - The tangency portfolio

Motivation

So far we have focused on assets where the future stream of cash flows were certain. For many other types of assets only the initial price paid is known while the future cash flows are uncertain. In order to model this, we need to introduce stochasticity to the setup.

We will start the analysis considering the case where uncertainty is resolved after a single period. We are interested in an investor who wishes to invest her money in a portfolio of assets taking into account the mean return of the portfolio and the variance of the return. Hence, the investor's utility function does not take into account quantities such as skewness, kurtosis etc.

The framework was first proposed in 1952 by Harry Markowitz in the seminal Journal of Finance paper "Portfolio Selection". The paper is the foundation of modern portfolio theory and Markowitz was later awarded the Nobel price for his work.

Main assumptions

- One-period model. All uncertainty is resolved after a single period.
- Investors only care about the mean and variance of the asset returns.
- Investors all agree on the expected returns, the variance of returns, and how returns covary with each other.
- All information is available and accessible at no cost.
- No transactions costs.
- Assets are infinitely divisible.
- Agents are rational and risk-averse.

- n : The number of risky assets.
- P_{it} : The price of asset $i \in \{1, \dots, n\}$ at time $t \in \{0, 1\}$.
- $R_i = \frac{P_{i1}}{P_{i0}}$: Total return of asset i .
- $r_i = \frac{P_{i1} - P_{i0}}{P_{i0}}$: Rate of return (or just the return) of asset i .
- $\bar{r}_i = \mathbb{E}[r_i]$: The expected return of asset i .
- $\sigma_i^2 = \mathbb{E}[(r_i - \bar{r}_i)^2]$: The variance of r_i .
- $\sigma_{ij} = \mathbb{E}[(r_i - \bar{r}_i)(r_j - \bar{r}_j)]$: The covariance between the returns of asset i and j .
- $\rho_{ij} = \frac{\sigma_{ij}}{\sigma_i \sigma_j} \in [-1, 1]$: The correlation between the returns of asset i and j .
- w_i : The fraction of investor's wealth invested in asset i . Hence, we have that $\sum_{i=1}^n w_i = 1$.

In matrix format

We will see that in many cases it is convenient to do computations in matrix form. Hence, we put the variables in matrix form.

- $\mathbf{r} = (r_1, \dots, r_n)^\top$: Vector of random returns.
- $\bar{\mathbf{r}} = (\bar{r}_1, \dots, \bar{r}_n)^\top$: Vector of expected returns.
- $\Sigma = \begin{pmatrix} \sigma_1^2 & \cdots & \sigma_{1n} \\ \vdots & \ddots & \vdots \\ \sigma_{n1} & \cdots & \sigma_n^2 \end{pmatrix}$: The covariance matrix of the returns.
- $\mathbf{w} = (w_1, \dots, w_n)^\top$: Vector of portfolio weights.
- $\mathbf{1}$: $n \times 1$ vector of ones.

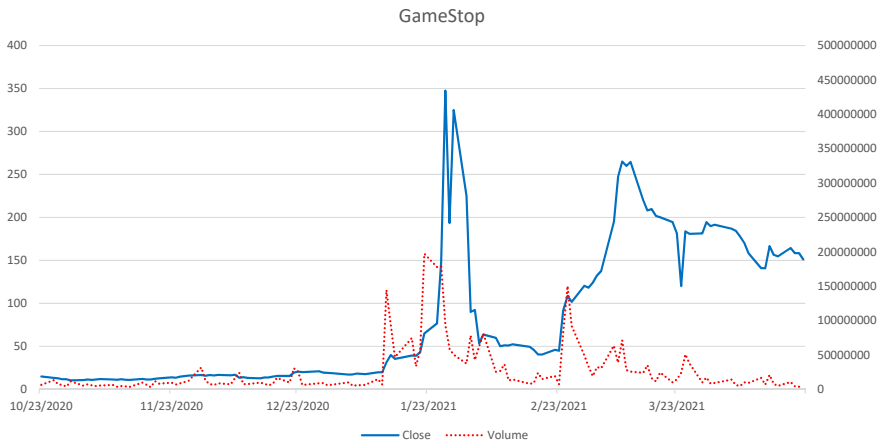
Short sales

Often it is the case that the “optimal” portfolio will contain short positions in assets. Short selling an asset refers to the procedure of selling an asset that you do not own. In practice, one do this by borrowing an asset (in exchange for collateral), which you then immediately sell. At a later date, you buy back the asset in the financial market and return it to the original owner (in exchange for the collateral).

- You profit $P_0 - P_1$ if the price has dropped over the period and loose if the price has increased.
- In contrast to a long position where maximum loss equals 100%, the potential loss is unlimited!
- We can also define a return as $r = -\frac{P_1 - P_0}{P_0}$.

Example on blackboard.

The GameStop case



- From November 23, 2020 - April 22, 2021: Minimum price USD 10.47 on November 30, 2020. Maximum price USD 347.51 on January 27, 2021.

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- Citron Research was one of the hedge funds that lost a lot of money.

Portfolio return

Let the initial wealth of the investor be denoted X_0 . Then the portfolio weight equals $w_i = \frac{X_{i0}}{X_0}$, where X_{i0} is the dollar amount invested in asset i at time 0. We have the following relations:

- The return on the portfolio equals

$$r_p = \frac{\sum_{i=1}^n r_i X_{i0}}{X_0} = \sum_{i=0}^n r_i w_i = \mathbf{w}^\top \mathbf{r}. \quad (1)$$

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- The expected return on the portfolio equals:

$$\bar{r}_p = \sum_{i=0}^n \bar{r}_i w_i = \mathbf{w}^\top \bar{\mathbf{r}}. \quad (2)$$

- The variance of the portfolio return equals (Blackboard)

$$\sigma_p^2 = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \sigma_{ij} = \mathbf{w}^\top \Sigma \mathbf{w}. \quad (3)$$

Example of two asset portfolio on blackboard

Portfolio return variance and covariance

- The variance of the portfolio return equals (Blackboard)

$$\sigma_p^2 = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \sigma_{ij} = \mathbf{w}^\top \Sigma \mathbf{w}. \quad (3)$$

- Similarly, we have the formula for the covariance between two portfolios p and p'

$$\sigma_{p,p'} = \sum_{i=1}^n \sum_{j=1}^n w_i^p w_j^{p'} \sigma_{ij} = \mathbf{w}_p^\top \Sigma \mathbf{w}_{p'} \quad (4)$$

Example of two asset portfolio on blackboard

Diversification - independent assets

Diversification, refers to the task of reducing overall portfolio variance by combining your assets in a “clever” way.

- Consider the case of n uncorrelated assets all sharing the same mean return $\bar{r}_i = m$ and variance $\sigma_i^2 = \sigma^2$. Now take an equally weighted portfolio in the n assets, $w_i = 1/n$. The portfolio return equals

$$r_p = \frac{1}{n} \sum_{i=1}^n r_i, \quad (5)$$

with mean

$$\bar{r}_p = \frac{1}{n} \sum_{i=1}^n m = m. \quad (6)$$

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$$\sigma_p^2 = \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{\sigma^2}{n}. \quad (7)$$

- Notice how the portfolio variance can be reduced all the way down to zero for $n \rightarrow \infty$ without lowering the expected return.

Diversification - Example: Dependent assets

We stay in a setting as in previous slides, except now the n assets are positively correlated with covariance

$$\sigma_{ij} = 0.3\sigma^2. \quad (8)$$

Now the equally weighted portfolio variance takes the form

$$\begin{aligned} \sigma_p^2 &= \sum_{i=1}^n \sum_{j=1}^n \frac{1}{n^2} \sigma_{ij} = \frac{1}{n^2} \left(\sum_{i=j} \sigma_{ij} + \sum_{i \neq j} \sigma_{ij} \right) \\ &= \frac{1}{n^2} (n\sigma^2 + 0.3(n^2 - n)\sigma^2) = \frac{\sigma^2}{n} + 0.3\sigma^2 \left(1 - \frac{1}{n}\right) \\ &= \frac{0.7\sigma^2}{n} + 0.3\sigma^2. \end{aligned}$$

Hence, not possible to lower the portfolio variance below $0.3\sigma^2$.

- **Unsystematic risk:** Risk that is connected to a specific asset and uncorrelated to other assets. This type of risk can be diversified away.

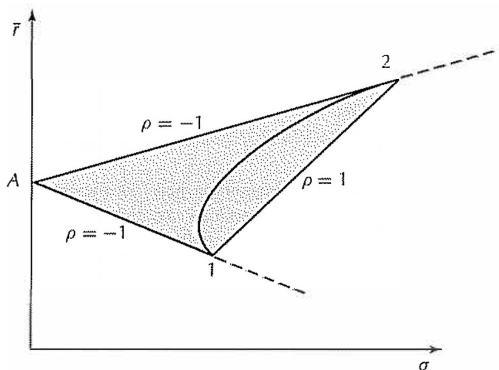
Some terminology

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- **Systematic risk:** Risk that all the assets are exposed to and therefore not possible to diversify away.
- The minimum portfolio variance depends on the structure of the risks between the available assets (i.e. it depends on the covariance matrix Σ). The more positively correlated the assets are, the less can the portfolio variance be reduced by diversification.

Portfolios in a $(\bar{r}, \sigma)$ -diagram

Consider a portfolio of two assets, 1 and 2, with weights $w_1 = 1 - \alpha$ and $w_2 = \alpha$.



Discuss

Introducing more assets - The feasible set

Now we introduce n assets and plot them all in a $(\bar{r}, \sigma)$ -diagram. If we form all arbitrary portfolios of them with $\sum_{i=1}^n w_i = 1$ we get the feasible set.

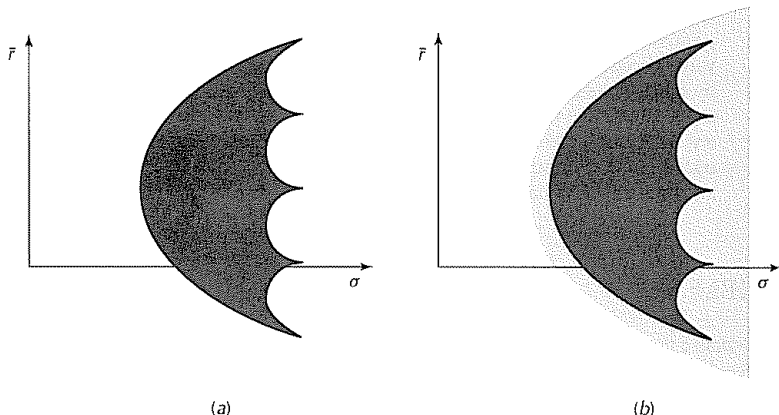


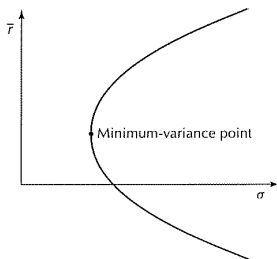
FIGURE 6.10 Feasible region. The feasible region is the set of all points representing portfolios made from n original assets. Two such regions can be defined: (a) no shorting and (b) shorting allowed

Remarks to the Figure

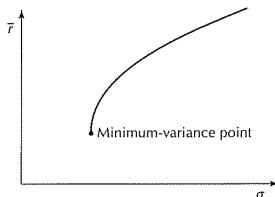
- When you have at least three assets available, the feasible set will be a two-dimensional region.
- The feasible set is convex to the left.

Some terminology

- **The critical frontier:** The set of portfolios defined by the minimum variance for a given return is termed the minimum-variance set or the critical frontier. A risk-averse investor will for a given return always choose the portfolio with the lowest variance.



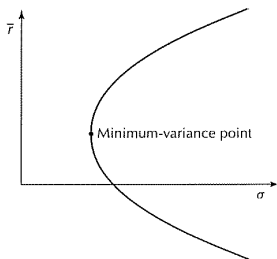
(a) Minimum-variance set



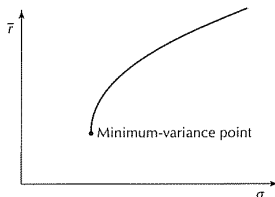
(b) Efficient frontier

Some terminology

- **The critical frontier:** The set of portfolios defined by the minimum variance for a given return is termed the minimum-variance set or the critical frontier. A risk-averse investor will for a given return always choose the portfolio with the lowest variance.
- **The efficient frontier:** A rational (greedy) investor will always choose the portfolio with highest return when variances are the same. The set of portfolios defined by the highest expected return for a given variance is termed the efficient frontier.



(a) Minimum-variance set



(b) Efficient frontier

The Markowitz framework

In the setting presented, an investor is exposed to the following optimization problem when deciding upon a portfolio allocation with a given required expected return $\bar{r}$:

$$\min_{\mathbf{w}} \frac{1}{2} \sigma_p^2 = \min_{\mathbf{w}} \frac{1}{2} \mathbf{w}^\top \Sigma \mathbf{w}, \quad (9)$$

subject to

$$\sum_{i=1}^n w_i \bar{r}_i = \mathbf{w}^\top \bar{\mathbf{r}} = \bar{r}_p$$

$$\sum_{i=1}^n w_i = \mathbf{w}^\top \mathbf{1} = 1.$$

Solution on Blackboard

Remarks:

- The $\frac{1}{2}$ in front of the objective function is for convenience. It does not change the optimal allocation.
- We will include a risk-free asset at a later stage.

Solution in matrix format

If Σ is invertible, the solution to (9) can be written as

$$\mathbf{w}^* = \lambda_1 \Sigma^{-1} \bar{\mathbf{r}} + \lambda_2 \Sigma^{-1} \mathbf{1} \quad (10)$$

$$= \frac{a \bar{r}_p - b}{d} \Sigma^{-1} \bar{\mathbf{r}} + \frac{c - \bar{r}_p b}{d} \Sigma^{-1} \mathbf{1}, \quad (11)$$

where (first two inequalities due to Σ being positive semidefinite)

$$a = \mathbf{1}^\top \Sigma^{-1} \mathbf{1} > 0 \quad (12)$$

$$b = \mathbf{1}^\top \Sigma^{-1} \bar{\mathbf{r}} \quad (13)$$

$$c = \bar{\mathbf{r}}^\top \Sigma^{-1} \bar{\mathbf{r}} > 0 \quad (14)$$

$$d = ac - b^2 > 0 \quad (15)$$

Remarks:

- $\mathbf{w}^*$ is on the critical frontier and is the portfolio with lowest variance for a given return $\bar{r}_p$.
- One can check that the second order conditions for the solution to be a minimum is full filled. We skip it here.

- The return of the optimal portfolio is

$$r_p = (\mathbf{w}^*)^\top \mathbf{r}.$$

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Blackboard

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Blackboard

- Hence, σ_p^2 is a parabola in $\bar{r}_p$.
- The efficient frontier is the top part of the critical frontier.

The minimum variance portfolio (MVP)

If we skip the condition on the required return in the optimization problem (9) we get the *minimum variance portfolio*. Alternatively, we can find the MVP by finding the minimum of the parabola in (16)

$$0 = \frac{\partial \sigma_p^2}{\partial \bar{r}_p} = \frac{2a\bar{r}_p - 2b}{d} \quad (17)$$

We have that (Blackboard)

$$\begin{aligned}\bar{r}_{MVP} &= \frac{b}{a} \\ \sigma_{MVP}^2 &= \frac{1}{a} \\ \mathbf{w}_{MVP} &= \frac{1}{a} \Sigma^{-1} \mathbf{1}.\end{aligned}$$

Example of three asset universe

Short selling constraints

Short selling constraint can be included by adding the condition

$$w_i \geq 0, \quad \forall i = 1, \dots, n \quad (18)$$

to the conditions in the optimization problem (9). However, the problem becomes harder to solve and one will in most cases need a software program to solve it.

The two-fund separation theorem

Theorem

Any portfolio on the efficient frontier can be attained by investing in two efficient portfolios.

Proof on Blackboard

- Very strong result! Investors only need to identify two optimal portfolios (mutual funds) in order to be able to invest in all efficient portfolios.

The two-fund separation theorem

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Any portfolio on the efficient frontier can be attained by investing in two efficient portfolios.

Proof on Blackboard

- Very strong result! Investors only need to identify two optimal portfolios (mutual funds) in order to be able to invest in all efficient portfolios.
- Bear in mind though that the result depends on a lot of strict assumptions. For instance, 1. investors are only concerned about mean and variance, 2. all investors agree upon mean and variances, 3. single period model.

Adding a risk-free asset

We include a risk-free asset and label it as asset 0. We will see how the analysis lead to a simple shape of the efficient frontier. We have

- $\bar{r}_0 = r_0 = r_f$.
- $\sigma_0^2 = 0$.
- $\sigma_{i0} = 0, \quad \forall i = 1, \dots, n$.
- $w_0 = 1 - \mathbf{w}^\top \mathbf{1}$.

We assume that we can borrow and lend unlimited in the risk-free asset.

Return and variance when a risk-free asset is included

Assume you put $1 - w_0$ in some risky asset with expected return $\bar{r}$ and variance σ^2 . The residual w_0 you invest in the risk-free asset with return r_f . Then we have that

$$\bar{r}_p = w_0 r_f + (1 - w_0) \bar{r} \quad (19)$$

$$\sigma_p^2 = (1 - w_0)^2 \sigma^2 \quad \Leftrightarrow \quad \sigma_p = (1 - w_0) \sigma \quad (20)$$

Hence, both the mean and the standard deviation vary linearly with w_0 .

The one-fund theorem

Theorem

There is a single efficient fund F of risky assets such that any efficient portfolio is a combination between the fund F and the risk-free asset.

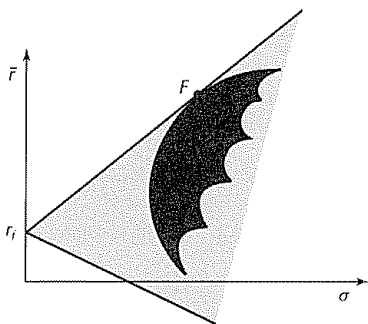


FIGURE 6.14 One-fund theorem. When both borrowing and lending at the risk-free rate are allowed, there is a unique fund F of risky assets that is efficient. All points on the efficient frontier are combinations of F and the risk-free asset.

Remark: Main insight leading to the topic of next week, the Capital Asset Pricing Model (CAPM).

We can also write up the problem in terms of the risky portfolio weights

$$\bar{r}_p = w_0 r_f + \mathbf{w}^\top \bar{\mathbf{r}} \quad (21)$$

$$\sigma_p^2 = \text{Var} \left(w_0 r_f + \mathbf{w}^\top \bar{\mathbf{r}} \right) = \mathbf{w}^\top \Sigma \mathbf{w} \quad (22)$$

Denote by $r_i^e = r_i - r_f$ the excess return of asset i . In vector notation we have that

$$\bar{\mathbf{r}}^e = \bar{\mathbf{r}} - r_f. \quad (23)$$

The return of the portfolio can be formulated in terms of excess returns

Blackboard

$$\bar{r}_p^e = \mathbf{w}^\top \bar{\mathbf{r}}^e. \quad (24)$$

Optimization problem with risk-free asset

Using the results of the previous slide we have that the portfolio return constraint in the Markowitz optimization problem can be reformulated in terms of an excess return constraint

$$\min_{\mathbf{w}} \frac{1}{2} \sigma_p^2 = \min_{\mathbf{w}} \frac{1}{2} \mathbf{w}^\top \Sigma \mathbf{w}, \quad (25)$$

subject to

$$\mathbf{w}^\top \bar{\mathbf{r}}^e = \bar{r}_p^e. \quad (26)$$

Notice how the weights need not add to one since the residual is just placed in the risk-free asset.

Solve on the blackboard

The optimization problem can be solved to yield that

$$\mathbf{w}^* = \frac{\bar{r}_p^e}{(\bar{\mathbf{r}}^e)^\top \Sigma^{-1} \bar{\mathbf{r}}^e} \Sigma^{-1} \bar{\mathbf{r}}^e, \quad (27)$$

with portfolio variance

$$\sigma_p^2 = \frac{(\bar{r}_p^e)^2}{(\bar{\mathbf{r}}^e)^\top \Sigma^{-1} \bar{\mathbf{r}}^e}. \quad (28)$$

Rearranging and taking the square root

$$\bar{r}_p^e = \pm \sigma_p \left((\bar{\mathbf{r}}^e)^\top \Sigma^{-1} \bar{\mathbf{r}}^e \right)^{\frac{1}{2}} \quad (29)$$

We observe that in the $(\sigma_p, \bar{r}_p)$ diagram this critical frontier consists of two straight lines connected in the point $(0, r_f)$.

- Like in the “no risk-free” asset case any portfolio on the critical frontier can be formed by two separate portfolios on the frontier.

The tangency portfolio

If we choose the portfolio with no investment in the risk free asset we have the “tangency” portfolio. Hence, in this case we have $\mathbf{w}^\top \mathbf{1} = 1$ and $w_0 = 0$.

- Entering $\mathbf{1}^\top \mathbf{w} = 1$ in (27) we get (Blackboard)

$$\bar{r}_{tan}^e = \frac{(\bar{\mathbf{r}}^e)^\top \Sigma^{-1} \bar{\mathbf{r}}^e}{\mathbf{1}^\top \Sigma^{-1} \bar{\mathbf{r}}^e}. \quad (30)$$

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- Insert in (27) (Blackboard)

$$\mathbf{w}_{tan} = \frac{\Sigma^{-1} \bar{\mathbf{r}}^e}{\mathbf{1}^\top \Sigma^{-1} \bar{\mathbf{r}}^e}. \quad (31)$$

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- One-fund theorem: All efficient portfolios can be formed by the risk-free asset and the tangency portfolio!
- Continue the three asset example.

The mean and variance assumption

The assumption that investors only care about the mean return and the variance of returns can be shown to hold if one of the following two cases hold:

- 1 The returns are normally distributed with

$$\mathbf{r} \sim N(\bar{\mathbf{r}}, \Sigma) \quad (32)$$

- 2 The utility function of investors are quadratic meaning that

$$U(r_p) = r_p - br_p^2. \quad (33)$$

Conclusion

In the present slide set, we focused on a situation of an investor who wished to optimally allocate a given wealth among a universe of risky assets. In particular, we took the prices of the available assets as given.

In the next slide set, we will turn the question around and consider a model for pricing of risky assets given a covariance structure in the market. The analysis will lead to the so-called CAPM model of Treynor, Sharpe, Lintner, and Mossin. Get ready for more Nobel price winning material.

Investment and Finance

The Capital Asset Pricing Model

Jakob Bloch

Aarhus University
School of Business and Social Sciences
Department of Economics and Business Economics

Out line of this week's lectures

- Motivation and assumptions
- The market portfolio
- Equilibrium pricing
- The capital asset pricing model
- The security market line
- Systematic and unsystematic risks
- Performance evaluation
- Critique of the CAPM

Motivation

In the previous slide set we focused on the optimal allocation of wealth in a financial market of assets with and without a risk-free asset. In order to do this, we took the characteristics of the assets for given. In particular, the expected returns of the assets were given.

In this slide set we will turn the argument around and study a framework for computing the expected return of risky assets. Knowledge of expected returns can then be used to compute prices of risky assets.

We will stay in the mean-variance setting and study the Capital Asset Pricing Model (CAPM) proposed by Treynor, Sharpe, Lintner, and Mossin in different papers published in the 60ties. Together with Markowitz, Sharpe was the recipient of the Nobel price in financial economics in 1990.

Assumptions

We stay in the assumptions of the previous slide set.

- One period model
- Investors are mean-variance optimizers.
- Investors are homogeneous in their beliefs meaning that they agree on the probabilistic structure of the assets.
- The existence of a unique borrowing and lending rate r_f .
- No transactions costs.
- The market is in equilibrium.

- The assumptions on the previous slide implies that all investors will agree on the location of the efficient frontier and they will all invest in portfolios on the efficient frontier.

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- Investors might position themselves differently on the efficient frontier based on their risk-preferences. However, we showed that all efficient portfolios can be attained as a combination of the risk-free asset and the tangency portfolio. Hence, all investors invest a fraction of their wealth in the tangency portfolio and the rest in the risk-free asset.

Optimal portfolios

- The assumptions on the previous slide implies that all investors will agree on the location of the efficient frontier and they will all invest in portfolios on the efficient frontier.
- Investors might position themselves differently on the efficient frontier based on their risk-preferences. However, we showed that all efficient portfolios can be attained as a combination of the risk-free asset and the tangency portfolio. Hence, all investors invest a fraction of their wealth in the tangency portfolio and the rest in the risk-free asset.
- Can we say something about the tangency portfolio in equilibrium?

The market portfolio

We use the following notation:

- n is the number of risky assets.
- P_{j0} is the price of asset j at time 0.
- X_j is the number of shares/units of asset j in the economy.

We define the market portfolio as the portfolio with weights w_M^j given by the supply of asset j relative to the total value of the market W_M

$$w_M^j := \frac{X_j P_{j0}}{\sum_{k=1}^n X_k P_{k0}} = \frac{X_j P_{j0}}{W_M} \quad \text{for } j = 1, \dots, n. \quad (1)$$

Equilibrium pricing - I

Assume we have I investors and denote by W_i and w_i^j the wealth and portfolio weight in asset j of investor i .

- The total wealth in the economy equals

$$W = \sum_{i=1}^I W_i. \quad (2)$$

Equilibrium pricing - I

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- In equilibrium the total wealth in the economy should be equal to the total value of the market $W = W_M$ and the value of the individual asset j satisfy

$$\sum_{i=1}^I w_i^j W_i = X_j P_{j0} = w_M^j W_M. \quad (3)$$

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$$\sum_{i=1}^I w_i^j W_i = X_j P_{j0} = w_M^j W_M. \quad (3)$$

- Hence, the market portfolio weight is a linear combination of the weights of the investors

$$w_M^j = \sum_{i=1}^I w_i^j \frac{W_i}{W_M}. \quad (4)$$

- We conclude that the market portfolio is on the efficient frontier since all the investor's portfolios are on the efficient frontier.

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- From the definition of the market portfolio weights, we have that

$$\mathbf{w}_M^\top \mathbf{1} = 1. \quad (5)$$

Equilibrium pricing - II

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Equilibrium pricing - II

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- The previous two bullet points imply that the market portfolio is the tangency portfolio! This is the main insight behind the CAPM theory.
- In equilibrium, market prices adjust such that the weights in the market portfolio equals the weights in the tangency portfolio $\rightarrow$ in equilibrium market clears and demand meets the supply.

The capital market line

In the lectures on the Markowitz framework we argued that the efficient frontier could be depicted in a $(\sigma, \bar{r})$ -diagram as a straight line with intersection at $(0, r_f)$ and running through the tangency portfolio. Substituting the tangency portfolio with the market portfolio we arrive at the *Capital Market Line* (CML).

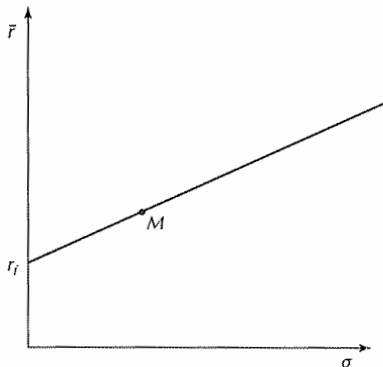


FIGURE 7.1 Capital market line. Efficient assets must all lie on the line determined by the risk-free rate and the market portfolio

We can put the capital market line on a formula which states the relation between the standard deviation of an *efficient* portfolio and the expected return

$$\bar{r}_p = r_f + \frac{\bar{r}_M - r_f}{\sigma_M} \sigma_p. \quad (6)$$

- The slope of the CML is given by $\frac{\bar{r}_M - r_f}{\sigma_M}$ and is referred to as the price of risk as it tells you how much expected return you get for taking on risk in the form of σ_p .

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- The CML only tells us something about the return/prices of *efficient* portfolios!
- We are also interested in the returns/prices of assets that are not on the efficient frontier → CAPM to the rescue.

The Capital Asset Pricing Model

Theorem

In equilibrium, the expected return $\bar{r}_i$ of an asset i satisfies

$$\bar{r}_i - r_f = \beta_i (\bar{r}_M - r_f), \quad (7)$$

where

$$\beta_i = \frac{\sigma_{iM}}{\sigma_M^2}. \quad (8)$$

Proof on blackboard

- β_i is the covariance between asset i and the market portfolio normalized by the variance of the market.
- Example 7.3 on blackboard

The CAPM pricing formula has some important consequences:

- The expected excess return on a asset i is a linear function of the asset's β . The slope of the function equals the excess return on the market portfolio $\bar{r}_M - r_f$ and we can label this as the risk-premium.

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- The risk of an asset can be decomposed into systematic risk and unsystematic risk. Only the systematic risk measured by the asset's *beta* is relevant for pricing. The unsystematic risk is not priced as this part can be diversified away.

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- The risk of an asset can be decomposed into systematic risk and unsystematic risk. Only the systematic risk measured by the asset's *beta* is relevant for pricing. The unsystematic risk is not priced as this part can be diversified away.
- Remember, the assumptions of the model imply that *all* investors are diversified and invest in the market portfolio and the risk-free asset.

β and expected return

- If $\beta_i > 1$ we have that $\bar{r}_i > \bar{r}_M$. The asset is risky in the sense that it is worth a lot in upturns and worth a little in downturns. Hence, not a good diversifier and investors are only willing to pay a “low” price today $\rightarrow$ “high” return.

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- If $\beta_i = 1$ we have that $\bar{r}_i = \bar{r}_M$. In all practical aspects, the asset is the market.

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- If $\beta_i = 1$ we have that $\bar{r}_i = \bar{r}_M$. In all practical aspects, the asset is the market.
- If $\beta_i = 0$ we have that $\bar{r}_i = r_f$. Even if the stock is risky in the sense that it has a high variance, it has expected return equal to the risk-free rate. Since $\beta_i = 0$ it implies that all risk is unsystematic and can be diversified away.

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- If $\beta_i < 0$ we have that $\bar{r}_i < r_f$. An asset with high variance can still have expected return less than the risk-free rate. When the market is in an upturn the asset is worth a little while in downturns it is worth a lot. Hence, very attractive for diversification and investors are willing to accept a low return $\rightarrow$ think of the asset as an insurance.

β of a portfolio

Consider a portfolio $\mathbf{w} = (w_1, \dots, w_n)^\top$ where the individual assets have betas $\boldsymbol{\beta} = (\beta_1, \dots, \beta_n)^\top$ and recall that the portfolio return has the form

$$r_p = \sum_{i=1}^n w_i r_i = \mathbf{w}^\top \mathbf{r}. \quad (9)$$

The beta of the portfolio can be computed as

$$\beta_p = \frac{\sigma_{pM}}{\sigma_M^2} = \frac{\sum_{i=1}^n w_i \sigma_{iM}}{\sigma_M^2} = \sum_{i=1}^n w_i \beta_i = \mathbf{w}^\top \boldsymbol{\beta}. \quad (10)$$

Hence, the beta of a portfolio is just the weighted average of the betas of the individual assets.

The security market line

If we plot the expected return of asset i against β_i or σ_{iM} , we get two alternative representations of the *security market line*

$$\bar{r}_i = r_f + \beta_i(\bar{r}_M - r_f) = r_f + \sigma_{iM} \frac{\bar{r}_M - r_f}{\sigma_m^2}. \quad (11)$$

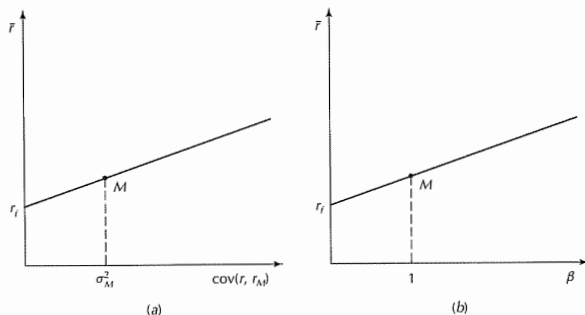


FIGURE 7.3 Security market line. The expected rate of return increases linearly as the covariance with the market increases or, equivalently, as β increases

- While the capital market line relates returns to the standard deviations of *efficient* portfolios, the security market line relates the expected return of *all* assets to the covariance of the asset with the market. Hence, if a given asset does not lie on the security market line, we can talk about the asset being mispriced. **Blackboard drawing**

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- If we take the difference between the expected returns of two assets and apply the SML we get

$$\bar{r}_i - \bar{r}_j = (\beta_i - \beta_j)(\bar{r}_M - r_f). \quad (12)$$

Hence, differences in expected returns are entirely driven by differences in asset betas.

- While the capital market line relates returns to the standard deviations of *efficient* portfolios, the security market line relates the expected return of *all* assets to the covariance of the asset with the market. Hence, if a given asset does not lie on the security market line, we can talk about the asset being mispriced. **Blackboard drawing**
- If we take the difference between the expected returns of two assets and apply the SML we get

$$\bar{r}_i - \bar{r}_j = (\beta_i - \beta_j)(\bar{r}_M - r_f). \quad (12)$$

Hence, differences in expected returns are entirely driven by differences in asset betas.

- In comparison to the CML, the SML highlights that the relevant risk quantity is β and not σ .

Decomposing the variance of an asset

Take the CAPM equation as starting point and write the return of asset i as (notice: actual returns and not expected returns)

$$r_i = r_f + \beta_i(r_M - r_f) + \epsilon_i, \quad (13)$$

where ϵ_i is a term that ensures the equality.

- The CAPM equation implies that $\mathbb{E}[\epsilon_i] = 0$ and that $\text{cov}(\epsilon_i, r_M) = 0$. **Blackboard**

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- We label $\beta_i^2 \sigma_m^2$ as the systematic risk of asset $i \rightarrow$ captures the part of the risk that correlates with the market.
- We label $\text{var}(\epsilon_i)$ the unsystematic (idiosyncratic) risk of asset i . Captures the risk that is independent of the market and can be diversified away.

Consider an asset i with expected return according to the CAPM

$$\bar{r}_i = r_f + \beta_i(\bar{r}_M - r_f), \quad (15)$$

and variance

$$\sigma_i^2 = \text{var}(\beta_i r_M + \epsilon_i) = \beta_i^2 \sigma_m^2 + \text{var}(\epsilon_i).$$

Can we form a portfolio that has same expected return but lower variance?

- Take a portfolio that invests $(1 - \beta_i)$ in the risk-free asset and β_i in the market portfolio.

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- From the Markowitz slide set we know that the new portfolio is efficient. Hence, not possible to form a portfolio with lower variance and same expected return.
- We can always form a portfolio that diversifies away $\text{var}(\epsilon_i)$. Hence, investors are not rewarded to take unsystematic risk.

Systematic and unsystematic risk illustrated

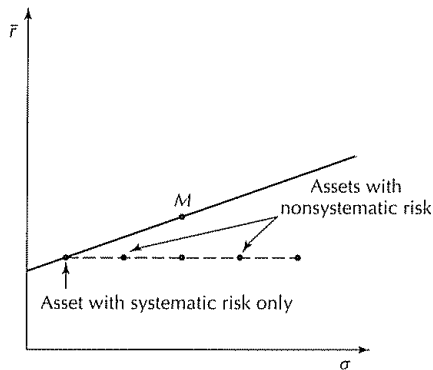


FIGURE 7.4 Systematic and nonsystematic risk. An asset on the capital market line has only systematic risk. Assets with nonsystematic risk fall to the right of the capital market line.

Luenberger presents two different approaches to performance evaluation.

- ① Jensen's alpha
- ② The Sharpe index

Jensen's alpha

Denote by $\hat{r}_i$ the historical average of the returns of asset i and by $\hat{\beta}_i$ the empirical beta of asset i . Jensen's alpha is defined from (termed J in Luenberger)

$$\hat{r}_i - r_f = \alpha_i + \hat{\beta}_i(\hat{r}_M - r_f). \quad (18)$$

According to CAPM, the α should be zero. An alpha different from zero can essentially occur for four reasons:

- 1 Not enough data meaning that quantities are not precisely estimated.
- 2 Maybe wrong choice for the market portfolio when computing $\hat{r}_m$.
- 3 Consider the case that $\alpha_0 > 0$. Asset i is a “good asset” (although this should not be possible according to CAPM assumptions).
- 4 The CAPM model is a flawed representation of the market.

The Sharpe index

The Sharpe index of an asset i is computed based on historical data as

$$S_i = \frac{\hat{r}_i - r_f}{\hat{\sigma}_i}. \quad (19)$$

Hence, in a (σ, r) -diagram the Sharpe index is the slope of the line running through r_f and the asset i .

- In theory, an asset should not be able to achieve a higher Sharpe index than the market. Hence, $S_i > S_M$ could be interpreted as either a “good” asset or as a model critique.

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- In theory, an asset should not be able to achieve a higher Sharpe index than the market. Hence, $S_i > S_M$ could be interpreted as either a “good” asset or as a model critique.
- A $S_i < S_M$ does not necessarily mean that the asset is “bad”. It just says the the asset is not efficient in itself.

CAPM as a pricing formula

How can we reformulate the CAPM equation as a pricing equation?
Rewrite the expected return on left hand side in terms of prices

$$\bar{r}_i = \mathbb{E} \left[\frac{P_{i1} - P_{i0}}{P_{i0}} \right] = \frac{\mathbb{E} [P_{i1}]}{P_{i0}} - 1 = r_f + \beta_i (\bar{r}_M - r_f). \quad (20)$$

Rearranging results in

$$P_{i0} = \frac{\mathbb{E} [P_{i1}]}{1 + r_f + \beta_i (\bar{r}_M - r_f)}. \quad (21)$$

- When the future payoff is stochastic, a premium $\beta_i (\bar{r}_M - r_f)$ should be added to the discount rate in order to get the correct price.

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- Interpretation: A risk averse investor will charge a risk-premium for investing in risky assets, i.e. the investor pays a lower price today.

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- Interpretation: A risk averse investor will charge a risk-premium for investing in risky assets, i.e. the investor pays a lower price today.
- It can be shown that the CAPM relation implies linear pricing rules. However, we skip it here.

Critique of the CAPM

A number of empirical studies have documented that the CAPM is too simplistic to represent real markets.

- The empirical connection between $\bar{r}_i$ and β_i is too flat compared to the SML implied by the CAPM (Fama and French 2004). Hence, empirically low beta stocks earn too high a return while high beta stocks earn too low a return.

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- More factors than just the market excess return is needed to model returns of assets (next slide set).

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- Roll's critique: In theory, the market portfolio is comprised of all risky assets in the world. Hence, the CAPM model is impossible to implement and to test in practice. A theory that can not be tested is not a "good" theory.
- In addition, one can argue against many of the assumptions also presented in the slide set on the Markowitz framework. For instance, one period model, homogeneous beliefs, mean-variance optimizing investors, etc.

Conclusion

In the current slide set, we studied a model where market returns depend on a single factor, namely the excess return of the market portfolio. If we relax many of the assumptions behind the CAPM, the theory does not hold. However, it might still be the case that we can find one or more factors suitable for modeling returns of assets.

In the next slideset, we will look at factor models and expand the number of factors to be greater than one. Furthermore, we will study the Arbitrage Pricing Theory (APT) developed by Stephen Ross in his paper “The arbitrage theory of capital asset pricing” published in 1976.

Investment and Finance

Factor Models and Arbitrage Pricing Theory

Jakob Bloch

Aarhus University
School of Business and Social Sciences
Department of Economics and Business Economics

Out line of this week's lectures

- Motivation
- Single-factor model
- Portfolio returns
- Multifactor models
- Arbitrage pricing theory
- Estimation of parameters

The theory developed on previous slides are based upon rather strict assumptions, such as the availability of asset parameters in the form of expected returns, variances, covariances etc. or assumptions like the availability of a market portfolio.

In this slideset we will relax some of these assumptions and see how one can build models based on observable factors, and in this way reduce the number of parameters significantly.

Single-factor model

In a single-factor model, we assume that a single factor f is the driver of the n asset returns

$$r_i = a_i + b_i f + \epsilon_i, \quad \text{for } i = 1, \dots, n. \quad (1)$$

- a_i asset specific constant.

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- b_i factor loadings determining the sensitivity of the return of asset i to the factor f .
- ϵ_i error terms with zero mean, $\mathbb{E}[\epsilon_i] = 0$, and assumed uncorrelated both with f , $\text{Cov}(f, \epsilon_i) = 0$, and error terms ϵ_j of other assets $\text{Cov}(\epsilon_j, \epsilon_i) = 0$.

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- σ_{ϵ_i} standard deviation of ϵ_i (assumed known).
- The model (1) can be viewed as a simple linear regression of observed returns of asset i on observations of the factor f .

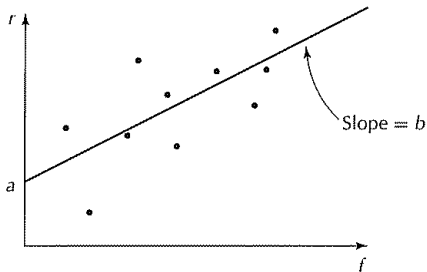


FIGURE 8.1 Single-factor model. Returns are related linearly to the factor f , except that random errors are added to the return

Characteristics

We can compute the quantities relevant for the mean-variance framework explicitly (**Maybe some of them on Blackboard**)

- Expected returns: $\bar{r}_i = a_i + b_i \bar{f}$.
- Variance of returns: $\sigma_i^2 = b_i^2 \sigma_f^2 + \sigma_{\epsilon_i}^2$.
- Covariance of returns: $\sigma_{ij} = b_i b_j \sigma_f^2$ for $i \neq j$.
- Covariance of returns with factor: $\text{cov}(r_i, f) = b_i \sigma_f^2 \Rightarrow b_i = \frac{\text{cov}(r_i, f)}{\sigma_f^2}$.

In comparison to the Markowitz approach, particularly the expression for the covariance corresponds to a significant decrease in model parameters.

The whole covariance matrix is determined by the factor loadings b_i , the variance of the factor and the variance of the error terms - a total of $2n + 1$ parameters. In the Markowitz approach we had $n(n + 1)/2$ parameters in the covariance matrix.

The return of a portfolio

Assume that the n asset returns are modeled by a single-factor model as in (1) and consider a portfolio with weights $\mathbf{w} = (w_1, \dots, w_n)^\top$ and $\sum_{i=1}^n w_i = 1$. We find that

$$\begin{aligned} r_p &= \sum_{i=1}^n w_i r_i \\ &= \sum_{i=1}^n w_i a_i + \sum_{i=1}^n w_i b_i f + \sum_{i=1}^n w_i \epsilon_i \\ &= a_p + b_p f + \epsilon_p, \end{aligned}$$

where

$$a_p = \sum_{i=1}^n w_i a_i, \quad b_p = \sum_{i=1}^n w_i b_i, \quad \epsilon_p = \sum_{i=1}^n w_i \epsilon_i.$$

The portfolio return parameters

Clearly, both a_p and b_p are new parameters equal to weighted averages of the individual asset parameters. The stochastic error term ϵ_p has features

$$\mathbb{E}[\epsilon_p] = \mathbb{E}\left[\sum_{i=1}^n w_i \epsilon_i\right] = \sum_{i=1}^n w_i \mathbb{E}[\epsilon_i] = 0 \quad (2)$$

$$\text{Cov}(f, \epsilon_p) = \text{Cov}\left(f, \sum_{i=1}^n w_i \epsilon_i\right) = \sum_{i=1}^n w_i \text{Cov}(f, \epsilon_i) = 0 \quad (3)$$

$$\begin{aligned} \sigma_{\epsilon_p}^2 &= \text{Var}(\epsilon_p) = \mathbb{E}[\epsilon_p^2] \\ &= \mathbb{E}\left[\left(\sum_{i=1}^n w_i \epsilon_i\right) \left(\sum_{j=1}^n w_j \epsilon_j\right)\right] = \mathbb{E}\left[\sum_{i=1}^n w_i^2 \epsilon_i^2\right] \\ &= \sum_{i=1}^n w_i^2 \sigma_{\epsilon_i}^2. \end{aligned} \quad (4)$$

Portfolio diversification

The portfolio return variance is given by

$$\sigma_p^2 = b_p^2 \sigma_f^2 + \sigma_{\epsilon_p}^2. \quad (5)$$

If we take the expression for the variance of the portfolio error term (4) and put an equal weight in each asset, $w_i = \frac{1}{n}$, and for $\sigma_{\epsilon_i}^2 < s^2$ for all i , we get that

$$\sigma_{\epsilon_p}^2 = \frac{1}{n^2} \sum_{i=1}^n \sigma_{\epsilon_i}^2 \rightarrow 0 \quad \text{for } n \rightarrow \infty \quad (6)$$

- Together with $\mathbb{E}[\epsilon_p] = 0$ equation (6) implies that $\epsilon_p \rightarrow 0$ for $n \rightarrow \infty$.

Maybe example on blackboard

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- Together with $\mathbb{E}[\epsilon_p] = 0$ equation (6) implies that $\epsilon_p \rightarrow 0$ for $n \rightarrow \infty$.
- The risk associated with error terms ϵ_i can be diversified away and we label this as idiosyncratic risk.
- The risk associated with $b_i f$ is not diversifiable and we label this as systematic risk.

Maybe example on blackboard

Multifactor models

Having only one factor to model asset returns might be too simplistic. If more flexibility is needed one can add more factors to the return equation for asset i

$$r_i = a_i + \sum_{k=1}^m b_{ki} f_k + \epsilon_i, \quad \text{for } i = 1, \dots, n. \quad (7)$$

We impose similar assumptions to the framework as in the one-factor case. Namely,

- $\mathbb{E}[\epsilon_i] = 0$
- $\text{Cov}(f_k, \epsilon_i) = 0$ for all $i = 1, \dots, n$ and $k = 1, \dots, m$.
- $\text{Cov}(\epsilon_j, \epsilon_i) = 0$ for all $i \neq j$.

- If we have enough observations via time series of the factors together with the returns of a given asset, we can estimate the various factor loadings, b_{ki} , and the intercept a_i .

Comments to multifactor models

- If we have enough observations via time series of the factors together with the returns of a given asset, we can estimate the various factor loadings, b_{ki} , and the intercept a_i .
- More factors can/should be included if one finds that the error terms of a single-factor model are large and are correlated with the chosen factor and other error terms $\rightarrow$ sign of a model misspecification.

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- More factors can/should be included if one finds that the error terms of a single-factor model are large and are correlated with the chosen factor and other error terms $\rightarrow$ sign of a model misspecification.
- For US stocks it is normal to include between 3 and 15 factors.

The CAPM as a factor model

Take a single-factor model for the excess return on stock i , where we choose the factor to be the excess return on the market

$$r_i - r_f = \alpha_i + \beta_i(r_M - r_f) + \epsilon_i, \quad (8)$$

where we use the notation α_i and β_i instead of a_i and b_i .

- Taking expectation yield

$$\bar{r}_i - r_f = \alpha_i + \beta_i(\bar{r}_M - r_f), \quad (9)$$

which looks like the CAPM model for $\alpha_i = 0$.

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which looks like the CAPM model for $\alpha_i = 0$.

- We have that

$$\sigma_{iM} = \text{Cov}(r_i, r_M) = \beta_i \sigma_M^2 \Rightarrow \beta_i = \frac{\sigma_{iM}}{\sigma_M^2}. \quad (10)$$

One-factor model vs. CAPM model

- The CAPM model predicts that $\alpha_i = 0$, while (8) makes no assumption like this.

One-factor model vs. CAPM model

- The CAPM model predicts that $\alpha_i = 0$, while (8) makes no assumption like this.
- CAPM theory makes no assumptions on the covariance structure of assets, but assumes the market portfolio is efficient. One-factor model in equation (8) assumes a strong correlation structure, but imposes no assumption about efficiency of market portfolio.

Arbitrage Pricing Theory (APT)

The factor models are not founded on solid theory. Instead, they can be thought of as purely statistical relations and there is no attempt made to connect the model parameters across assets. Hence, it is relevant to ask if we can choose the parameters a_i and b_i independent of each other?

Arbitrage pricing theory gives the answer without imposing strict assumptions like the CAPM framework. The assumptions are:

- 1 Investors prefer greater return over less return.
- 2 Markets do not admit arbitrage.
- 3 The universe of financial assets is of infinite size. In practice this assumption is not too far from reality.

Single-factor APT

Consider the case of two risky assets with (note that there are no error terms $\rightarrow$ think of i and j as well-diversified portfolios.)

$$r_i = a_i + b_i f \quad (11)$$

$$r_j = a_j + b_j f \quad (12)$$

' Assuming that market is free of arbitrage we find that (Blackboard)

$$\bar{r}_i = \lambda_0 + b_i \lambda_1 = r_f + b_i \lambda_1, \quad (13)$$

where the last equality assumes the existence of a risk free asset.
Moreover,

$$\lambda_1 = \frac{a_i - \lambda_0}{b_i} + \bar{f}. \quad (14)$$

- a_i and b_i are linearly dependent.
- Only b_i determines the expected returns as other parameters λ_0 and λ_1 is fixed by the market conditions.

Theorem

Assume the market is free of arbitrage and assume the existence of n well-diversified portfolios with returns governed by $m < n$ factors

$$r_i = a_i + \sum_{k=1}^m b_{ik} f_k \quad \text{for } i = 1, \dots, n. \quad (15)$$

Then the expected return of the assets equals

$$\bar{r}_i = \lambda_0 + \sum_{k=1}^m b_{ik} \lambda_k \quad \text{for } i = 1, \dots, n. \quad (16)$$

APT in matrix format

We can rewrite the equations of the Theorem in matrix format

$$\bar{\mathbf{r}} = \begin{bmatrix} \bar{r}_1 \\ \bar{r}_2 \\ \vdots \\ \bar{r}_n \end{bmatrix} = \begin{bmatrix} 1 & b_{11} & \dots & b_{1m} \\ 1 & b_{21} & \dots & b_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & b_{n1} & \dots & b_{nm} \end{bmatrix} \begin{bmatrix} \lambda_0 \\ \lambda_1 \\ \vdots \\ \lambda_m \end{bmatrix} \quad (17)$$

$$= \begin{bmatrix} \mathbf{1} & \mathbf{B} \end{bmatrix} \boldsymbol{\lambda} \quad (18)$$

- In order to compute the parameters $\lambda_0, \dots, \lambda_m$ we need at least $m + 1$ linearly independent assets. If we use these $m + 1$ assets, we can invert the matrix $\begin{bmatrix} \mathbf{1} & \mathbf{B} \end{bmatrix}$.

Example on Blackboard

APT in matrix format

We can rewrite the equations of the Theorem in matrix format

$$\bar{\mathbf{r}} = \begin{bmatrix} \bar{r}_1 \\ \bar{r}_2 \\ \vdots \\ \bar{r}_n \end{bmatrix} = \begin{bmatrix} 1 & b_{11} & \dots & b_{1m} \\ 1 & b_{21} & \dots & b_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & b_{n1} & \dots & b_{nm} \end{bmatrix} \begin{bmatrix} \lambda_0 \\ \lambda_1 \\ \vdots \\ \lambda_m \end{bmatrix} \quad (17)$$

$$= \begin{bmatrix} \mathbf{1} & \mathbf{B} \end{bmatrix} \boldsymbol{\lambda} \quad (18)$$

- In order to compute the parameters $\lambda_0, \dots, \lambda_m$ we need at least $m + 1$ linearly independent assets. If we use these $m + 1$ assets, we can invert the matrix $\begin{bmatrix} \mathbf{1} & \mathbf{B} \end{bmatrix}$.
- λ_k can be interpreted as the *price of risk* for the factor f_k . For a given asset i with factor loading b_{ik} we get return in excess of the risk-free rate equal to $b_{ik}\lambda_k$.

Example on Blackboard

Take a two-factor model for the return on asset i

$$r_i = a_i + b_{i1}f_1 + b_{i2}f_2 + \epsilon_i. \quad (19)$$

Consider the covariance of a diversified portfolio i with the market return (hence $\epsilon_i = 0$)

$$\text{Cov}(r_i, r_M) = b_{i1}\text{Cov}(f_1, r_M) + b_{i2}\text{Cov}(f_2, r_M).$$

Divide both sides by the market return variance

$$\beta_i = b_{i1}\beta_{f_1} + b_{i2}\beta_{f_2}. \quad (20)$$

- Assets have different betas because they have different factor loadings.

Choosing the factors

It is not a trivial task to choose which and how many factors to include in order to model returns. The factors should be chosen based on:

- 1 Statistical analysis
- 2 Theory
- 3 Fundamental/macroeconomic factors

Once a number of factors have been chosen, they should both explain a big part of the variation in the observed returns and the assumptions of the model should be satisfied. I.e. $Cov(f_k, \epsilon_i) = 0$ and $Cov(\epsilon_i, \epsilon_j) = 0$.

Estimation of parameters

In practice, the parameters $\bar{r}_i$, σ_i^2 , and σ_{ij} are often estimated based on historical data. Assume you have a history of historical returns for the assets in question, r_{it} for $i = 1, \dots, n$ and $t = 1, \dots, T$. You can obtain unbiased estimates using the following formulas:

$$\hat{r}_i = \frac{1}{T} \sum_{t=1}^T r_{it} \rightarrow \bar{r}_i \quad (21)$$

$$\hat{\sigma}_i^2 = \frac{1}{T-1} \sum_{t=1}^T (r_{it} - \hat{r}_i)^2 \rightarrow \sigma_i^2 \quad (22)$$

$$\hat{\sigma}_{ij} = \frac{1}{T-1} \sum_{t=1}^T (r_{it} - \hat{r}_i)(r_{jt} - \hat{r}_j) \rightarrow \sigma_{ij} \quad (23)$$

It can be shown that if the returns are normally distributed, the standard deviation of the estimates of the expected return and the variance equal

$$stdev(\hat{r}_i) = \frac{\sigma_i}{\sqrt{T}} \quad (24)$$

$$stdev(\hat{\sigma}_i^2) = \frac{\sqrt{2}\sigma_i^2}{\sqrt{T-1}}. \quad (25)$$

- You need many observations to estimate the mean, while the variance estimate is more precise.

Example 8.3: A statistical try

First, simulate monthly return data from a normal distribution with mean $\bar{r} = 1\%$ and $\sigma = 4.33\%$

Next, estimate the expected return and standard deviation of simulated return series.

TABLE 8.3
Monthly Rates of Return and Estimation of Mean (Expressed as Percent)

	Year of return								Overall
	1	2	3	4	5	6	7	8	
Jan	-8.65	2.61	6.39	-4.52	1.28	4.49	-1.44	3.30	
Feb	8.61	-2.38	-1.22	2.30	14	7.58	-4.34	3.75	
Mar	5.50	-3.28	1.12	-3.96	-2.63	5.02	1.24	3.95	
Apr	2.04	7.45	3.69	-84	3.15	-51	8.92	-3.13	
May	7.51	7.96	28	.35	-47	-19	-46	-31	
Jun	-2.50	-9.37	3.61	6.96	7.04	1.18	8.28	-89	
Jul	2.28	-7.27	-1.45	4.23	3.68	1.61	-5.33	-6.39	
Aug	1.85	-5.30	6.83	.21	2.74	2.62	-1.01	-60	
Sep	5.86	5.69	2.32	.14	-2.08	-2.32	3.77	-76	
Oct	1.37	5.24	-3.79	-6.48	1.73	-3.08	4.18	1.92	
Nov	3.17	2.94	-52	-1.11	6.18	5.42	-2.27	-3.97	
Dec	9.23	1.94	2.77	2.86	.38	2.93	4.91	5.18	
Mean	3.02	.52	1.67	.01	1.76	2.06	1.37	.17	1.32
σ	5.01	5.88	3.21	3.81	2.98	3.24	4.66	3.55	4.12

Each column represents a year of randomly generated returns. The true mean values are all 1%, but the estimates deviate significantly from this value

Discuss

Conclusion

This concludes the first part of the course.

We have studied a framework for analyzing fixed cash flow stream and we have seen how to extract relevant rates and discount factors from bond markets. Moreover, we have introduced uncertainty in the form of one-period models for asset returns.

In many applications, it is relevant to include more periods in the model and sometimes it is even desirable to use continuous time models. All this will be studied in the second half of the course.

Investment and Finance

State-space Models and Risk-neutral Pricing

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Introduction

- Lecturer: Gyuri Venter
 - Associate professor at the Department of Economics and Business Economics
 - Research: Asset pricing with market frictions, liquidity, and information economics.
 - Email: venter@econ.au.dk
 - Office: 2630-K113, Department of Economics and Business Economics.
 - PhD in Finance, London School of Economics, 2012.
 - Taught (among others) Investments and Derivatives at CBS, Risk Management and PhD level Asset pricing at WBS.
- Lectures: Tuesday 10-12, Aud. D4; Wednesday 10-12, Aud. D1.
Please tell me if you have suggestions for improvement; e.g., if it is too slow/too fast, comments on the structure, etc.
- Exercises: Uploaded every week; part two starts on week 44.

Introduction

Second half of the course will focus on modelling asset prices and pricing and hedging of derivatives.

We will cover:

- Risk-neutral pricing and state-space models.
- Financial derivatives.
- Stochastic processes.
- Option pricing models.
- Stochastic interest rate models.

Introduction

In this part of the course, we will mainly consider financial markets where the payoffs of the financial assets in the market are stochastic.

That means we do not know the cash flow as is the case, e.g., for bonds.

We have already incorporated uncertainty in the CAPM model, for example.

We will try to describe the uncertainty with more specific models, by using stochastic processes.

Most of the second part of the course will be concerned with modelling and pricing of *financial derivatives*, which we will introduce next week.

This week

In this topic we will cover

- Pricing under uncertainty.
- Utility functions.
- A consumption-based pricing equation.
- Relation to the mean-variance approach.
- Probability measures.
- State-space models.
- Pricing in state-space models.
- Risk-neutral pricing.

Preliminaries

Some notation:

- X will usually denote a random variable.
- P is usually a real-valued price function. For example, $P(X)$ is a number describing the "price" of the random variable X .
- $\mathbb{P}$, $\mathbb{E}$, $\mathbb{V}$ (or Var) denote probability, expected value and variance, respectively.
- t usually denotes time.
- T usually denotes the time to maturity of some contract, bond etc.
- We will introduce subscripts to denote evolution over time, for example P_t is the price function *at time* t , which is different from P_s when $s \neq t$. (We will come back to this when discussing stochastic processes later)

Reminder from Lecture 1

- *Arbitrage*: a situation where money can be earned (now or in the future) at no cost and with no risk involved.
- Example: At bank A you can deposit money at an annual rate of 12% and at bank B you can borrow money at 10%.
- You can arbitrage this by borrowing 100 USD at bank B and deposit the money at bank A. After one year you can collect $112 - 110 = 2$ USD.
- You can also scale this up!
- Arbitrage opportunities should not occur in financial markets and if they do, they should disappear rapidly. Also known as “no free lunch.”
- *Law of one price*: Identical future cash flows should cost the same today. The law of one price basically tells us that an investor should not care whether \$1 dollar comes from one place or another (perfectly reasonable from an economic perspective). Note: in fact this does not always hold, e.g. “green bonds.”

Pricing with stochasticity

Let X denote the value at time T of some financial security. If X is deterministic, then by *no arbitrage* the price of this security is simply

$$P(X) = \frac{1}{(1 + r_f)^T} X$$

where r_f is the per-period risk free rate (for this horizon).

If X is stochastic, a naive guess of the price could be

$$P(X) = \frac{1}{(1 + r_f)^T} \mathbb{E}[X]$$

This is a good guess, but not entirely correct. (This price is too high!)

The reason being that investors are *risk-averse*: A loss of 100 dollars typically hurts more than a gain of 100 dollars benefits investors.

Pricing with stochasticity (cont'd)

You have already seen a model that attempts to fix this, namely CAPM:

$$\frac{\mathbb{E}[X] - P(X)}{P(X)} \equiv \mathbb{E}[r] = r_f + \beta(\mathbb{E}[r_M] - r_f) \text{ and } P(X) = \frac{1}{(1 + \mathbb{E}[r])^T} \mathbb{E}[X]$$

The second term is a correction due to the risk inherent in investing in X . Recall that usually this correction is positive, since

$$\beta = \frac{\text{cov}(X, r_M)}{\text{Var}[r_M]} > 0$$

If variance/st. dev. is the correct measure of risk, we know the correction to apply.

Can this theory be reconciled with investors maximizing their (risk-averse) utility?

SBF on Choice Under Uncertainty

SBF  @SBF_FTX · 2020年12月10日 ...

10) What about a wackier bet? How about you only win 10% of the time, but if you do you get paid out 10,000x your bet size? (For now, let's assume you only get to do this bet once.)

2 5 54

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11) Kelly suggests you only bet \$10k: you'll almost certainly lose. And if you kept doing this much more than \$10k at a time, you'd probably blow out.

That this bet is great expected value; you win 1,000x your bet size, way better than the first one! It's just very risky.

4 4 56

SBF  @SBF_FTX · 2020年12月10日 ...

12) In many cases I think \$10k is a reasonable bet. But I, personally, would do more. I'd probably do more like \$50k.

Why? Because ultimately my utility function isn't really logarithmic. It's closer to linear.

8 47 107

Utility functions

Definition (Utility function)

A utility function is a function $u : \mathbb{R} \rightarrow \mathbb{R}$ that is

- **Continuous:** Investments that yield similar returns leads to similar utility.
- **Monotone:** More money, less problems.
- **(Typically) Concave:** Investors are *risk averse*.

Under suitable assumptions on investor preferences¹, investors maximize their *expected* utility

$$\max \mathbb{E}[u(X)]$$

where X is the stochastic payoff of the investment.

¹Von Neumann-Morgenstern axioms.

Comment: Payoffs vs returns

The return of a financial asset X from period t to $t + 1$ is either

- Total (gross) return: $R = \frac{X_{t+1}}{X_t}$
- Rate of return or net return: $r = \frac{X_{t+1} - X_t}{X_t} = R - 1$

In most of the course (and usually in finance) we work with the rate of return.

In this part of the course we will also be working a lot with *payoffs*.

The time $t + 1$ payoff is simply the net cash flow at time $t + 1$, X_{t+1} .

Pricing with stochasticity

Let us now consider a simple one-period economy, where X is the stochastic payoff of some investment at time 1, with price P_0 today. (So forget about the t 's for now)

X is the *payoff* of the asset.

An investor then has the utility function

$$U(c_0, c_1) = u(c_0) + \delta \mathbb{E}[u(c_1)]$$

where c_0 and c_1 are consumption today and tomorrow. Note that c_1 depends on how much we have tomorrow, so it is stochastic.

δ is a discount factor, that determines the investor's preference for consumption now vs later ($\approx$ "patience").

The asset pricing equation

Let $\bar{c}_0$ and $\bar{c}_1$ denote the level of consumption if we buy no units of the payoff X and let ω denote the amount of units of X that we buy.

$$\max u(c_0) + \delta \mathbb{E}[u(c_1)]$$

s.t.

$$c_0 \leq \bar{c}_0 - \omega P_0$$

$$c_1 \leq \bar{c}_1 + \omega X$$

Solving this using the first order conditions (**on blackboard**) we arrive at the following price of X today:

$$P_0 = \mathbb{E} \left[\delta \frac{u'(c_1)}{u'(c_0)} X \right] \quad (1)$$

This is the *Euler equation*, our fundamental asset pricing equation.

The asset pricing equation (cont'd)

Note how the price is still an expectation, but is weighted with the factor $\Lambda = \delta \frac{u'(c_1)}{u'(c_0)}$.

Assume that we have many assets, $X_1, X_2, \dots, X_N$. One may think that each asset should be priced according to the risk inherent to that asset

$$P(X_i) \stackrel{?}{=} \frac{1}{1 + r_f + rx_i} \mathbb{E}[X_i]$$

where rx_i is a risk correction term (excess return) for asset X_i .

The Euler equation (1) tells us that this is not needed! All prices are calculated as

$$P(X_i) = \mathbb{E}[\Lambda X_i]$$

Thus Λ is often called the *stochastic discount factor*.

The asset pricing equation (cont'd)

Consider now the returns of the X_i 's defined as $R_i = \frac{X_i}{P(X_i)}$. Think of a return as a payoff with price one.

Dividing equation (1) by P_i for each asset then yields

$$1 = \mathbb{E}[\Lambda R_i]$$

Using that $\mathbb{E}[YZ] = \text{cov}(Y, Z) + \mathbb{E}[Y]\mathbb{E}[Z]$ we then have

$$1 = \text{cov}(\Lambda, R_i) + \mathbb{E}[R_i]\mathbb{E}[\Lambda] \quad (2)$$

Recall that the risk free asset has $\text{cov}(R_f, \Lambda) = 0$ and $R_f = 1 + r_f$. This implies

$$\mathbb{E}[\Lambda] = \frac{1}{1 + r_f}$$

The asset pricing equation (cont'd)

Equation (2) tells us that the price of a return is exactly the discounted expected return, plus a risk adjustment:

$$\mathbb{E}[r_i] - r_f = -(1 + r_f) \cdot \text{cov}(r_i, \Lambda) \quad (3)$$

where $r_i = R_i - 1$.

So the risk premium on the asset X_i depends on the correlation between its return and Λ . We also see that idiosyncratic risk does not affect the price.

In the CAPM model, Λ is a linear function of market returns.

The mean-variance (Markowitz) approach can be shown to be obtained from all of this in two cases:

- Investor utility is quadratic: $u(x) = ax - \frac{1}{2}bx^2$ with $a > 0, b \geq 0$.
- Returns are normally distributed, i.e., $R_i \sim N(\mu, \sigma^2)$.

Generally there is no reason to believe either of these cases to be true.

A digression: Probability measures

You are used to talking about and doing calculations with probabilities. But let us consider what is needed in order to talk about probability. We will consider only the discrete case.

Definition (Probability measure)

A probability measure is a function $\mathbb{P}(A) \in [0, 1]$, where $A \subseteq \Omega$ is an event from the entire sample space Ω , satisfying

- 1 $\mathbb{P}(\emptyset) = 0$ and $\mathbb{P}(\Omega) = 1$
- 2 $\mathbb{P}(\bigcup_{i \in I} A_i) = \sum_{i \in I} \mathbb{P}(A_i)$ where I is an index set and the A_i 's are disjoint.

Any function $\mathbb{P}$ satisfying the above definition leads to a valid interpretation of probabilities.

A digression: Probability measures (cont'd)

To make things more concrete, consider a discrete random variable X with pmf under the measure $\mathbb{P}$ given by

$$\pi(x) = \begin{cases} \pi_i, & \text{if } x = x_i, i \in I \\ 0, & \text{otherwise} \end{cases}$$

Then we could also define a different measure $\mathbb{Q}$ such that the pmf of X under $\mathbb{Q}$ is

$$q(x) = \begin{cases} q_i, & \text{if } x = x_i, i \in I \\ 0, & \text{otherwise} \end{cases}$$

where the only criteria is that $\sum_{i \in I} \pi_i = \sum_{i \in I} q_i = 1$ and $\pi_i, q_i > 0$.

A digression: Probability measures (cont'd)

When $\mathbb{P}$ and $\mathbb{Q}$ are not identical, i.e. $q_i \neq \pi_i$ for at least one $i \in I$, they may have different expected values of X . Recall:

$$\mathbb{E}^{\mathbb{P}}[X] = \sum_{i \in I} x_i \pi_i, \quad \mathbb{E}^{\mathbb{Q}}[X] = \sum_{i \in I} x_i q_i$$

Then we can relate the expected values under the different measures by

$$\mathbb{E}^{\mathbb{Q}}[X] = \sum_{i \in I} q_i x_i = \sum_{i \in I} \pi_i \frac{q_i}{\pi_i} x_i = \mathbb{E}^{\mathbb{P}} \left[\frac{q(X)}{\pi(X)} X \right] \quad (4)$$

This type of result will pop up later. It is very useful and it underpins some very deep mathematical ideas.

State-space models: Motivation

We will now consider a model where the stochastic payoffs of assets are described via the *states of the world*.

It is probably realistic to think that the value of a stock or financial asset is affected by the various states of the world. Consider for example when the state "COVID pandemic" was realized.

It is of course difficult to characterize every possible state we can end up in, but we could also bundle up states. A very simple model could simply be "good day" vs "bad day", where the price of a stock either increases or decreases.

The setup

The *sample space* is denoted Ω , and contains all of the possible states the world can end up in. We will work with the case $|\Omega| < \infty$.

We simply consider an economy with two periods: $t \in \{0, 1\}$ (today and tomorrow).

- S : The number of possible states at $t = 1$.
- s : State number; $s = 1, \dots, S$.
- π_s : The probability of ending up in state s .
- n : Number of financial assets.
- P_i : The price of asset i at time $t = 0$.
- d_s^i : The value at time $t = 1$ of asset i in state s .

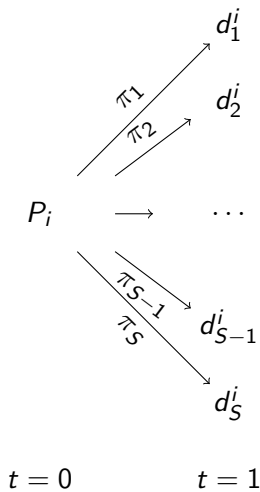
The setup

In matrix notation:

$$\pi = \begin{pmatrix} \pi_1 \\ \vdots \\ \pi_S \end{pmatrix}, \quad P = \begin{pmatrix} P_1 \\ \vdots \\ P_n \end{pmatrix}, \quad d^i = \begin{pmatrix} d_1^i \\ \vdots \\ d_S^i \end{pmatrix}$$

$$D = \begin{pmatrix} - & d^{1'} & - \\ & \vdots & \\ - & d^{n'} & - \end{pmatrix}$$

The model, graphically



State prices

Definition (State prices)

A vector $\psi = (\psi_1, \dots, \psi_S) \in \mathbb{R}^S$ is called a vector of state prices for the system (P, D) if $\psi_j > 0$ for $j = 1, \dots, S$ and if

$$P = D\psi \quad (5)$$

The definition of state prices equation by equation:

$$P_i = (d^i)' \psi = \sum_{s=1}^S d_s^i \psi_s, \quad i = 1, \dots, n$$

ψ prices the assets by "discounting" the uncertain cash flows. Note the similarity of equation (5) to the usual bond pricing equation.

State prices (cont'd)

ψ_s is interpreted as the price at $t = 0$ for receiving 1 unit of money in state s at $t = 1$.
I.e. the price of the payoff

$$e^s = (0, \dots, 0, \underbrace{1}_s, 0, \dots)'$$

The asset e^s is usually called the elementary asset or the *Arrow-Debreu asset* or *contingent claim*. Thus ψ is also sometimes called *Arrow-Debreu prices*.

Theorem (Fundamental theorem of asset pricing, FTAP)

The following statements are equivalent

- 1 The market defined by (P, D) is arbitrage free.
- 2 There exists a $\psi \in \mathbb{R}_+$ that solves

$$P = D\psi$$

State prices (cont'd)

As you have seen before, absence of arbitrage is equivalent to being able to actually price assets in the market. However, a unique set of prices is not guaranteed only by this.

Theorem (Completeness)

Assume that the market defined by (P, D) is arbitrage free. Then the following two statements are equivalent

- 1 *The market defined by (P, D) is complete.*
- 2 *There exists a unique set of state prices ψ such that*

$$P = D\psi$$

So from linear algebra we know that the market is complete if and only if D is invertible, which in particular requires that $n \geq S$.

Example – Arbitrage, completeness

Security	Price	States		
		1	2	3
A	100	110	110	110
B	100	90	100	170

- Three states, two securities.
- Any securities redundant? Is there arbitrage? Is the market complete?

Example – Arbitrage, completeness (cont'd)

Security	Price	States		
		1	2	3
A	100	110	110	110
B	100	90	100	170
C		100	105	140

- Three states, three securities.
- Any securities redundant? Is there arbitrage? Is the market complete?

Risk-neutral pricing

Assume that the market (P, D) is arbitrage free. Then there exists a vector of state prices ψ .

Consider an asset with payment stream

$$d^0 = (1, 1, \dots, 1)'$$

i.e., the risk-free asset. Then we know that the price ψ_0 of this asset is given by

$$\psi_0 = \sum_{s=1}^S \psi_s = \frac{1}{1 + r_f} \quad (6)$$

Now define the normalized state prices as

$$q_s = \frac{\psi_s}{\psi_0}$$

Risk-neutral pricing (cont'd)

Clearly we have $q_s > 0$ and $\sum_{s=1}^S q_s = 1$. So we may interpret the q_s 's as probabilities!

Let us call the associated probability measure $\mathbb{Q}$.

We can then rewrite the price of asset i as

$$P_i = \sum_{s=1}^S \psi_i d_s^i = \psi_0 \sum_{s=1}^S q_s d_s^i = \psi_0 \mathbb{E}^{\mathbb{Q}} [d^i] = \frac{1}{1 + r_f} \mathbb{E}^{\mathbb{Q}} [d^i]$$

The price of asset i is an expectation, discounted by r_f !

So, in words: The price today of *any* asset is equal to the discounted expected value under the measure $\mathbb{Q}$.

This is called *risk-neutral pricing*.

Risk-neutral pricing (cont'd)

The previous result leads to a different formulation of no-arbitrage conditions:

Theorem (Absence of arbitrage)

The following statements are equivalent.

- 1 The market given by (P, D) is arbitrage free.
- 2 There exists a probability measure $\mathbb{Q} = (q_1, \dots, q_S)$ such that

$$P_i = \frac{1}{1 + r_f} \mathbb{E}^{\mathbb{Q}} [d^i]$$

Let $\mathbb{P} = (\pi_1, \dots, \pi_S)$ be the probability measure corresponding to the "real world" probabilities. Then we may write

$$P_i = \frac{1}{1 + r_f} \mathbb{E}^{\mathbb{Q}} [d^i] = \frac{1}{1 + r_f} \mathbb{E}^{\mathbb{P}} \left[\frac{q(d^i)}{\pi(d^i)} d^i \right] =: \mathbb{E}^{\mathbb{P}} [\Lambda d^i]$$

Risk-neutral pricing (cont'd)

So we have recovered the fundamental asset pricing equation with stochastic discount factor

$$\Lambda = \frac{1}{1 + r_f} \frac{q(d^i)}{\pi(d^i)}$$

Note: above we always assumed absence of arbitrage. Therefore we implicitly assume the existence of the probability measure $\mathbb{Q}$.

We often do not care about the "real-world" probabilities $\mathbb{P}$. They do not appear in our risk-neutral pricing formulas!

Risk-neutral pricing is extremely important when pricing derivatives, which is what we will be doing from the next lecture and on.

Risk-neutral pricing (cont'd)

We have derived the risk-neutral probabilities and their connection to no arbitrage in a very specialized setting. We may wonder: Does the risk-neutral pricing principle always apply?

The answer is yes! E.g., Delbaen and Schachermayer (1994) show that any financial market is free of arbitrage if and only if there exists a risk-neutral $\mathbb{Q}$ -measure.

Therefore any financial asset in any arbitrage-free financial model can be priced according to

$$P_0 = d_{0,T} \mathbb{E}^{\mathbb{Q}} [P_T]$$

- **Upside:** Prices = expected values.
- **Downside:** May be hard to characterize the measure $\mathbb{Q}$.

Pricing in state-space models—Summary

Let us sum up how to price a new asset in our state-space model:

- Determine the payoff at time $t = 1$ of the asset, i.e., the d_s 's.
- Find a set of strictly positive state prices ψ based on traded assets.
- The existence and positivity of the state-prices are directly related to no existence of arbitrage.
- The uniqueness of strictly positive state prices is related to market completeness.
- "Discount" the payoffs at time $t = 1$ by the state prices

$$P = \sum_{s=1}^S \psi_s d_s$$

- Alternatively, construct the risk-neutral probabilities: $q_s = \frac{\psi_s}{\psi_0}$ and compute

$$P = \psi_0 \sum_{s=1}^S q_s d_s$$

Next week and future lectures

- We will introduce derivatives.
- Forwards, futures, swaps.
- Options.
- We will determine their payoffs, and we will be interested in pricing them.
- Any questions?
- Thanks for your attention!

Investment and Finance

Derivatives

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Introduction

Last time we considered a model where the payoff of a financial asset depends on the realized future state of the world.

Today we shall move on to a class of financial assets that are somewhat special, but supremely important, namely *financial derivatives*.

The market for financial derivatives worldwide is absolutely huge. Some estimate its size as far more than the global world GDP (although the size is difficult to estimate).

In this lecture we shall introduce some general concepts and some of the most fundamental derivatives.

Derivatives are financial weapons of mass destruction!

– Warren Buffet

This week

In this topic we will cover

- Derivatives: What are they?
- Forwards.
 - Pricing.
 - Cost of carry.
- Futures.
 - Pricing and relation to forwards.
 - Hedging with futures.
- Swaps.
 - Commodity swaps.
 - Interest rate swaps.

Financial derivatives

A financial derivative is a financial instrument, whose value/payoff depends on the value of some *underlying* asset.

The value of the contract is *derived* from the value of the underlying – hence the name.

The most common types of derivatives are

- Forwards
- Futures
- Swaps
- Options

We will consider the first three types this week, while options are saved for later.

Pricing derivatives

A derivative contract consists of three parts:

- An underlying instrument, S .
- A time to maturity T , and possible settlement dates $0 \leq t_1 < \dots < t_n$.
- A pre-specified payoff structure.

Depending on how complicated the payoffs in the contract are, the derivative can be very difficult to correctly price.

We mainly have three ways to price derivatives:

- A replicating strategy.
- No-arbitrage arguments.
- Risk-neutral pricing.

Pricing derivatives (cont'd)

Replicating strategies and no-arbitrage arguments provide model-free prices, but may not be easy or possible to come up with.

For *risk-neutral valuation*, recall that we need to compute (at least almost like this):

$$P_0 = d_{0,T} \mathbb{E}^{\mathbb{Q}} [f(S(T))]$$

where f is the real-valued payoff function that mathematically describes the payoff structure.

Note that we may think of a derivative as a composite function of S . This is worth keeping in mind later.

Terminology and assumptions

Generally a derivative contract requires two parts

- Buyer – long position.
- Seller – short position.

We assume *perfect capital markets*:

- Markets are liquid - no price impact of trading.
- No transaction costs.
- No restrictions on shorting.

Notation

So far we have mostly worked with one-period models. In general, we simply work with time $t \in [0, \infty)$. Therefore, the notation becomes a bit heavier as we carry around the time index:

- T is the maturity of a derivative contract.
- P_t is the *price* of the derivative at time $t \leq T$ (when it makes sense to talk about it).
- V_t is the *value* of the derivative at time $t \leq T$.
- S_t or $S(t)$ is the price of the underlying asset at time $t \geq 0$.
- $d_{t,T}$ is the discount factor from time t to T .
- $B_{t,T}$ is the zero coupon bond price with maturity T at time $t \leq T$.

Forward contracts

A forward contract is a derivative contract in which the buyer is *obliged* to buy a *given amount* of the underlying security for a *given price* K at a *certain future date* T . (And the seller is obliged to sell at that price.)

Forwards trade in *over-the-counter markets*.

They are particularly popular for commodities, currencies, and interest rates, and can be used to hedge foreign exchange or interest rate risk.

No money changes hand at inception.

Why would we want to buy a forward contract?

Forwards: Commodity example

A farmer is currently growing 1 ton of wheat for harvest and sale in September next year.

- Wheat currently sells for GBP 100 per ton;
- No current trend (expected increase or decrease) in wheat prices;
- Wheat prices can change unexpectedly.
- The farmer needs to plan his September budget now, because of upcoming expenses.

A cereal maker plans to purchase 1 ton of wheat in September to make cereal.

- The cereal maker needs to set sales price for September now, for marketing purposes.

The farmer and cereal maker can *both* hedge their September cash flows by signing a forward contract → we need a *forward price*.

A forward on a non-dividend-paying underlying

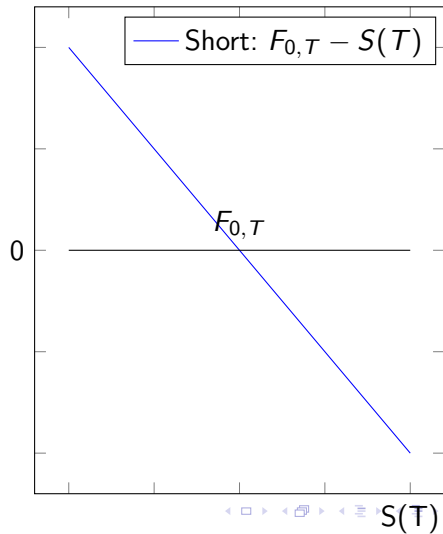
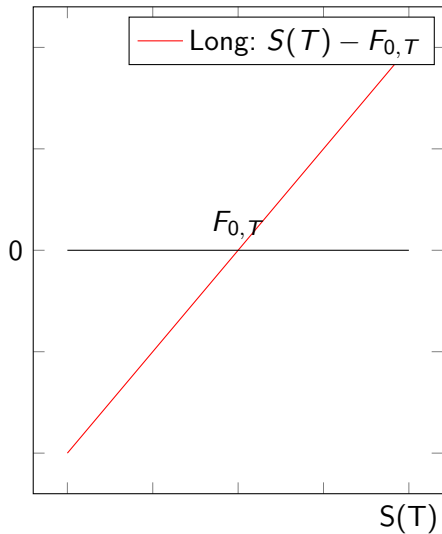
Consider a forward contract with forward price $F_{0,T}$ and maturity T .

The underlying asset does not pay dividends and its market price at maturity is S_T .

The forward contract generates the following payoffs:

- Buyer's (long position) gain at maturity is $S_T - F_{0,T}$, while seller's (short position) gain is $-(S_T - F_{0,T})$.
- We see that the actual payoff, depending on the future value of the underlying, may be positive or negative.
- If the price of underlying at time T falls below $F_{0,T}$ buyer loses and seller gains, and vice versa if underlying is above F .

Forwards, graphically



Price of a forward

Definition (Forward price)

The forward price $F_{0,T}$ is defined to be the delivery price that solves

$$V_0 = 0$$

Theorem

In absence of arbitrage, the theoretical forward price (without cost of carry) is

$$F_{0,T} = \frac{S_0}{d_{0,T}}$$

where $d_{0,T}$ is the discount factor between period 0 and T .

Note: no assumption about how the price S_t evolves—it is model-free!

Price of a forward: Proof by no arbitrage

Forward price $F_{0,T}$ set at date 0 so that it costs nothing to enter into this contract.

Consider the replicating portfolio, created at time 0:

- buy one share of stock and
- borrow $d_{0,T}F_{0,T}$.

Payoffs at T :		
underlying price	Forward	Replicating portfolio
S_T	$S_T - F_{0,T}$	$S_T - F_{0,T}$

Value of forward at time 0 equals the value of replicating portfolio:

$$V_0 = S_0 - d_{0,T}F_{0,T}.$$

Price of a forward: Proof by no arbitrage (cont'd)

Forward price $F_{0,T}$ set at date 0 so that it costs nothing to enter into this contract. The value of the forward contract at time 0 is given by,

$$V_0 = S_0 - d_{0,T}F_{0,T}.$$

The condition $V_0 = 0$ implies $F_{0,T} = S_0/d_{0,T}$. Choosing a different delivery price $\tilde{F}_{0,T} \neq F_{0,T}$ would mean a transaction would take place today.

Note that forward prices depend and change with the asset price:

- The value of a forward position created in the past at F varies as $V_t = S_t - d_{t,T}F$.
- After trading the forward, the value V_t moves away from zero as S_t evolves. At time t the value is

$$V_t = d_{t,T}(F_{t,T} - F_{0,T})$$

where we used that for a new forward contract at t with maturity T , the fair price should be $F_{t,T} = S_t/d_{t,T}$.

Cost of carry

So far we have not considered much about the underlying security S . In practice S can be many different things, and sometimes there are costs associated with owning the underlying security up until time T .

Consider for example a forward on oil. The oil drums have to be stored until maturity, which costs money. These costs are called *cost of carry*.

Assume that a cost of carry c_k is incurred at dates $k \in \{0, \dots, M\}$ with $M < T$. Then we should subtract these costs to get the forward price:

$$F_{0,T} = \frac{S_0}{d_{0,T}} + \sum_{k=0}^M \frac{c_k}{d_{k,T}}$$

Does this formula make sense?

Cost of carry (cont'd)

Many commodities have an associated cost of carry due to storage, insurance and so forth.

However, these costs may also be negative. Consider for example a bond that pays out coupons or a stock that pays dividends. In this case, the buyer should pay less for the forward as they do not receive the dividends prior to time T .

Example: Forward with maturity $T = 3$ on a five-year coupon bond with yearly coupon $c = 5\%$, price P .

Convenience yield

When considering commodities for consumption, it may not be feasible to short sell.

This means our strategy of no-arbitrage based on selling the underlying asset short does not work for finding the forward price.

Thus we can at best conclude that

$$F_{0,T} \leq \frac{S_0}{d_{0,T}} + \sum_{k=0}^M \frac{c_k}{d_{k,T}} \quad (1)$$

Does this then represent an arbitrage opportunity for the holders of the commodity?
→ not necessarily.

The reason is that selling the commodity today and buying it back later means you do not control it for that period of time.

Convenience yield (cont'd)

In other words: There is some value in owning the underlying asset and not having it "out there" to buy back later.

Define the intrinsic value of owning one unit of the underlying as y .

y is called the *convenience yield* and is defined to be such that the inequality (1) is an equality:

$$F_{0,T} = \frac{S_0}{d_{0,T}} + \sum_{k=0}^M \frac{c_k - y}{d_{k,T}}$$

Futures

A forward contract is a so called "over-the-counter" (OTC) product. That means the trade is privately negotiated between two parties and the contract is only settled once.

This means that investors are very exposed to the risk that the transaction does not take place due to a default.

A *futures contract* is a standardized forward contract that is traded on a stock exchange (so not OTC). This means futures contracts are regulated by a central government body and that there is a guarantee that the contract is settled.

The key difference between a forward and a futures contract is the fact that futures contracts are *settled daily*, called *marking to market*.

Margin for futures trades

To deal with credit risk, futures exchanges require buyers and sellers to deposit funds in a *margin account*.

- Funds in the margin account guarantee that investors have enough funds to honor their obligations.
- Moreover, if an investor does not have enough money in the margin account, she receives a *margin call* and needs to urgently top up the margin account to cover any losses.

Since the clearing house as the central counterparty has a very high rating and is heavily insured, counterparty risk in futures contracts is very low.

Not surprisingly, futures are in general much more liquid than forwards.

Still, as we will see, with constant r_f , *futures and forward prices are the same*.

Futures (cont'd)

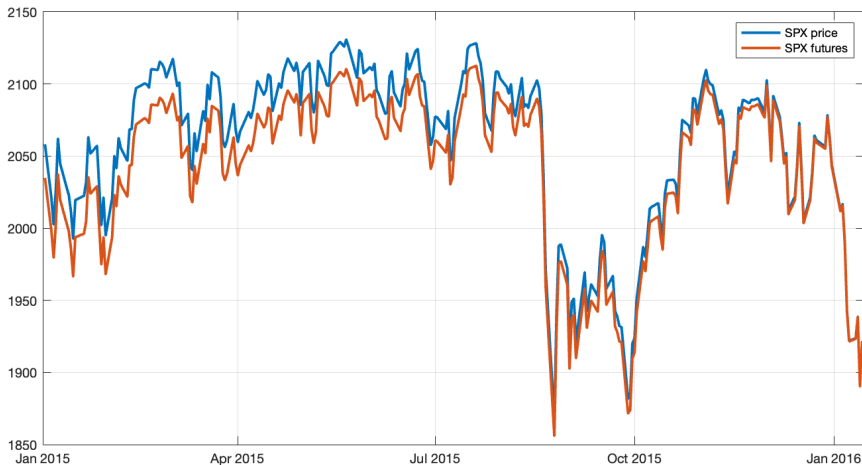
The marking to market of a futures contract means that it is regularly "reset" at pre-determined intervals.

Let $G_{0,T}$ denote the price of a futures contract with maturity T today. As with forwards, the value of the contract today is zero.

Tomorrow, at time $t = 1$, the price has changed to $G_{1,T}$. If $G_{1,T} > G_{0,T}$, the seller transfers $G_{1,T} - G_{0,T}$ to the buyer and vice versa if $G_{1,T} < G_{0,T}$.

The new futures price is then $G_{1,T}$, continuing this way until maturity.

Futures price vs spot price of S&P 500 index



Forward and futures prices—RN valuation

We have seen that the price of any asset is given by the discounted risk-neutral expectation:

$$P_0 = d_{0,T} \mathbb{E}^{\mathbb{Q}} [P_T],$$

where $d_{0,T} = \prod_{i=0}^{T-1} d_{i,i+1}$, as long as the risk-free rate that governs the discount factor is *deterministic*.

If the interest rate evolves stochastically over time, one can show that we can still write the following, with the discounting inside the expectation sign:

$$P_0 = \mathbb{E}^{\mathbb{Q}} [d_{0,T} P_T]$$

For example, then holds that the ZCB price $B_{t,T}$ is given as

$$B_{0,T} = \mathbb{E}^{\mathbb{Q}} [d_{0,T}]$$

Forward and futures prices—RN valuation (cont'd)

Taking the risk-free rate—and hence the discount factor—to be stochastic, we obtain the following forward price

$$F_{0,T} = \frac{\mathbb{E}^{\mathbb{Q}} [d_{0,T} S(T)]}{B_{0,T}} = \mathbb{E}^{\mathbb{Q}} [S(T)] + \frac{\text{Cov}^{\mathbb{Q}} (d_{0,T}, S(T))}{B_{0,T}} \quad (2)$$

In addition, we can show the following:

Theorem (Futures price)

The futures price $G_{0,T}$ with asset $S(t)$ as underlying is given by

$$G_{0,T} = \mathbb{E}^{\mathbb{Q}} [S(T)]$$

As a result, if interest rates are deterministic, then $G_{0,T} = F_{0,T}$ (as Cov above is 0). Otherwise, when the price of an asset is positively correlated with the discount factor, the forward price is $>$ the futures price.

Side comment: The price of a futures contract under CAPM

Assuming that CAPM is true, we know that

$$\frac{\mathbb{E}^{\mathbb{P}}[S(1)] - S(0)}{S(0)} = r_f + \beta \left(\mathbb{E}^{\mathbb{P}}[r_M] - r_f \right)$$

Reshuffling we get

$$G_{0,1} = S(0)(1 + r_f) = \mathbb{E}^{\mathbb{P}}[S(1)] - S(0)\beta \left(\mathbb{E}^{\mathbb{P}}[r_M] - r_f \right)$$

So again we see how the risk-neutral expectation is (usually) lower than the expectation under the physical measure:

$$G_{0,1} = \mathbb{E}^{\mathbb{Q}}[S(1)] < \mathbb{E}^{\mathbb{P}}[S(1)] \quad (3)$$

The objective expectation is adjusted with a risk-premium.

Futures on commodities

Recall: Investors can use futures contracts to lock in the price (= *hedging*) of some asset or commodity that it needs in the future. Thus the futures contract serves as an insurance against price deviations.

Futures contracts on commodities such as oil and gas are a very large market.

The oil crisis in the 1970's led to the oil futures market becoming very large.

The market for gas futures is more speculative, with most contracts never actually “going physical.”

During the COVID crisis, in the end of April 2020, the price of oil futures to be settled in May even went negative! Partial explanation: low demand for oil during lockdown combined with the cost of storing the oil.

Hedging with futures

Suppose a company has a contractual obligation to deliver a product at time T . The product requires some commodity to make.

- The price of the required amount of commodities today is P , so expected profits Π are $\Pi = R - P$ where R is sales revenue.
- If the commodity price goes up tomorrow, total cost becomes $\tilde{P}$ and profits may turn negative.
- Solution: Buy futures with delivery price P on the commodity for the price of G , giving profits $\Pi = R - P - G$.
- The futures price should thus reflect the risk (or expectations) regarding price changes.

This type of hedge with an equal and opposite position in the futures market is a "perfect hedge".

Hedging with futures (cont'd)

Often it is not possible to hedge perfectly, due to circumstances in the market.

In order to have a measure for how good a hedging strategy is, we may introduce the basis b_t defined as

$$b_t = S(t) - G_{t,T^*}$$

where

- T is the maturity of the futures contract.
- $S(t)$ is the price of the underlying at time t .
- $T^* \geq T$ is the maturity of the futures contract used for hedging.
- $S^*(t)$ is the price of the underlying of the futures contract at time t . The underlying asset does not have to be the asset we are hedging against.
- G_{t,T^*} is the price of a futures on S^* with maturity T^* .

Hedging with futures (cont'd)

The hedging strategy is a perfect hedge if $S = S^*$ and $T = T^*$. In that case we have

$$G_{T,T^*} = S(T) \implies b_T = 0$$

A common hedging method is to buy/sell a future and then sell/buy the future again at time T . So if we want to sell an asset S at time T , the sales price of the hedging strategy is

$$G_{0,T^*} + S(T) - G_{T,T^*} = G_{0,T^*} + b_T$$

When the hedge is not perfect, the future cash flow is uncertain. I.e. b_T is stochastic.

This gives rise to the notion of *basis risk*, which is generally lower than the price risk due to futures and spot prices usually moving in tandem.

Hedging with futures (cont'd)

Consider a portfolio of one unit of S and h units of futures contracts for hedging. Then from time 0 to T the change in value of the portfolio is

$$V_T - V_0 = S(T) - S(0) + h(G_{T,T^*} - G_{0,T^*})$$

To determine h we would have to minimize the inherent risk in this change in value.

Suppose we equate variance with risk. Then we minimize

$$\min_h \mathbb{V}[V_T - V_0] = \min_h \mathbb{V}[S(T)] + h^2 \mathbb{V}[G_{T,T^*}] + 2h \text{Cov}(S(T), G_{T,T^*})$$

The optimal h is thus

$$h = -\frac{\text{Cov}(S(T), G_{T,T^*})}{\mathbb{V}[G_{T,T^*}]}$$

Proof on board

Swaps

The last type of derivatives that we want to consider in this lecture is the *swap*.

A swap is an agreement between two parties to swap future cash flows. In the agreement is specified payment dates and how to compute the payments.

Most commonly, a variable cash flow (floating leg) is swapped for a fixed cash flow (fixed leg). These are called *plain vanilla swaps*. Examples of underlying instruments are

- Interest rates
- Currencies
- Commodities
- Credit

The swap is yet another way to minimize risk, by trading away possible variable costs for fixed ones.

Swaps (cont'd)

A plain vanilla swap has the following setup:

- Two parties: A and B.
- Underlying instrument S .
- Contract maturity T .
- At each pre-specified period $0 \leq t_1, \dots, t_n = T$, party A pays party B a fixed price X .
- Party B pays a variable amount $f(S(t_i))$, $i = 1, \dots, n$, which is determined based on the value of the underlying $S(t_i)$.

The cash flow of the swap is thus

$$(f(S(t_1)) - X, f(S(t_2)) - X, \dots, f(S(t_n)) - X)$$

Leading to the value of the swap today:

$$V_0 = \sum_{i=1}^n d_{0,t_i} (f(S(t_i)) - X)$$

Swaps (cont'd)

Definition (Fair price of swap)

Given a plain vanilla swap on the underlying S with payment periods $t_1, \dots, t_n$ with variable payments $f(S(t_i))$ and fixed payments X , the *fair price* of the swap is the value of X that solves

$$V_0 = 0$$

This definition is equivalent to saying that the fair price solves

$$V_{\text{floating}} = V_{\text{fixed}}$$

Commodity swaps

In a *commodity swap*, the variable price is simply the spot price of the underlying commodity.

- At each pre-specified period t_i , party A pays party B a fixed price X .
- Party B pays the variable amount $S(t_i)$ at each period, $i = 1, \dots, n$.
- Thus party A is always able to buy the underlying asset at spot price, but pays only a fixed price to do so.

Illustration on blackboard

Theorem (Fair price of commodity swap)

The fair price at time 0 of a commodity swap with payment periods $t_1, \dots, t_n$ and underlying S is given by

$$X = \frac{nS(0)}{\sum_{i=1}^n d_{0,t_i}}$$

Proof on blackboard

Interest rate swaps

In an *interest rate swap*, the underlying is a pre-specified interest rate. Typically the reference interest rate is the so called LIBOR rate.

- A fictional notional H is chosen along with a reference interest rate r .
- At each pre-specified period t_i , party A pays party B a fixed interest rate c based on this notional.
- At time t_{i-1} party B pays the variable amount $r_{i-1,i}$ which is the interest rate between periods $i - 1$ and i . Thus the variable interest rate is determined one period ahead and the first payment is $c - r_{0,t_1}$.

Theorem (Fair price of interest rate swap)

The fair fixed rate, c , at time 0 of an interest rate swap with periods $t_0, \dots, t_n$ and reference interest rate r is given by

$$c = \frac{1 - d_{0,t_n}}{\sum_{i=1}^n d_{0,t_i}}$$

Swaps (cont'd)

The market for swaps is very large, both for hedging and for speculation.

The largest markets for swaps are interest rate swaps and currency swaps.

Interest rate swaps are commonly offered from banks to their loan takers.

A big factor in valuing swaps is the counterparty risk: The risk that any of the involved parties default and is unable to pay.

A big issue in counterparty risk is also the credit default swaps, which are somewhat special and controversial. They have sometimes been compared to "insuring your neighbours house against a fire".

Summary

In this topic we have introduced a very important class of financial contracts: derivatives

A derivative is a contract whose final payoff depends on an underlying instrument, and thus the ultimate cash flow is uncertain.

Due to the dependence on the underlying asset, derivative contracts can be very useful to hedge risk associated with the underlying asset.

We have introduced three fundamental derivatives: forward, futures and swaps and considered how to price and hedge with these derivatives.

The theory and research on derivatives is vast, and derivative contracts may quickly become extremely complex to put a price on.

Investment and Finance

Stochastic Processes

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Motivating stochastic processes

We have introduced three major types of derivatives: Forwards, futures, and swaps.

The pricing of these derivatives was “straightforward”, mainly due to the fact that we know that a certain stream of payments must take place—we just do not know the size of the payment.

One of the most important derivatives is the *option*. In this contract the holder has the *right*, but *not the obligation* to buy the underlying asset at a pre-specified price—hence the name “option”.

The option contract complicates things, since now there is uncertainty about whether or not there will be a transaction at all.

To determine the price of an option, we can no longer rely only on trading strategies. We need a *pricing model*.

Option pricing models

An option pricing model is a mathematical model that allows us to calculate the price of options.

Usually this is done by modelling the movements in the price of the underlying asset explicitly.

Given that we still require the model to be stochastic, a natural way to model the underlying process is by *stochastic processes*.

The theory of stochastic processes is vast, deep and can get very complicated. To fully understand the concepts we require a lot of measure and integration theory.

We do not have time for that here, so we will have to make do with a less rigorous, but hopefully intuitive, treatment.

Outline of this topic

This week we will cover

- Stylized facts of financial returns.
- Stochastic processes: Discrete vs continuous.
- Binomial lattices.
- Distribution of the price process.
- Brownian motion.
- Stochastic differential equations.
- Itô's lemma.
- The geometric Brownian motion.

Stylized facts

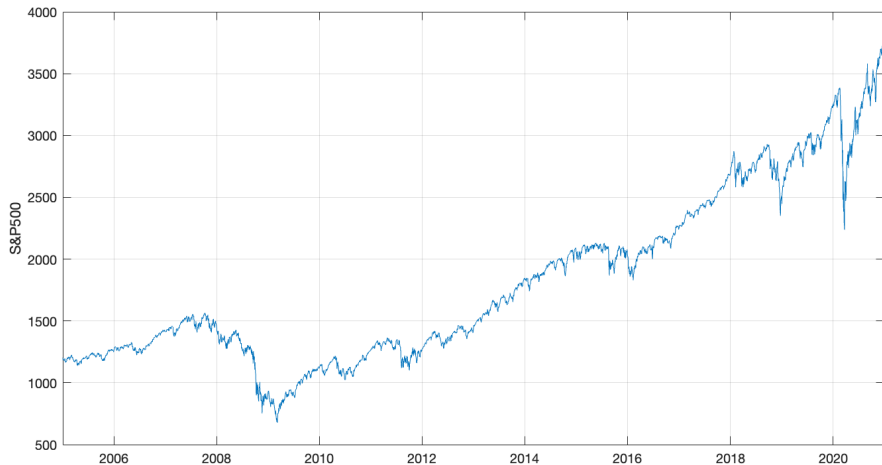
Financial returns and their statistical properties over time have been analyzed for many years. Over the years, some empirical facts have been established.

These are called the *stylized facts* of financial returns. Some of them are:

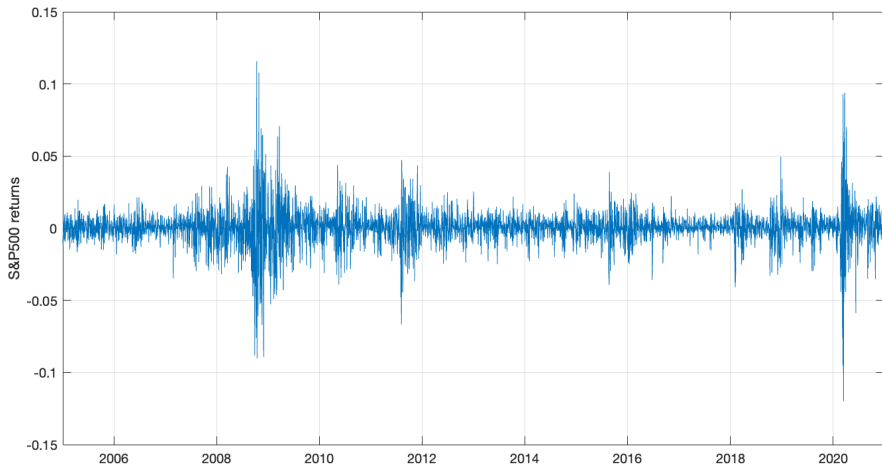
- Returns are uncorrelated on time-scales longer than ≈ 15 minutes.
- Returns are not independent.
- Returns have *heavy tails*.
- We see large down-movements more often than equally large up-movements.
- Over large time-scales, the distribution of returns is approximately normal.
- At all time-scales, returns display high variability.
- Leverage effect: If returns on a stock vary a lot, the price of the stock is lower.

Note that we usually work with the logarithm of returns. **Why?**

The S&P 500 index over time



Log-returns of S&P 500 over time



Stochastic processes

A stochastic process $(X(t))_{t \in T}$ is a collection of random variables, indexed by the set T . Usually this is interpreted as time.

We have two cases:

- A discrete-time stochastic process: $T = \mathbb{N}$.
- A continuous-time stochastic process: $T = \mathbb{R}_+$.

Random variables are specified via their distributions. Stochastic *processes* are specified via the distribution of their increments: $X_t - X_s$, $s < t$.

We will use stochastic processes to model the price process of an asset $S(t)$.

The binomial model

Consider first the discrete case. A very simple stochastic process could be the binomial model, where the increments of the price are described as

$$S(t + \Delta t) = \begin{cases} uS(t), & \text{With probability } p \\ dS(t), & \text{With probability } 1 - p \end{cases}$$

where $u > 1$ and $d < 1$. Meaning the stock price either goes up or down in the next period.

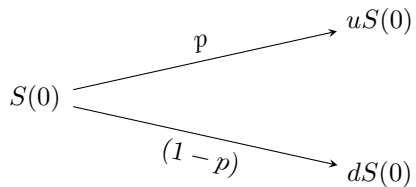
This can be thought of as a state space model where the states are “up” and “down”, or “good day” and “bad day”.

Of course the question is: What should u and d be?

Notice that we have assumed a dependency of the price next period on the price today.

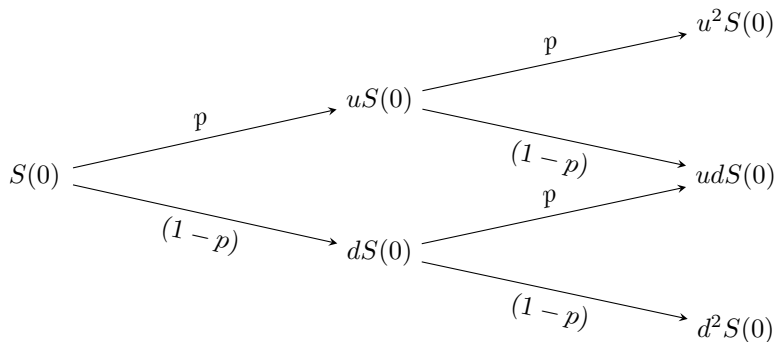
The binomial model, visually

It is easier to think of the binomial model by visualising it:



The binomial model, visually (cont'd)

It is easier to think of the binomial model by visualising it:



Note that it is *recombining* by design since $u \cdot d = d \cdot u$.

The binomial model (cont'd)

The binomial model is obviously a simple model, but it illustrates some important concepts well.

We usually consider the model up to an endpoint. Assume that we consider the price over the interval $[0, T]$ and over N partitions, i.e. $\{0, \frac{T}{N}, \frac{2T}{N}, \dots, T\}$.

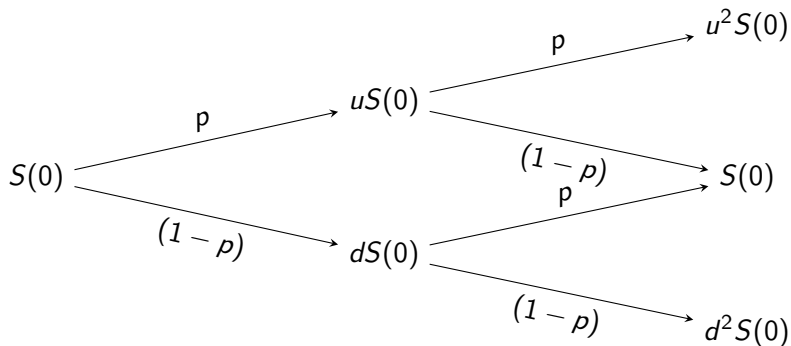
Denote by K the total number of upticks from time 0 to time T . Then we have

$$\mathbb{P}\left(S(T) = S(0)u^K d^{N-K}\right) = \binom{N}{K} p^K (1-p)^{N-K}$$

hence the name: Binomial model.

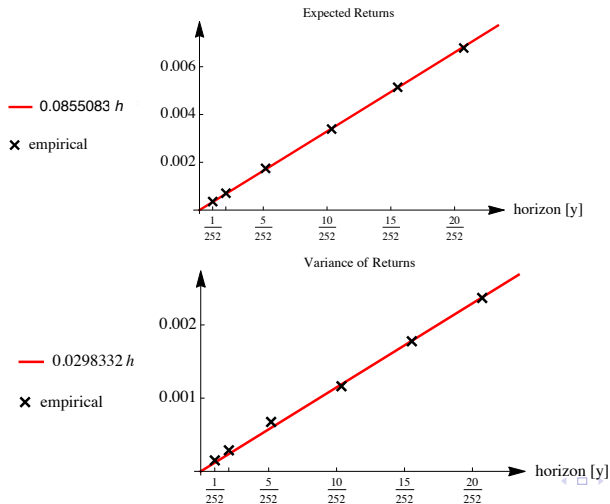
The binomial model (cont'd)

It is often convenient to have the binomial model without trend. This means $d = \frac{1}{u}$.



Calibrating the model to data

S&P 500 returns (1995-2015) from `finance.yahoo.com`



Calibrating the model to data (cont'd)

The model depends on *three parameters*: p, u, d .

One way to choose these could be such that the the mean and variance of the log-returns per unit of time are matched in the model and in the data.

The log-return from t to $t + \Delta t$ is $\log \left(\frac{S(t+\Delta t)}{S(t)} \right)$.

$$\hat{\mu} \cdot \Delta t = \mathbb{E} \left[\log \left(\frac{S(t + \Delta t)}{S(t)} \right) \right], \quad \hat{\sigma}^2 \cdot \Delta t = \mathbb{V} \left[\log \left(\frac{S(t + \Delta t)}{S(t)} \right) \right]$$

where $\hat{\mu}$ and $\hat{\sigma}^2$ are the empirical mean and variance from data:

$$\hat{\mu} = \frac{1}{N} \sum_{k=1}^N \log \left(\frac{S(t_k)}{S(t_{k-1})} \right), \quad \hat{\sigma}^2 = \frac{1}{N-1} \sum_{k=1}^N \left[\log \left(\frac{S(t_k)}{S(t_{k-1})} \right) - \hat{\mu} \right]^2$$

Calibrating the model to data (cont'd)

From the preceding equations we get

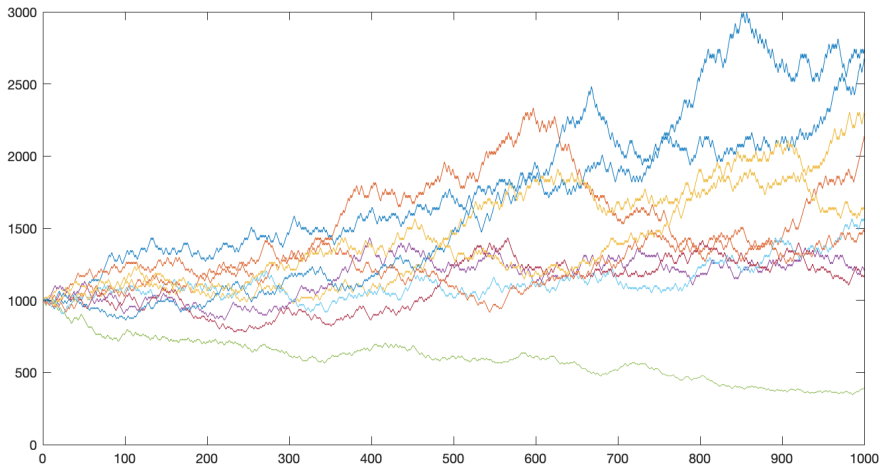
$$\begin{aligned}\hat{\mu} \cdot \Delta t &= p \cdot \log(u) + (1 - p) \cdot \log(d) \\ \hat{\sigma}^2 \cdot \Delta t &= (\log(u) - \log(d))^2 p(1 - p)\end{aligned}$$

So we have two equations and three unknowns. If we choose, e.g., $d = \frac{1}{u}$ we can solve the model.

For $\Delta t \rightarrow 0$, we approximately get

$$p = \frac{1}{2} + \frac{1}{2} \frac{\hat{\mu}}{\hat{\sigma}} \sqrt{\Delta t}, \quad u = e^{\hat{\sigma} \sqrt{\Delta t}}, \quad d = e^{-\hat{\sigma} \sqrt{\Delta t}}$$

10 simulated price paths in calibrated binomial model (to S&P 500)



Extending the model

A downside of the binomial model is that the model is static, in the sense that all periods are fixed.

To illustrate: In the one-period model, what happens if we have to consider an option with maturity $\frac{T}{2}$?

If we add a new period, we need to re-calibrate and re-calculate all the nodes in the new model.

Instead we could try to formulate a model in continuous time.

Extending the model (cont'd)

The binomial model has a useful property in that it is multiplicative:

$$S(t + \Delta t) = S(t)\varepsilon_{t+\Delta t} \implies \frac{S(t + \Delta t)}{S(t)} = \varepsilon_{t+\Delta t}$$

Or equivalently: The log-price is additive:

$$\log(S(t + \Delta t)) = \log(S(t)) + \log(\varepsilon_{t+\Delta t})$$

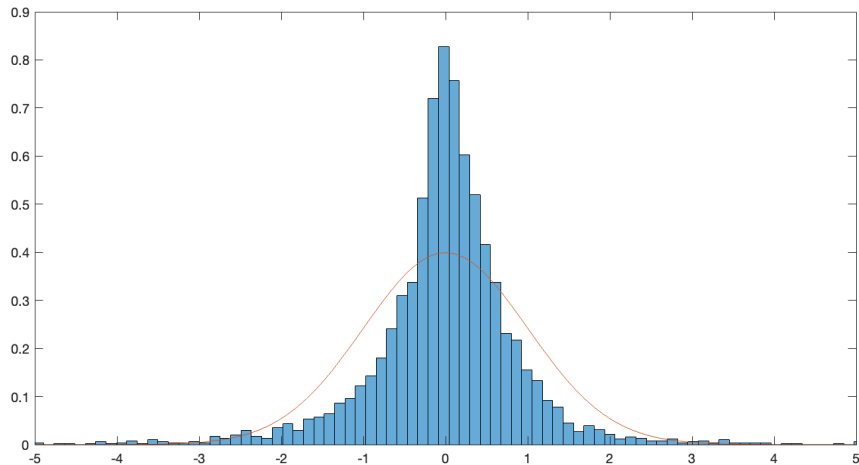
where $\varepsilon_{t+\Delta t}$ is a random variable equal to u or d with probability p and $(1 - p)$.

Realistically, $\varepsilon_{t+\Delta t}$ should be able to take on more values – preferably a continuum.

We will assume that log-returns are normally distributed.

$$\log(\varepsilon_{t+\Delta t}) \stackrel{i.i.d}{\sim} N(\mu\Delta t, \sigma^2\Delta t)$$

Log-returns vs fitted normal distribution



The distribution of the price process: Log-normal

Given normally distributed log-returns, S_t has *log-normal distribution*.

The log-normal distribution has pdf given by

$$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\log(x)-\mu)^2}{2\sigma^2}}, \quad x > 0,$$

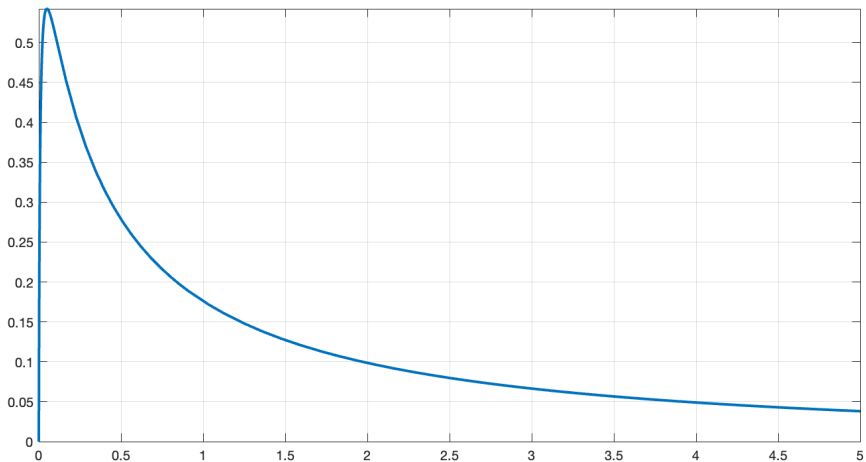
and mean and variance

$$\mathbb{E}[S(t)] = S(0)e^{(\mu+\frac{1}{2}\sigma^2)t}, \quad \mathbb{V}[S(t)] = (e^{\sigma^2t} - 1) e^{2\mu+\sigma^2t}$$

A useful feature of specifying the distribution of log-returns is that the price is always positive:

$$S(t + \Delta t) = e^{\log(S(t)) + \log(\varepsilon_t)} > 0 \quad \forall t \quad (1)$$

The log-normal distribution



The Brownian motion

We have extended the log-returns to take a continuum of values. We would also like to extend the price process to continuous *time*.

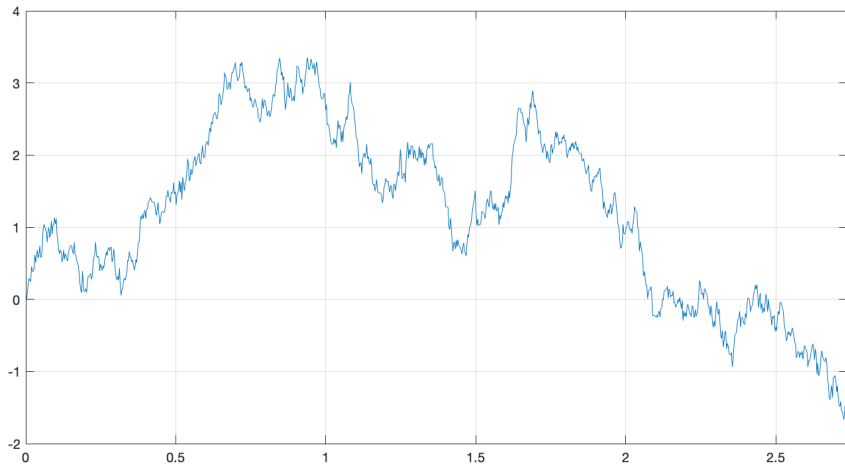
For this, we need to take the limit as $\Delta t \rightarrow 0$ which requires a surprisingly large amount of work. For this we need the *Brownian motion*.

Definition

A stochastic process $(W_t)_{t \in [0, \infty)}$ is called a *Brownian motion* or *Wiener process* if

- $W_0 = 0$.
- For $0 = t_0 < t_1 < \dots < t_n$ the increments $W_{t_n} - W_{t_{n-1}}, \dots, W_{t_1} - W_{t_0}$ are independent.
- For $0 < s < t$ we have $W_t - W_s \sim N(0, (t - s))$ and $W_t - W_s \sim W_{t-s}$.
- $W(t)$ is continuous in t .

Simulated path of Brownian motion



The Brownian motion (cont'd)

Note that the variance of the increments of the Brownian motion is scaled by the time:

$$\mathbb{V}[W_t - W_s] = t - s$$

The Brownian motion has many interesting properties. It is *continuous*, but

- *not monotone* on any interval.
- *not differentiable* at any point.

It is, however, possible to define a *stochastic integral* from the Brownian motion:

$$\int_0^t f(s) \, dW_s := \lim_{|\Delta t| \rightarrow 0} \sum_{k=1}^n f(t_k) (W_{t_k} - W_{t_{k-1}}) \quad (2)$$

In particular we have: $\int_t^T dW_s = W_T - W_t$.

Stochastic differential equations

Let $\varepsilon_{t+\Delta t} \sim N(\mu\Delta t, \sigma^2\Delta t)$ and consider the multiplicative (log-additive) model:

$$\log(S(t + \Delta t)) = \log(S(t)) + \varepsilon_{t+\Delta t} = \log(S(t)) + \mu\Delta t + \sigma(W_{t+\Delta t} - W_t)$$

Rearrange:

$$\frac{\log(S(t + \Delta t)) - \log(S(t))}{\Delta t} = \mu + \sigma \frac{(W_{t+\Delta t} - W_t)}{\Delta t}$$

Letting $\Delta t \rightarrow 0$ above, we would normally obtain the derivative of $\log(S(t))$ in t . However, W is not differentiable!

Instead, using the stochastic integral (2), we may define

$$\begin{aligned} \log(S(t + \Delta t)) &= \log(S(t)) + \mu\Delta t + \sigma(W_{t+\Delta t} - W_t) \\ &= \log(S(t)) + \int_t^{t+\Delta t} \mu \, ds + \sigma \int_t^{t+\Delta t} dW_s \end{aligned} \tag{3}$$

Stochastic differential equations (cont'd)

The definition of the log-price movements in equation (3) is mathematically rigorous. However, we often write it in a more intuitive way, as a stochastic differential equation.

Definition (Stochastic differential equation)

Let $(W_t)_{t \in [0, \infty)}$ be a Brownian motion and let $X(t)$ be given as (sometimes called an Itô process)

$$X(t) = X(0) + \int_0^t a(s, X(s)) \, ds + \int_0^t b(s, X(s)) \, dW_s$$

Then we write $X(t)$ as a stochastic differential equation:

$$dX(t) = a(t, X(t))dt + b(t, X(t))dW_t$$

We sometimes refer to the symbol $dX(t)$ as the *dynamics* of $X(t)$.

Stochastic differential equations (cont'd)

The representation of $X(t)$ by its dynamics is useful due to the intuitive “differential” form.

For the log-price in equation (3), we have dynamics

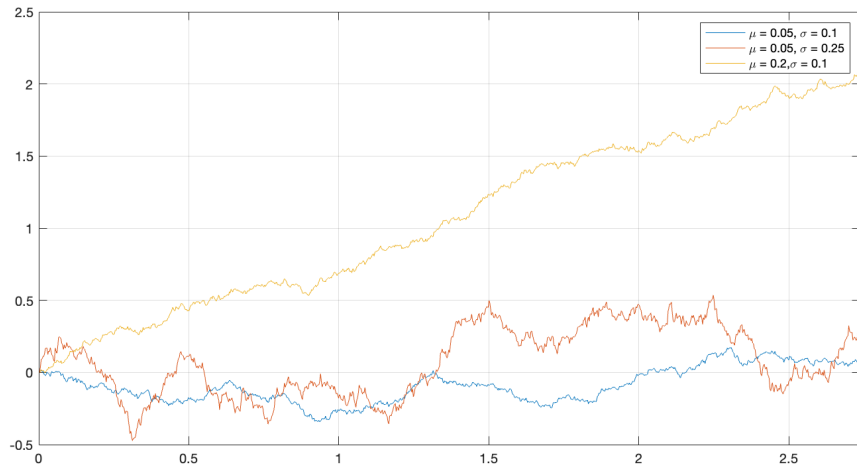
$$d \log(S(t)) = \mu dt + \sigma dW_t$$

Think of it as: Small changes in the log-price are equal to a small constant plus noise.

- The constant μ is called the *drift*.
- The constant σ is called the *volatility*.

The drift pushes the process up/down, while the volatility determines how wildly the process deviates around the drift.

Simulated log-prices as in equation (3)



Itô's lemma

Given a stochastic differential equation of the form

$$dX(t) = a(t, X(t)) dt + b(t, X(t)) dW_t$$

we are often interested in how a function of $X(t)$, $f(t, X(t))$, then evolves.

Turns out this is not trivial.

If $W(t)$ was differentiable (*which it is not!*), we could simply use the chain rule

$$df(t, X(t)) \stackrel{?}{=} \frac{\partial f}{\partial t} \cdot dt + \frac{\partial f}{\partial x} \cdot dX \tag{4}$$

Itô's lemma (cont'd)

Due to the Brownian motion not being so well-behaved, the ordinary chain rule does not apply. Instead we have *Itô's lemma*.

Theorem (Itô's lemma)

Let $X(t)$ be described by the stochastic differential equation

$$dX(t) = a(t, X(t)) dt + b(t, X(t)) dW_t$$

and let $f(t, x)$ be a function that is continuously differentiable in t and twice continuously differentiable in x . Then $f(t, X(t))$ has dynamics

$$df(t, X(t)) = \left\{ \frac{\partial f}{\partial t} + a(t, X(t)) \frac{\partial f}{\partial X} + \frac{1}{2} b(t, X(t))^2 \frac{\partial^2 f}{\partial X^2} \right\} dt + b(t, X(t)) \frac{\partial f}{\partial X} dW_t$$

Itô's lemma (cont'd)

The notation in theorem (3) can be a little confusing, so we specify:

$$\frac{\partial f}{\partial X} = \frac{\partial f(t, x)}{\partial x} \Big|_{x=X(t)}, \quad \frac{\partial^2 f}{\partial X^2} = \frac{\partial^2 f(t, x)}{\partial x^2} \Big|_{x=X(t)}$$

From Itô's lemma we see that the new process $f(t, X(t))$ has

- Drift: $\frac{\partial f}{\partial t} + a(t, X(t)) \frac{\partial f}{\partial X} + \frac{1}{2} b(t, X(t))^2 \frac{\partial^2 f}{\partial X^2}$.
- Volatility: $b(t, X(t)) \frac{\partial f}{\partial X}$.

Itô's lemma tells us, that we should add the correction $\frac{1}{2} b(t, X(t))^2 \frac{\partial^2 f}{\partial X^2}$ to the usual derivatives of the chain rule.

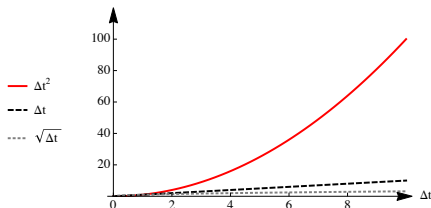
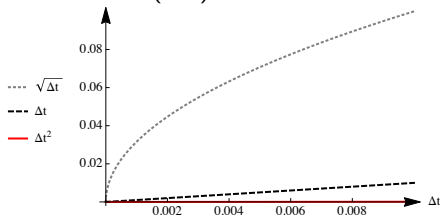
Notice also that the stochasticity in df is still from the same Brownian motion, W_t .

Itô's lemma (cont'd)

For very small Δt s, can think of a Wiener process as

$$W_{t+\Delta t} \approx \begin{cases} w_t + \sqrt{\Delta t} & \text{with probability } \frac{1}{2} \\ w_t - \sqrt{\Delta t} & \text{with probability } \frac{1}{2} \end{cases}$$

When Δt is small, $(\Delta t)^2 \ll \Delta t \ll \sqrt{\Delta t}$:



The *second-order effect* of a factor that moves like a Wiener process is thus *as large as the first order effect* of a smooth factor.

Itô's lemma $\approx$ *Taylor expansion for Wiener processes.*

Geometric Brownian motion

Given log-price dynamics $d \log(S(t)) = \mu dt + \sigma dW_t$, what is the price dynamics?

$S(t) = f(t, S(t)) = e^{\log(S(t))}$. Use Itô's lemma:

$$\frac{\partial f}{\partial t} = 0, \quad \frac{\partial f}{\partial \log(S)} = S(t), \quad \frac{\partial^2 f}{\partial (\log(S))^2} = S(t)$$

Insert into the dynamics

$$dS(t) = \left(\mu + \frac{1}{2} \sigma^2 \right) S(t) dt + \sigma S(t) dW_t$$

This process is called a *geometric Brownian motion*.

Geometric Brownian motion (cont'd)

So if the log-price is described by an Itô process, then the price is a geometric Brownian motion

$$dS(t) = \left(\mu + \frac{1}{2}\sigma^2 \right) S(t) dt + \sigma S(t) dW_t \quad (5)$$

The infinitesimal return $\frac{dS(t)}{S(t)}$ thus has drift $\mu + \frac{1}{2}\sigma^2$ and volatility σ .

You will show at the exercises that the solution to the stochastic differential equation (5) is

$$S(t) = S(0)e^{\mu \cdot t + \sigma W_t}$$

From this we may obtain properties of $S(t)$ such as the expected value

$$\mathbb{E}[S(t)] = S(0)e^{(\mu + \frac{1}{2}\sigma^2)t}$$

Geometric Brownian motion (cont'd)

We can also start by assuming that the price process follows a geometric Brownian motion:

$$dS(t) = \mu S(t)dt + \sigma S(t)dW_t$$

How does the log-price look then? $\rightarrow$ Itô's lemma.

$$d \log(S(t)) = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t$$

$$\mathbb{E}[S(t)] = S(0)e^{\mu t}$$

Simulation

In these slides I have shown some simulations. It is often useful to be able to simulate the price process.

The basic idea is as follows

- Convert $[0, T]$ into small subintervals of length Δt .
- Initialize at $\log(S(0)) = \bar{S}$.
- Simulate a standard normal random variable, N —any programming software, even Excel, is able to simulate standard normal random variables.
- Define $\log(S(\Delta t)) = \log(S(0)) + \mu\Delta t + \sigma\sqrt{\Delta t}N$.
- Iterate until time T and calculate $S(T) = \exp(\log(S(T)))$.

Investment and Finance

Options

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Outline of this topic

This week we will cover

- Options and terminology
- Intrinsic value and time value
- The put-call parity
- Pricing European options
 - The one-period binomial model
 - The two-period binomial model
 - The N -period binomial model
- American options

Options

Last week we briefly mentioned options to motivate the need for stochastic processes.

An option on an underlying security S is a derivative contract that gives the holder the right, but not the obligation, to either buy or sell the underlying at a pre-specified price K at some future point(s) in time.

- **Call option:** We are allowed to buy the underlying.
- **Put option:** We are allowed to sell the underlying.

The point of the contract is that we only buy/sell if it is advantageous to do so. The *payoff* is never negative.

This also means that option prices are always positive.

Options (cont'd)

There are several types of options:

- **European option:** The option can only be exercised at maturity T .
- **American option:** The option may be exercised *at any point in time* before T —early exercise.

These two are the most commonly traded option types and are usually referred to as “vanilla” options.

European options are simpler and are usually the most traded on indexes while American options are usually the most traded for single stocks.

A European option payoff function can be represented as

$$\text{payoff}_{\text{call}}(T) = \max\{S(T) - K, 0\}, \quad \text{payoff}_{\text{put}}(T) = \max\{K - S(T), 0\}$$

Notice that the optionality introduces *non-linearity in the payoff*.

Option payoffs and terminology

Consider a European call option and denote the value at time t by C_t .

$$S(T) \leq K \implies \text{buy the underlying at market price} \implies C_T = 0$$

$$S(T) \geq K \implies \text{exercise the option and save money} \implies C_T = S(T) - K$$

Similar reasoning for put options, but the other way around.

We say that a call option at time t is *in-the-money*, *at-the-money* or *out-of-the-money* when

$$S(t) > K, \quad S(t) = K, \quad S(t) < K$$

respectively (and similar for put options).

Intrinsic value and time value

Consider a call option on S with strike K and maturity T . Its price at $t \in [0, T]$ is

$$P = IV + TV$$

with

- **Intrinsic value (IV):** $\max\{S(t) - K, 0\}$. The value if the option expired today.
- **Time value (TV):** There is still time $\rightarrow$ can still end up (more) ITM.

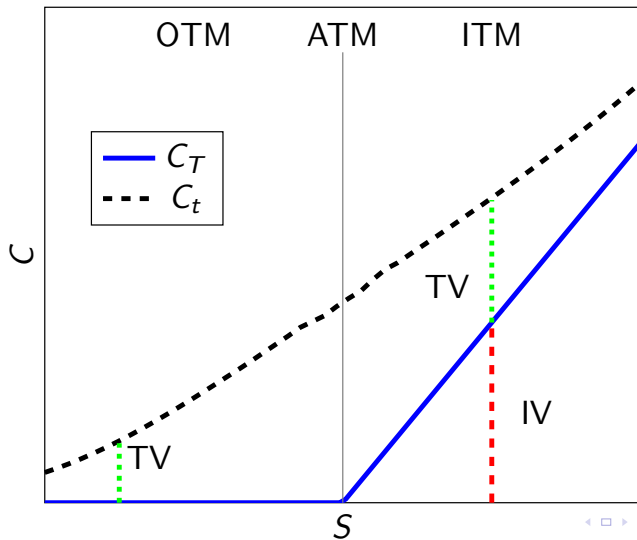
If $Pr(\text{ending up ITM}) > 0$, the option has strictly positive total value.

If $IV_t = 0$ at some point in time t , then the time value cannot be negative.

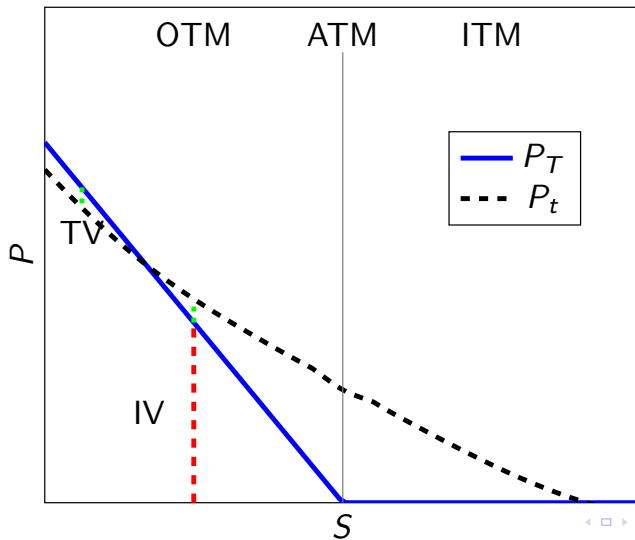
But if $IV > 0$, the time value of a European option can actually be negative! [Example?](#)

American options *always* have positive time value.

Value of European call



Value of European put



The put-call parity

Calls and puts are in many ways similar. For European options we have a very strong relation between their prices: The put-call parity.

$$C_t - P_t = S(t) - d_{t,T}K$$

Proof on board.

So if we know the price of a European call, we may easily obtain the corresponding put price (with same K).

The put-call parity is *not* true for American options! Here, the best we can do is

$$S(t) - K \leq C_t^a - P_t^a \leq S(t) - d_{t,T}K$$

Pricing options

Next step: to actually determine the price of options, with specific models.

A useful starting point: the *binomial model* as introduced in the previous topic.

Recall the setup:

- Time horizon: $[0, T]$ with N partitions.
- $\Delta t = \frac{T}{N}$.
- $S(t)$ is the price of the underlying at time $t = 0, \Delta t, \dots, T$.
- r is the net interest rate and $d_{s,t}$ the discount factor between period s and t .
- At period $t + \Delta t$, the price has moved up or down from time t to: $uS(t)$ with probability p and $dS(t)$ with probability $1 - p$.

Pricing options: Binomial model

As we now want to price derivatives, we need our model to be arbitrage-free.

Theorem (Arbitrage in binomial model)

The binomial option pricing model is free of arbitrage (and complete) if and only if

$$d < \frac{1}{d_{t,t+\Delta t}} < u$$

Proof:

- If $\frac{1}{d_{t,t+\Delta t}} \leq d < u$: Buy the underlying, borrow at the risk-free rate.
- If $\frac{1}{d_{t,t+\Delta t}} \geq u > d$: Short the underlying, put the money in the bank.

One-period binomial model

Let us start by considering the one-period model— $N = 1$, $\Delta t = T$.

- Starting price: $S(0) = S_0$; at time T : $S(T) = uS_0$ or $S(T) = dS_0$.

Denote by C_u and C_d the payoff of a European call if we go up or down

$$C_u = \max\{uS_0 - K, 0\}, \quad C_d = \max\{dS_0 - K, 0\}$$

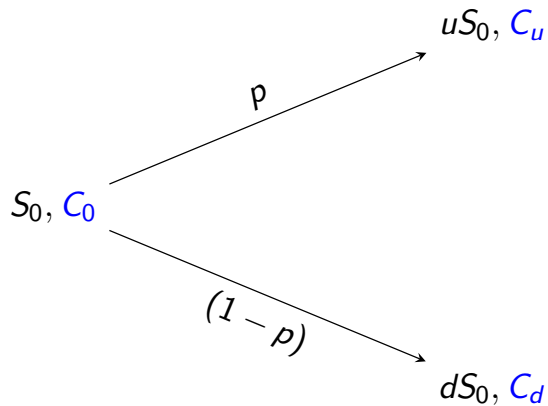
These are simply the realized payoffs, which are of course unknown at time 0

$$C_T = \max\{S_T - K, 0\}$$

$$\mathbb{P}(C_T = C_u) = p, \quad \mathbb{P}(C_T = C_d) = 1 - p$$

Two ways to price: *hedging* or *replication and risk-neutral valuation*.

One-period model



Pricing by hedging: One-period model

Assume that we have access to a bank account paying the risk-free rate r (compounded however you like), and, as a result, a discount factor $d_{0,T}$.

Let us create a portfolio of one short unit of the option and x long units of the underlying to **hedge the risk**:

$$V_u = -C_u + xuS_0, \quad V_d = -C_d + xdS_0$$

The hedge is perfectly effective if its payoff is the same in both states at time T , $V_u = V_d$, which implies

$$x = \frac{C_u - C_d}{(u - d)S_0}$$

Absence of arbitrage $\implies$ this portfolio must then earn the riskfree rate:
 $(-C_0 + xS_0) = d_{0,T}(C_u + xuS_0)$, which implies...

Pricing by hedging: One-period model (cont'd)

Theorem (Option price in the one-period model)

The price of a European call C_0 in the one-period binomial model is given by

$$C_0 = \frac{C_u - C_d}{(u - d)S_0} S_0 + d_{0,T} \frac{uC_d - dC_u}{(u - d)}$$

The derivative can be perfectly hedged by shorting

$$\Delta \equiv x = \frac{1}{S_0} \frac{C_u - C_d}{u - d}$$

units of the underlying.

The **Delta** (“ Δ ”) is the sensitivity of the option price to the underlying price:

- Short Δ units of the underlying when hedging the option.
- Buy Δ units of the underlying when replicating the option (see in the next slides).

Pricing by replication: One-period model

We want to find the call price today, C_0 , by a **replicating strategy** for the option: construct a portfolio with value C_u if the price goes up, C_d if the price goes down.

Absence of arbitrage $\implies$ portfolios with the same payoffs have the same setup costs.

Let v denote our portfolio and assume that it consists of x units of the underlying and B units in the bank. The value today of v is:

$$v_0 = xS_0 + B$$

At time T , the value is

$$v_u = xuS_0 + \frac{1}{d_{0,T}}B, \quad v_d = xdS_0 + \frac{1}{d_{0,T}}B$$

Replication then requires: $v_u = C_u$ and $v_d = C_d$.

Pricing by replication: One-period model (cont'd)

Theorem (Option price in the one-period model)

The price of a European call C_0 in the one-period binomial model is given by

$$C_0 = \frac{C_u - C_d}{(u - d)S_0} S_0 + d_{0,T} \frac{uC_d - dC_u}{(u - d)} = d_{0,T} \left(\frac{\frac{1}{d_{0,T}} - d}{u - d} C_u + \frac{u - \frac{1}{d_{0,T}}}{u - d} C_d \right)$$

We can rewrite the call price as

$$C_0 = d_{0,T} [qC_u + (1 - q)C_d]$$

where

$$q = \frac{\frac{1}{d_{0,T}} - d}{u - d} \tag{1}$$

Risk-neutral valuation: One-period model

The no-arbitrage condition $d < \frac{1}{d_{0,T}} < u$ ensures that

$$q \in (0, 1)$$

So the call price C_0 is equal to the discounted q -expectation of the payoff:

$$C_0 = d_{0,T} \mathbb{E}^{\mathbb{Q}}[C_T],$$

where the measure $\mathbb{Q}$ is associated to the risk-neutral probabilities q and $1 - q$.

So the option price does not depend on the objective up-down probabilities p and $1 - p$, only the risk-neutral ones, q and $1 - q$.

(Actually, p influences the option price indirectly through S_0 .)

Expected Return on the Underlying

Q: Why has the expected return on the underlying, as captured by p , no effect on the arbitrage-free price of the derivative?

A1: Think about hedging. If the probability p of a positive return is higher,

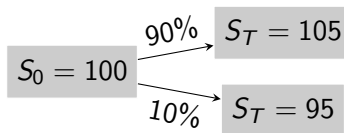
- The expected payoff of the option I sold is higher.
- The expected return of the hedging portfolio is higher as well.
- My overall portfolio is still risk-free.

A2: Think about replication.

- There is a replicating portfolio for the derivative.
- This replicating portfolio gives the same payoff as the derivative no matter which scenario occurs.
- Changing p without changing S_0 does not affect the setup costs of the replicating portfolio.

Example: European call option pricing

- $T = 6$ months, $K = 99$, $r = 0\%$ per year
- Dynamics of the underlying:



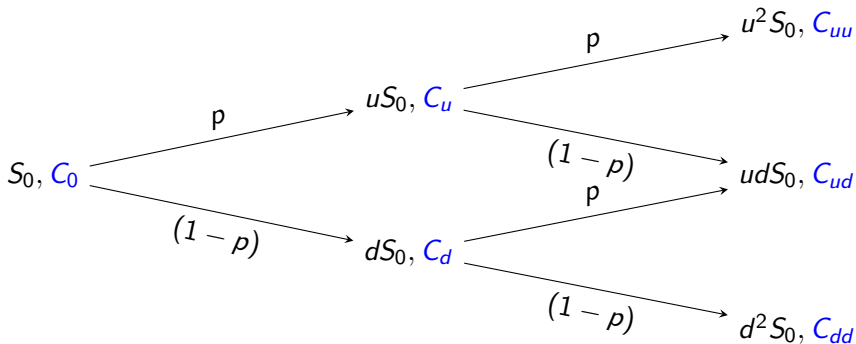
- The call trades at $c = 4$.

Questions:

- What is the Δ of the option?
- What is the risk-neutral probability q ?
- Is there an arbitrage strategy?

Option pricing: Two-period model

What happens if we extend the model to two periods?



Option pricing: Two-period model (cont'd)

General principle is the same: Start from time T and use the replicating strategy one step backwards each time.

Time T values of the call option:

$$C_{uu} = \max\{u^2 S_0 - K, 0\}, \quad C_{ud} = \max\{udS_0 - K, 0\}, \quad C_{dd} = \max\{d^2 S_0 - K, 0\}$$

Calculate option values at time $T - \Delta t$ as

$$C_u = d_{\Delta t, T} \mathbb{E}^Q [C_T \mid S(\Delta t) = uS_0] = d_{\Delta t, T} (qC_{uu} + (1 - q)C_{ud})$$

$$C_d = d_{\Delta t, T} \mathbb{E}^Q [C_T \mid S(\Delta t) = dS_0] = d_{\Delta t, T} (qC_{ud} + (1 - q)C_{dd})$$

Finally go all the way back to C_0 :

$$C_0 = d_{0, \Delta t} (qC_u + (1 - q)C_d) = d_{0, T} (q^2 C_{uu} + 2q(1 - q)C_{ud} + (1 - q)^2 C_{dd})$$

Note: q is now calculated using $d_{0, \Delta t}$ instead of $d_{0, T}$. (Also for no arbitrage.)

Option pricing: N -period model

With N periods, same idea: work iteratively backwards in the model.

Theorem (European option price in N -period model)

In an arbitrage free N -period binomial model with constant risk-neutral probabilities q and $1 - q$, the price of a European call, C_0 , is given by

$$\begin{aligned} C_0 &= d_{0,T} \mathbb{E}^{\mathbb{Q}} [C_T] \\ &= d_{0,T} \sum_{k=0}^N \binom{N}{k} q^k (1-q)^{N-k} \max \left\{ S_0 u^k d^{N-k} - K, 0 \right\} \end{aligned}$$

A put option can be priced in the exact same way, but using the put payoff instead.

Letting $\Delta t \rightarrow 0$, the option price converges to the Black-Scholes price, as we shall discuss next week.

Pricing American options

So far we have only considered the problem of pricing European options. American options are significantly more complicated.

In the binomial model, however, American options are not too hard to price.

Recall: For American options, it is advantageous to exercise early if the intrinsic value of the option goes above the price (negative time value).

$$C_t^a \leq \max\{S_t - K, 0\}$$

The added optionality to the American option implies that it should be more expensive than the corresponding European option.

Pricing American options (cont'd)

So far we have assumed that the underlying asset pays no dividends.

For a European call option with no dividends the following holds:

$$C_t \geq \max\{S(t) - K, 0\}, \quad t \in [0, T]$$

Proof on board

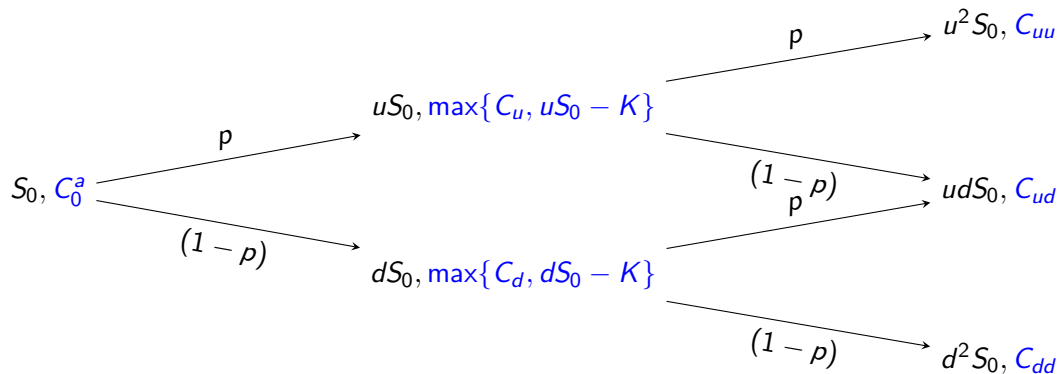
But since the American call price, C_t^a , is always greater than C_t , it is never advantageous to exercise early! Thus

$$C_t^a = C_t$$

Generally speaking (with dividends for example), we should check for early exercise. Thus we should replace C_t at each node with

$$C_t^a = \max\{C_t, S_t - K\}$$

Binomial tree for American options



Dividends

Sometimes the underlying instrument pays out dividends. For example for an option on a bond.

Generally there are two ways these are paid out

- Proportionally: The dividend at time t is a fraction of $S(t)$. So for example $\delta S(t)$ where δ is the dividends in percent.
- Absolute: The dividend D is a fixed payout and is subtracted from the current value $S(t)$. The price then evolves from $S(t) - D$.

The proportional approach means that the price process is simply multiplied by $(1 - \delta)$ and evolves as usual.

The absolute approach means that the tree is no longer recombining, which can be a problem.

Pricing of general derivatives

Consider a general derivative X on S that can only be exercised at time T and with payoff function $f(S)$.

Then we can price this derivative by risk-neutral valuation in the exact same way as we price European options:

$$\begin{aligned} X_0 &= d_{0,T} \mathbb{E}^{\mathbb{Q}}[X_T] \\ &= d_{0,T} \sum_{k=0}^N \binom{N}{k} q^k (1-q)^{N-k} f(u^k d^{N-k} S_0) \end{aligned}$$

where the risk-neutral probabilities q and $1 - q$ are calculated as in equation (1).

Rounding off: Why options?

So what is the use of an option contract?

Much like other derivatives, options are useful for hedging and risk management.

Options are like forwards, but less risky since there is no certain commitment. We do not *have* to exercise.

Options can also be used in various combinations to perform *speculative strategies*. Bull spreads, bear spreads, butterfly spreads etc.

Some brokers do not allow short selling at all. This can limit our ability to play the downside. However, we can always buy put options!

Summary

We have introduced the very first option pricing model based on the binomial lattice model.

We have seen how to price European calls and puts in this setup, by replication and by risk-neutral valuation.

We have also considered the case of American options, which is more involved.

Pricing formulas for other derivative contracts follow analogously to what we have done. The binomial model is not at all restricted to only options.

Investment and Finance

The Black-Scholes-Merton model

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Motivation

We have discussed options and the need for stochastic processes to model asset prices.

Last time: discrete time setting by using a binomial lattice—pricing formulas and methods for both European and American options.

This week: in continuous time. Many advantages compared to the discrete case, but also more work.

Main framework: the most famous option pricing model of all time—the Black-Scholes(-Merton) model.

Created by Fisher Black and Myron Scholes in 1973; Robert Merton derived something closely related at the same time. Scholes and Merton won the Nobel prize for it.

Overview

Model	Authors			States	Year
Binomial or CRR	Cox 	Ross 	Rubinstein 	• discrete time • discrete state	1979
BSM	Black 	Scholes 	Merton 	• continuous time • continuous state	1973

- Same economic content in both models
- Binomial model is more accessible
- BSM gives closed-form solutions

Binomial (last week)

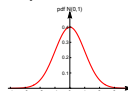
BSM (this week)

Basic
shock

Bernoulli
(coin toss)



Gaussian $\mathcal{N}(0, 1)$
(bell-shaped distribution)



Combine into a
process

Random Walk

Wiener Process z_t
(aka Brownian Motion)

Returns

- * underlying
- * portfolio

$$\frac{S_n}{S_{n-1}} = u^{Z_n} d^{1-Z_n} = e^{\sigma \sqrt{\Delta t} (2Z_n - 1)}$$

$$\Delta w_n = (w_n - \phi_{S,n}) (e^{r\Delta t} - 1) + \phi_{S,n} \frac{\Delta S_n}{S_n}$$

Stochastic Integrals

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t$$

$$dw_t = (w_t - \phi_{S,t}) r dt + \phi_{S,t} \frac{dS_t}{S_t}$$

Hedging

compare ΔS_n and Δf_n

compare dS_t and $d(f(t, S_t))$
Itô's Lemma

Replication

- Setup costs are $f_0 = e^{-rT} \hat{\mathbb{E}}_0[f_T]$
- Under RN $\mathbb{Q}$: expected rate of return on S_n is r

- Setup costs $f_0 = e^{-rT} \hat{\mathbb{E}}_0[f_T]$
- Under RN $\mathbb{Q}$: expected rate of return on S_t is r

Girsanov Theorem

Outline of this topic

This week we will cover:

- The BSM model setup
- The BSM equation
- The BSM formula
- Hedging in BSM
- The Greeks
- Implied volatility
- Risk-neutral pricing and the BSM formula
- Shortcomings of the model

Black-Scholes-Merton: Standing Assumptions and Question

The stock price satisfies

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t$$

for two constants μ and σ . There are no dividends.

There is a risk-free investment paying an interest rate r —the “bank account” then follows a non-stochastic differential equation:

$$dB_t = rB_t dt$$

No trading frictions:

Trading is continuous * no transaction costs * no taxes * short-selling is costless * securities are perfectly divisible * etc.

What is the price of a European call/put in this framework?

Black-Scholes-Merton: Pricing via Hedging Argument

Pricing argument: **Absence of arbitrage** $\implies$ the return on a risk-free portfolio is r .

An arbitrageur sells the derivative at time t and **hedges** his position by investing $\phi_{S,t}$ in the stock.

Assume that the value f_t of the derivative at time t is given by

$$f(t, S_t)$$

for a smooth function f of the time and the underlying price.

Portfolio dynamics:

asset	position	value	value change over dt
underlying	$\frac{\phi_{S,t}}{S_t}$	$\phi_{S,t}$	$\phi_{S,t} \frac{dS_t}{S_t}$
derivative	-1	$-f(t, X_t)$	$-d(f(t, X_t))$

Black-Scholes-Merton: Pricing via Hedging Argument (cont'd)

Denote the value of the portfolio at time t by

$$\Pi_t = -f_t + \phi_{S,t}$$

The change in this portfolio over the next instant dt is

$$d\Pi_t = -df_t + \phi_{S,t} \frac{dS_t}{S_t}$$

If this portfolio is **riskless**, we must earn a return of r on it, i.e.

$$d\Pi_t = r\Pi_t dt.$$

But how does $df_t = d(f(t, S_t))$ depend on dt and dS_t , and how much $\phi_{S,t}$ to invest in stocks?

Black-Scholes-Merton: Pricing via Hedging Argument (cont'd)

Itô's Lemma in the BSM world states that

$$d(f(t, S_t)) = \left[f_t(t, S_t) + f_S(t, S_t)\mu S_t + \frac{1}{2}f_{SS}(t, S_t)\sigma^2 S_t^2 \right] dt + f_S(t, S_t)\sigma S_t dW_t,$$

Substituting it in, the portfolio is riskless (no dW_t term) iff

$$\phi_{S,t} = f_S(t, S_t)S_t$$

Matching drifts (the dt terms), after some algebra, we obtain:

Proposition

The price $f(t, S_t)$ of the derivative satisfies

$$r f(t, S) = f_t(t, S) + r S f_S(t, S) + \frac{1}{2}\sigma^2 S^2 f_{SS}(t, S)$$

This equation is the [Black-Scholes-Merton differential equation](#).

Black-Scholes-Merton: Pricing via Hedging Argument (cont'd)

Notice that we have not used (yet) that this was a call option.

There are many solutions to the BSM equation—one for each derivative contract written on S_t . Numerical solution are always possible.

We always need **boundary conditions**. E.g., for a European call option,

$$\left\{ \begin{array}{lcl} r f(t, S) & = & f_t(t, S) + r S f_S(t, S) + \frac{1}{2} \sigma^2 S^2 f_{SS}(t, S) \\ f(T, S) & = & [S - K]^+ \\ f(t, 0) & = & 0 \\ \lim_{S \rightarrow \infty} \frac{f(t, S)}{S - K e^{-r(T-t)}} & = & 1 \end{array} \right.$$

Q: What do the three boundary conditions mean intuitively?

Black-Scholes-Merton Option Pricing Formula

Proposition

The price of a **European call option** is

$$c(S_0, r, \sigma, K, T) = S_0 N(d_1) - Ke^{-rT} N(d_2),$$

with

$$N(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = \text{cdf of a standard normal distribution}$$
$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{\sigma^2}{2}\right) T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

The price of the corresponding **put option** is

$$p(S_0, r, \sigma, K, T) = Ke^{-rT} N(-d_2) - S_0 N(-d_1).$$

Proof: Plug into the BSM equation and verify that it is a solution. $\square$

Proposition

The price of a European **call** option is

$$c(S_0, r, \sigma, K, T) = S_0 N(d_1) - Ke^{-rT} N(d_2),$$

with

$$N(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = \text{cdf of a standard normal distribution}$$
$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{\sigma^2}{2}\right) T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

The **Delta of the call** is

$$\Delta_{\text{call}} = \frac{\partial c}{\partial S_0}(S_0, r, \sigma, K, T) = N(d_1).$$

Proof: Direct calculation for the Δ . $\square$

BSM Deltas (cont'd)

Proposition

The price of a European **put** option is

$$p(S_0, r, \sigma, K, T) = Ke^{-rT} N(-d_2) - S_0 N(-d_1),$$

with the same definitions of N , d_1 , and d_2 as for the call.

The **Delta of the put** is

$$\Delta_{\text{put}} = \frac{\partial p}{\partial S_0}(S_0, r, \sigma, K, T) = N(d_1) - 1 = -N(-d_1).$$

Proof: Use the put-call parity. $\square$

Δ of a European call: Exercise

Data

- $T = \frac{3}{12}$, $\sigma = 25\%$, $r = 0.11\%$
- 3 call options: moneyness $\equiv \frac{S_0}{K} = \begin{cases} 1.2 & \text{in the money} \\ 1 & \text{at the money} \\ 0.8 & \text{out of the money} \end{cases}$

Calculate the Δ of the call option for the three moneyness values.

Δ of a European call: Exercise

Data

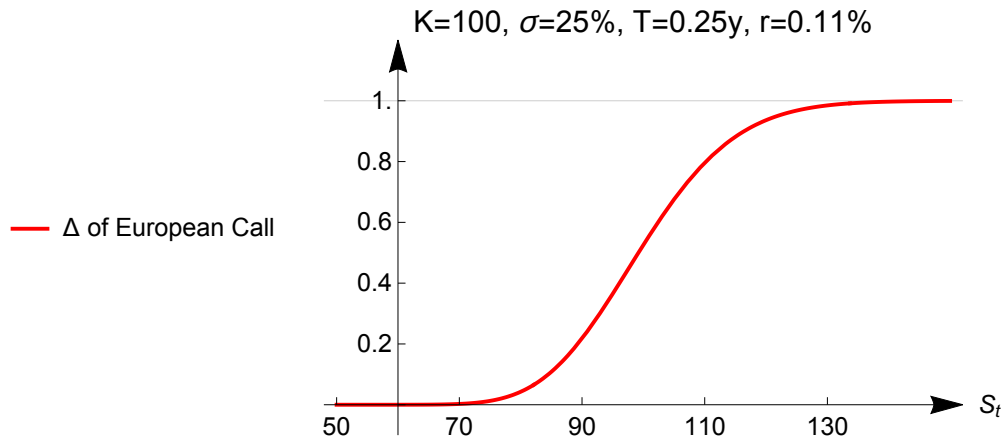
- $T = \frac{3}{12}$, $\sigma = 25\%$, $r = 0.11\%$
- 3 call options: moneyness $\equiv \frac{S_0}{K} = \begin{cases} 1.2 & \text{in the money} \\ 1 & \text{at the money} \\ 0.8 & \text{out of the money} \end{cases}$

Calculate the Δ of the call option for the three moneyness values.

Solution

$$\begin{aligned}\Delta = N(d_1) &= N\left(\frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{\sigma^2}{2}\right) T}{\sigma\sqrt{T}}\right) \\ &= N\left(\frac{\ln(\text{moneyness}) + \left(0.0011 + \frac{0.25^2}{2}\right) \frac{3}{12}}{0.25\sqrt{\frac{3}{12}}}\right) = \begin{cases} 0.94 & \text{if moneyness} = 1.2 \\ 0.53 & \text{if moneyness} = 1 \\ 0.04 & \text{if moneyness} = 0.8 \end{cases}\end{aligned}$$

Δ of a European call



Q: Why is it bounded? * Increasing? * Convergence toward 0/1?

Delta Hedging

Calculated the sensitivity $\Delta \equiv \frac{\partial}{\partial S} f(t, S_t)$ of the value of the derivative to the price of the underlying.

Long Δ shares of the underlying to hedge one short derivative: our portfolio is immune to first-order changes in the price of $S(t)$: *delta-neutral*.

Also short Δ shares of the underlying to hedge one long derivative.

Hedge is *locally perfect*: The option price is convex in S_t (its $\Gamma > 0$), so delta hedging only works for small changes in the underlying

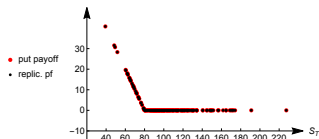
Keeping the hedge perfect requires *continuous rebalancing* (a dynamic hedging strategy), which is costly in practice due to transaction costs, so we rebalance at some fixed frequency (e.g. daily) or if the risk becomes too large.

Delta Hedging in the BSM framework

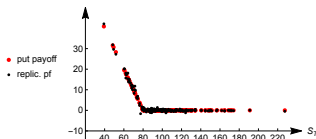
Δ still useful for realistic rebalancing frequencies

(Eu. put, $S_0 = 100$, $K = 80$, $\mu = 7\%$, $\sigma = 0.4$, $T = \frac{1}{2}$, $q = 1\%$, $r = 0$, 200 simulations)

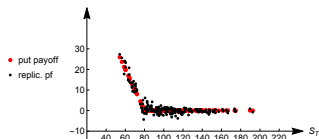
“continuous”



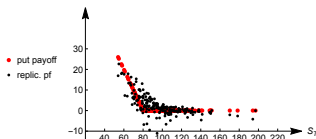
twice a day



weekly



monthly



Delta Hedging (cont'd)

Delta hedging can be used to hedge entire portfolios.

Say we have a portfolio of stocks and derivatives.

Let the portfolio price process be denoted V_t . Then we may want to hedge this portfolio against (small) changes in some stock or index, $S(t)$.

$$\Delta_V = \frac{\partial V}{\partial S} = 0$$

Assume we have some derivative on the stock with price process f_t . How many units x of the derivative do we buy to make our position delta-neutral?

$$\frac{\partial V}{\partial S} + x \frac{\partial f}{\partial S} \implies x = -\frac{\Delta_V}{\Delta_f}$$

Gamma (Γ)

- Curvature of the derivative price w.r.t. the underlying price.
- $\Gamma \equiv \frac{\partial^2}{\partial S^2} f(t, S_t) = \frac{\partial}{\partial S} \Delta(t, S_t)$
- Measures how fast Δ must be adjusted.
- Some like this analogy:

in your car	with your derivative
what time it is	S_t
where you are	f_t
how fast you are	Δ
how much you are accelerating	Γ

Gamma (Γ) (cont'd)

- The Γ of a European call is

$$\frac{n(d_1) e^{-qT}}{S_0 \sigma \sqrt{T}},$$

with n being the pdf of a standard normal random variable, and

$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r - q + \frac{\sigma^2}{2}\right) T}{\sigma \sqrt{T}}.$$

- Γ of a European put is the same (because of put-call parity)
- Γ of forwards, futures, and the underlying are all zero

Theta (Θ)

- Sensitivity of the derivative price to time to maturity
- $\Theta \equiv \frac{\partial}{\partial t} f(t, S_t)$
- Not about hedging, since you cannot hedge time

Theta (Θ) (cont'd)

- For a European call option,

$$\Theta = -\frac{S_0 n(d_1) \sigma e^{-qT}}{2\sqrt{T}} + qS_0 N(d_1) e^{-qT} - rKe^{-rT} N(d_2)$$

- Can check that

$$rc = \Theta + rS\Delta + \frac{1}{2}\sigma^2 S^2 \Gamma$$

for a European call.

- Actually,

$$rf = \Theta + rS\Delta + \frac{1}{2}\sigma^2 S^2 \Gamma \tag{1}$$

for any derivative price f and the associated Θ , Δ , and Γ .

- This is actually the same Black-Scholes-Merton differential equation expressed with Greeks.

Vega ($\mathcal{V}$)

- Sensitivity of the derivative price to volatility
- $\mathcal{V} \equiv \frac{\partial}{\partial \sigma} f(t, S_t)$
- Irrelevant in the BSM setting, since vol is constant, but possibly useful in reality.

Vega ($\mathcal{V}$) (cont'd)

- The $\mathcal{V}$ of a European call is

$$S_0 \sqrt{T} n(d_1) e^{-qT},$$

with n being the pdf of a standard normal random variable, and

$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r - q + \frac{\sigma^2}{2}\right) T}{\sigma \sqrt{T}}.$$

- $\mathcal{V}$ of a European put same as for call.
- $\mathcal{V}$ of the underlying is zero.

Rho (ρ)

- Sensitivity of the derivative price to the interest rate
- $\rho \equiv \frac{\partial}{\partial r} f(t, S_t)$
- Irrelevant in the BSM setting, but possibly useful in reality

Rho (ρ) (cont'd)

- The ρ of a European call is

$$KTe^{-rT}N(d_2),$$

with N being the cdf of a standard normal random variable, and

$$d_2 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r - q - \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}.$$

- The ρ of a European put is

$$-KTe^{-rT}N(-d_2),$$

Implied volatility

The BSM formula shows that the call option price c_t is a function of $S(t)$, K , $T - t$, r and σ .

$$c_t^{\text{BS}} = c(S(t), K, T - t, r, \sigma)$$

The first four parameters are directly observable, but the volatility parameter σ is not. On the other hand, we can observe the current market price c_t^{market} .

From these observable quantities, we may define the **implied volatility** σ_{IV} , which solves:

$$c(S(t), K, T - t, r, \sigma_{\text{IV}}) = c_t^{\text{market}}$$

So if we plug σ_{IV} into the BSM formula, we get the observed market price. It is volatility parameter that is *implied* by the market price.

Implied volatility (cont'd)

The implied volatility σ_{IV} is in one-to-one correspondence with the BSM option price c_t^{BS} as the BSM option price is increasing in σ .

The implied volatility is therefore very useful (and in fact often used) quantity that reflects option prices. Many traders quote implied volatilities instead of prices.

The implied volatility can be interpreted as the market's forecast of the future volatility of the underlying, σ .

Remember that volatility does not measure direction, just variability.

Note: We need a computer algorithm to find σ_{IV} . There is no analytical solution.

Black-Scholes-Merton: Pricing via Replication Argument

Pricing argument: Absence of arbitrage $\implies$ two portfolios with the same payoffs have the same setup costs.

- Want the fair price f_0 at $t = 0$ of a derivative with payoff f_T at T .
- If we can find a replicating strategy $(w_0, (\phi_{S,t})_{t=0}^T)$ satisfying

$$\begin{cases} dw_t &= (w_t - \phi_{S,t}) r dt + \phi_{S,t} \frac{dS_t}{S_t} \\ w_T &= f_T \end{cases}$$

then Absence of Arbitrage imposes $f_0 = w_0$.

Black-Scholes-Merton: Pricing via Replication Argument (cont'd)

The recipe in discrete time led to **risk-neutral pricing**, with the steps:

- Find the risk-neutral probability $\mathbb{Q}$.
- The expected rate of return on S_t under $\mathbb{Q}$ is r .
- The arbitrage-free price of the derivative is $f_0 = e^{-rT} \hat{\mathbb{E}}_0[f_T]$.
- The Δ of the derivative is

$$\Delta = \frac{\text{change in the value of the derivative}}{\text{change in the value of the underlying}} = \frac{\partial f_0}{\partial S_0}$$

- The investment in the replicating portfolio is thus

$$\phi_{S,t} = \Delta(t, S_t) S_t.$$

Q: How do we find $\mathbb{Q}$ in a BSM setting?

Black-Scholes-Merton: Pricing via Replication Argument (cont'd)

Proposition (Girsanov Theorem)

If z is a $\mathbb{P}$ -Wiener process (under the physical probability measure $\mathbb{P}$), there exists another probability measure under which

$$dz_t^{\mathbb{Q}} \equiv dz_t + \alpha dt$$

is a Wiener process as well.

Corollary

Choosing $\alpha = \frac{\mu - r}{\sigma} = \text{"market price of risk"}$ in Girsanov Theorem yields a probability $\mathbb{Q}$ under which

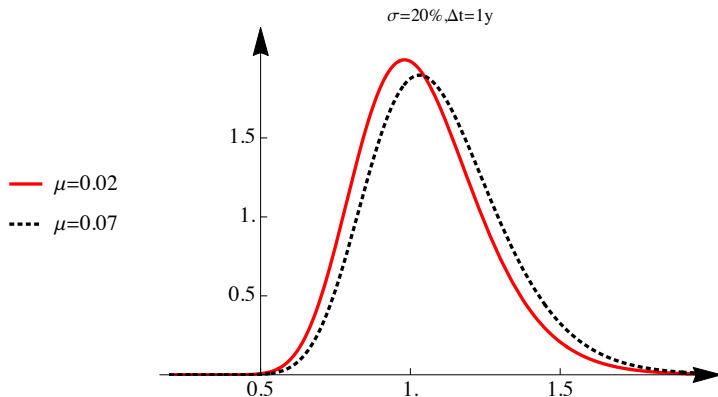
$$dW^{\mathbb{Q}} \equiv dW + \frac{\mu - r}{\sigma} dt$$

is a $\mathbb{Q}$ -Brownian motion. In particular, $\frac{dS_t}{S_t} = rdt + \sigma dW_t^{\mathbb{Q}}$ and the expected rate of return on S_t is r .

Just like in the binomial setting, we call $\mathbb{Q}$ the risk-neutral probability measure.

Black-Scholes-Merton under the Risk-Neutral Measure

Under the risk-neutral probability $\mathbb{Q}$, the stock is still as risky as under the physical probability $\mathbb{P}$ but yields a lower return:



Black-Scholes-Merton: Pricing via Replication Argument (cont'd)

Arbitrage-Free Derivative Pricing

The arbitrage-free price f_0 of a derivative at $t = 0$ is

$$f_0 = E^{\mathbb{Q}} [e^{-rT} f_T]$$

with

- f_T being the payoff of the derivative at maturity
- $E^{\mathbb{Q}}$ being the expectation under the **risk-neutral probability** $\mathbb{Q}$ under which

$$\frac{dS_t}{S_t} = rdt + \sigma dW_t^{\mathbb{Q}}$$

for a $\mathbb{Q}$ -Wiener process $W^{\mathbb{Q}}$.

- The delta of the call option is $\Delta = \frac{\partial f_0}{\partial S_0}$.

Black-Scholes-Merton: Pricing via Replication Argument (cont'd)

Why does the recipe work?

- The risk-free rate is r .
- The expected rate of return on S_t under $\mathbb{Q}$ is r .
- The $\mathbb{Q}$ -expected rate of return on any portfolio is r as well.
- As this also holds for our replication process: $E^{\mathbb{Q}}[w_T] = w_0 e^{rT}$.
- Recalling the definition of a replication process:

$$f_0 = w_0 = e^{-rT} E^{\mathbb{Q}}[w_T] = e^{-rT} E^{\mathbb{Q}}[f_T],$$

as stated above.

Risk-neutral pricing and BSM

So under $\mathbb{Q}$, the distribution of $S(t)$ is

$$\log(S(t)) \stackrel{\mathbb{Q}}{\sim} N\left(\log(S_0) + \left[r - \frac{1}{2}\sigma^2\right]t, \sigma^2 t\right)$$

Can we obtain the BSM formula from this? Of course!

$$\mathbb{E}_0^{\mathbb{Q}}[C_T] = \int_0^\infty \max\{s - K, 0\} \frac{1}{s} \frac{1}{\sqrt{2\pi\sigma^2 T}} e^{-\frac{1}{2} \frac{(\log(s) - \log(S_0) - r - \frac{1}{2}\sigma^2 T)^2}{\sigma^2 T}} ds$$

Do some brute-force calculations, multiply by e^{-rT} and voila:

$$c_0 = S_0 N(d_1) - Ke^{-rT} N(d_2)$$

Feynman-Kac Theorem: connection between expectations of Itô processes and partial differential equations, which explains why seemingly different approaches give same result.

Black-Scholes-Merton Option Pricing Formula

Proposition

The price of a **European call option** is

$$c(S_0, r, \sigma, K, T) = S_0 N(d_1) - Ke^{-rT} N(d_2),$$

with

$$N(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = \text{cdf of a standard normal distribution}$$
$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{\sigma^2}{2}\right) T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

The price of the corresponding **put option** is

$$p(S_0, r, \sigma, K, T) = Ke^{-rT} N(-d_2) - S_0 N(-d_1).$$

Proof: Calculate $e^{-rT} \hat{\mathbb{E}} \left[[S_T - K]^+ \right]$. $\square$

Practice: Forward Pricing with BSM RN Valuation

- If I go long one forward contract at $t = 0$, at time $t = T$ I will receive $S_T - F_{0,T}$.
- Pricing this contract yields

$$0 = e^{-rT} \mathbb{E}^{\mathbb{Q}} [S_T - F_{0,T}] = e^{-rT} \left(\mathbb{E}^{\mathbb{Q}} [S_T] - F_{0,T} \right).$$

- The $\mathbb{Q}$ -dynamics of the stock are

$$\frac{dS_t}{S_t} = rdt + \sigma dW_t \Leftrightarrow S_T = S_0 e^{(r - \frac{1}{2}\sigma^2)T + \sigma W_T}$$

- If $X \sim \mathcal{N}(a, b^2)$, then $\mathbb{E}^{\mathbb{Q}} [e^X] = e^{a + \frac{b^2}{2}}$. As a result,

$$\mathbb{E} [S_T] = \mathbb{E}^{\mathbb{Q}} \left[S_0 e^{(r - \frac{1}{2}\sigma^2)T + \sigma \sqrt{T}W} \right] = S_0 e^{rT}$$

because $W \sim \mathcal{N}(0, 1)$.

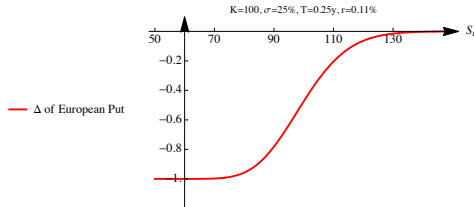
- We get the cost-of-carry relation

$$F_{0,T} = \mathbb{E}^{\mathbb{Q}} [S_T] = S_0 e^{rT}$$

Portfolio Insurance

Portfolio managers may want to acquire a put option on their portfolio: insurance against downside but keeps the upside. Could be expensive or unavailable—[replicate](#) it.

- Trade the assets on the side, or use index futures.



Buy when prices increase, sell when prices fall → trading [contributes to volatility](#), especially to downward movements.

This [portfolio insurance](#) strategy is sometimes blamed for the 1987 crash: the Dow Jones lost over 20% on October 19, 1987.

Pricing by Monte Carlo simulation

In general the risk-neutral pricing may be difficult, since for a general payoff function $g(x)$ we have to calculate

$$\mathbb{E}^{\mathbb{Q}}[g(S(T))] = \int g(x) \frac{1}{s} \frac{1}{\sqrt{2\pi\sigma^2(T-t)}} e^{-\frac{1}{2} \frac{(\log(s) - \log(S(t)) - r - \frac{1}{2}\sigma^2(T-t))^2}{\sigma^2(T-t)}} ds,$$

and this integral may not have an explicit solution.

For *solving the BSM equation*, we could apply finite-difference methods by *approximating the derivatives*.

Or we can use *Monte Carlo simulations* that use the representation of prices as expectations. It relies on the *law of large numbers*.

Pricing by Monte Carlo simulation (cont'd)

$$\mathbb{E}^{\mathbb{Q}}[g(S(T))] \approx \frac{1}{N} \sum_{k=1}^N g(\hat{S}_k(T))$$

where $\hat{S}_k(T)$ is the k 'th simulated value of $S(T)$ and N is large.

Naturally it requires that we are actually able to simulate $S(T)$ —see the discrete time lecture with normal shocks:

$$S(t + \Delta t) \approx S(t) + r\Delta t + \sigma\sqrt{\Delta t}W$$

where $W \sim N(0, 1)$.

- **Pros:** Often easy to implement, is asymptotically correct, allows for pricing *path-dependent* derivatives.
- **Cons:** Convergence is slow (requires very large N), mainly useful for European-type derivatives.

An exotic example

Consider a so-called European *lookback option* with strike K , which has payoff

$$c_T^{\text{LB}} = \max\left\{\sup_{0 \leq t \leq T} S(t) - K, 0\right\}$$

So the payoff depends on the entire path of the price process until time T , since we may “look back” in time to the highest value we have observed.

This may not be very easy to plug into our differential equation/risk-neutral expectation and solve $\rightarrow$ Monte Carlo simulation to the rescue:

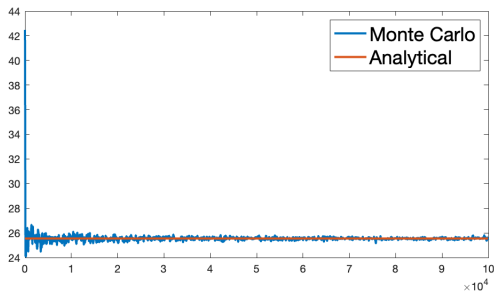
- Choose small increment Δt .
- Put $S^{\max} = S(0)$.
- Simulate $S(\Delta t)$. If $S(\Delta t) > S^{\max}$, set $S^{\max} = S(\Delta t)$, otherwise do nothing.
- Repeat until time T .
- Repeat all of the above many times and take the average payoff.

An exotic example (cont'd)

In the case of a European lookback option, we actually do have an explicit solution:

$$c_0^{\text{LB}} = c_0 + S_0 e^{-rT} \frac{\sigma^2}{2r} \left[e^{-rT} \Phi(d_1) - \left(\frac{S_0}{K} \right)^{-\frac{2r}{\sigma^2}} \Phi \left(d_1 - \frac{2r\sqrt{T}}{\sigma} \right) \right]$$

But then we can compare the Monte Carlo price to the real theoretical price:



BSM with dividends

Need adjustment—we may do it in two ways:

- Continuously paid out dividends at a constant rate ζ .
- A discrete lump-sum dividend at time τ of size D_τ .

With continuous dividends, one can show that the European call option formula becomes

$$c_t = S(t)e^{-\zeta(T-t)}\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2)$$
$$d_1 = \frac{\log\left(\frac{S(t)}{K}\right) + \left(r - \zeta + \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t}$$

With a one time dividend D_τ , we simply replace $S(t)$ in the usual formula with

$$S(t) - e^{-r(\tau-t)}D_\tau$$

The BSM model: Shortcomings

The BSM model is extremely important in all continuous-time finance.

The BSM model is both arbitrage-free and complete.

Does that mean we have solved all of finance? Sadly not.

Consider this: In the market we observe put options on the S&P500 index at closing time with maturity the *very next morning*, that are more than 5% out of the money.

These put options still trade at a positive price, meaning that investors believe that the price of this large index may drop by a large amount overnight. This is equivalent to a log-return of -1.26 .

With an estimated variance of log-returns of ca. $\sigma = 0.2$ and mean $\mu = 0.1$, the probability of ever exercising these puts is basically zero. (Too light tails.)

The BSM model: Shortcomings (cont'd)

The BSM model is not able to capture some of the stylized facts, such as heavy tails or time-varying volatility.

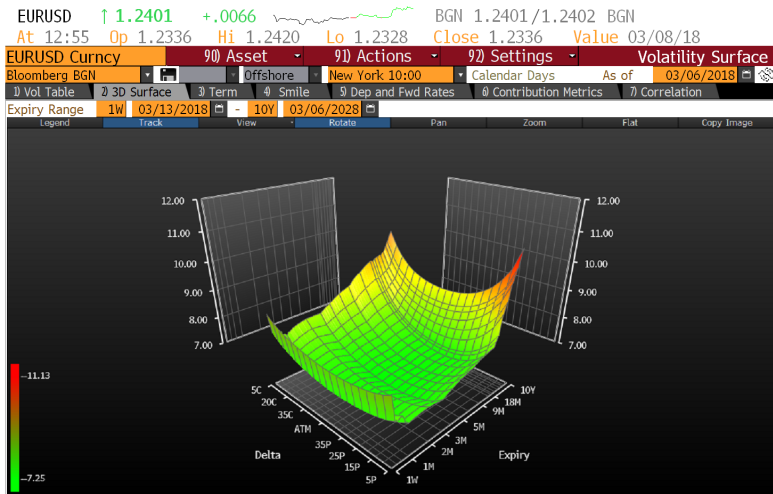
In fact, much of modern financial research is exactly on the volatility of σ and how this varies over time $\rightarrow$ topics such as local volatility models, stochastic volatility models and jump models.

A very popular stochastic volatility model is the Heston (1993) model where the volatility itself is a stochastic process, σ_t .

Alternatively, Merton (1976) suggested using jumps in S_t .

The literature on modelling financial assets and derivatives is vast, but the BSM model still serves an important role as a benchmark and for calculating implied volatilities.

The BSM model: Shortcomings (cont'd)



This is FX volatility surface by tenor and delta which is EUR Call/Put forward delta without premium adjustment.

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Investment and Finance

Stochastic Interest Rates

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Outline of this topic

This week we will cover:

- Stochastic interest rates and discount factors.
- The binomial model.
 - Pricing zero-coupon bonds: Backwards vs forwards.
- Callable bonds.
- Continuous time models.
 - Ho-Lee, Black-Derman-Toy, Vasicek, Cox-Ingersoll-Ross.
 - Zero coupon bond pricing in continuous time.

The basics of fixed income

Securities

- Securities promising **fixed, predetermined payments** at given points in time.
 - * Deposit accounts
 - * Bonds
- Securities whose payoffs depend on interest rates or the price of basic fixed income securities
 - * Bond futures, forwards, and options
 - * Caps, floors
 - * Swaps
 - * Swaptions

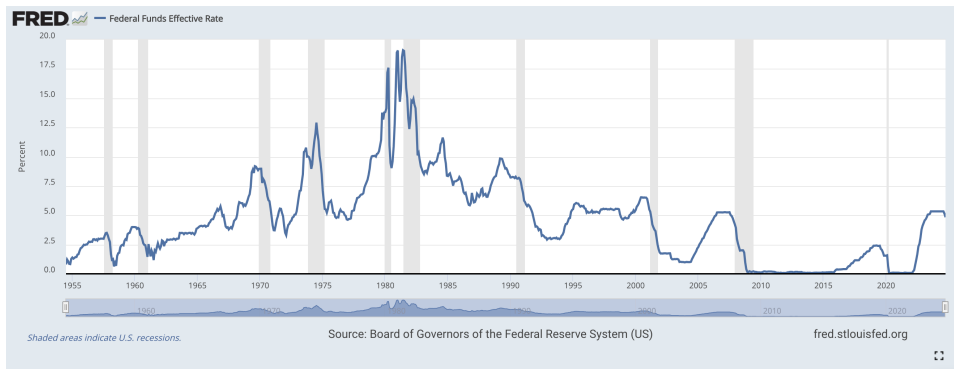
Risks: **Interest rates are stochastic**

- At any time for lots of maturities there exists a risk-free investment opportunity, but future rates are unknown.
- We assume no credit risk for now.

The basics of fixed income (cont'd)

So far interest rates have been deterministic.

Actual interest rates, e.g. the Federal funds rate, behave much more wildly:



The basics of fixed income (cont'd)

Corporations issue bonds much more often than equity.

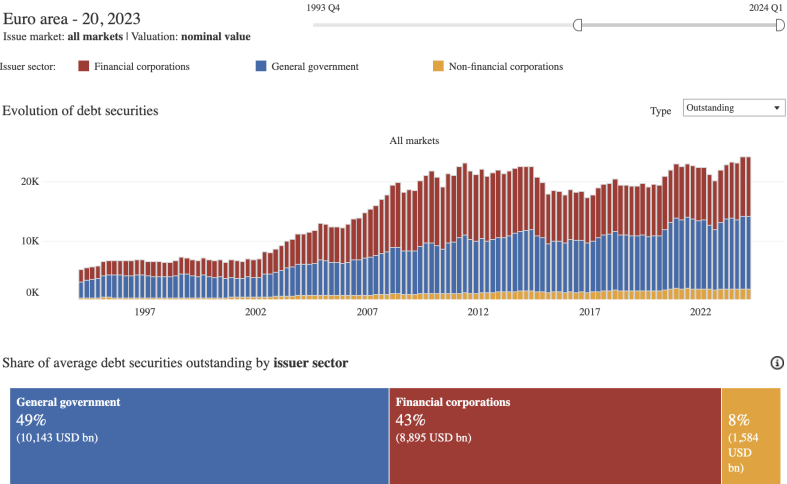
Financial institutions are mostly invested in fixed-income markets.

- Matching assets and liabilities
- Regulatory constraints

Fixed-income desks often much larger than equity desks.

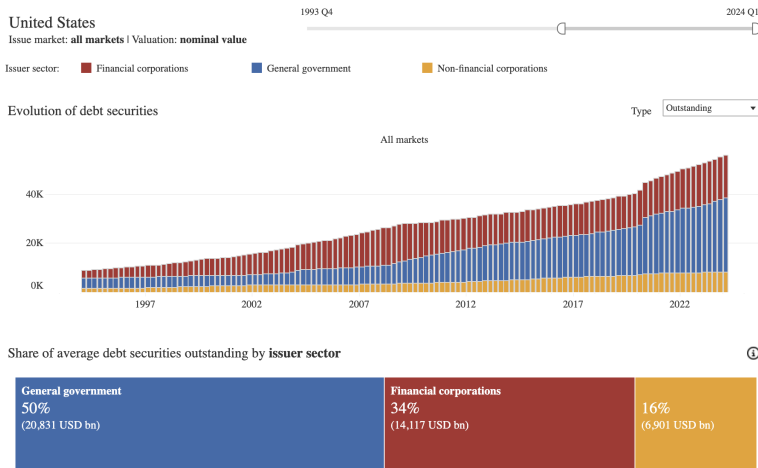
Also higher demand for fixed-income derivatives.

Total debt securities



Source: www.bis.org

Total debt securities



Source: www.bis.org

Fixed income derivatives on exchanges

Exchange-traded futures and options, by location of exchange

Notional principal, in billions of US dollars

2024-06		 View: Value		 Level: Level 4							
	Open interest			Daily average turnover							
	2023-Q4	2024-Q1	2024-Q2	2022	2023	2024-02	2024-03	2024-04	2024-05	2024-06	
▼ Futures											
▼ All markets	37,343	39,354	37,320	8,045	9,248	10,331	9,726	9,838	8,759	9,294	
➤ Interest rate	37,002	38,992	36,947	7,892	9,107	10,200	9,566	9,683	8,622	9,110	
Foreign exchange	342	362	373	153	140	131	160	155	137	184	
➤ North America	26,680	28,833	26,841	5,898	6,787	7,295	6,322	6,463	5,562	5,841	
➤ Europe	7,861	7,843	7,724	1,848	2,116	2,714	3,033	2,933	2,798	2,995	
➤ Asia and Pacific	1,610	1,502	1,622	198	239	230	300	286	285	320	
➤ Other Markets	1,192	1,176	1,133	101	106	91	72	157	114	138	
▼ Options											
▼ All markets	49,836	54,078	50,718	1,805	2,237	2,894	2,153	2,116	1,811	1,596	
➤ Interest rate	49,723	53,969	50,596	1,786	2,216	2,878	2,132	2,109	1,806	1,588	
Foreign exchange	113	109	121	19	21	16	21	7	6	8	
➤ North America	34,871	35,202	33,506	1,316	1,813	2,207	1,521	1,565	1,214	1,132	
➤ Europe	13,504	17,438	15,673	466	385	634	571	504	513	450	
➤ Asia and Pacific	13	12	1	15	15	12	15	1	0	0	
➤ Other Markets	1,448	1,427	1,538	8	23	41	46	46	84	14	

Source: www.bis.org

Fixed income derivatives OTC

Global OTC derivatives market

In billions of US dollars

2024-S1	View: Value				Level: Level 4			
	Notional amounts outstanding				Gross market value			
	2022-S2	2023-S1	2023-S2	2024-S1	2022-S2	2023-S1	2023-S2	2024-S1
All contracts	617,991	712,881	667,058	729,838	20,754	19,835	18,122	17,052
Foreign exchange contracts	107,579	118,467	118,004	129,885	4,846	4,310	4,197	3,668
By instrument								
Outright forwards and forex swaps	62,849	69,701	67,797	74,538	2,560	2,325	2,196	1,876
Currency swaps	31,802	34,541	36,184	38,005	2,016	1,718	1,767	1,526
Options	12,906	14,200	13,999	17,317	270	268	234	266
Other products	22	25	24	24				
Interest rate contracts	490,634	573,587	529,813	578,805	14,635	14,389	12,783	11,985
By instrument								
FRAs	44,978	61,779	56,023	62,804	465	404	403	328
Swaps	405,546	465,814	425,277	469,173	13,249	13,163	11,682	10,996
Options	39,887	45,758	48,288	46,602	920	822	698	661
Other products	223	236	224	226				
Equity-linked contracts	6,919	7,837	7,783	8,686	504	570	582	673
By instrument								
Forwards and swaps	3,469	3,688	3,830	4,335	277	265	258	261
Options	3,449	4,149	3,954	4,351	227	305	324	412

Source: www.bis.org

Stochastic interest rates (cont'd)

Suppose the interest rate $r(t)$ is deterministic, but non-constant and we hold an amount B_t in the bank at time t . At time $t + \Delta t$ we then have

$$B_{t+\Delta t} \approx B_t + r(t) \cdot \Delta t \cdot B_t$$

In the limit as $\Delta t \rightarrow 0$ this becomes exact, and we get the following

$$\frac{\partial B_t}{\partial t} = r(t)B_t \iff B_t = e^{\int_0^t r(s) ds} B_0$$

where B_0 is the initial level of wealth in the bank.

Written in our stochastic differential equation notation, we have

$$dB_t = r(t)B_t dt$$

Stochastic interest rates (cont'd)

If the interest rate process $r(t)$ is stochastic, by time t we end up with

$$B_t = e^{\int_0^t r(s) \, ds} B_0,$$

the realized discount factor is given by

$$d_{t,T} = e^{-\int_t^T r(s) \, ds},$$

and the *zero-coupon bond price* at time t for a zero coupon bond with maturity T is

$$B_{t,T} = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T r(s) \, ds} \right].$$

See special cases when $r(t)$ is deterministic or constant.

Question: How do we model interest rates consistent with no-arbitrage?

Which measure to choose?

We will always assume that our interest rate model is formulated directly under the risk-neutral measure, so no p 's, only q 's.

When pricing options, we derived the q 's from a replication argument; markets were complete, so one set of probabilities solving our problem $\implies$ we knew $\mathbb{Q}$.

Here we cannot buy an interest rate to replicate with—underlying is not traded, so we specify the model directly with the risk-neutral q probabilities.

Existence of the q 's means no arbitrage, but we still have to take care when specifying the model—cannot pick q 's arbitrarily.

A first idea...

Let us start out in discrete time, assuming three periods, $t \in \{0, 1, 2\}$. Assume that the term structure of interest rates is constant, but may be subject to parallel shifts.

The initial term structure is simply

$$s_{0,t} = r \in [0, \infty)$$

The price of a two-year zero coupon bond is

$$B_{0,2} = \frac{1}{(1+r)^2}$$

Now assume a parallel shift happens at time $t = 1$, so that $s_{1,t} = \tilde{r}$. Assume also that the size of the shift is unknown at time $t = 0$, but

$$\mathbb{E}^{\mathbb{Q}}[\tilde{r}] = r$$

A first idea... (cont'd)

Assuming such a shift happens, could we have

$$B_{0,2} = \frac{1}{(1+r)} \mathbb{E}^{\mathbb{Q}} \left[\frac{1}{1+\tilde{r}} \right] ? \implies \frac{1}{(1+r)} = \mathbb{E}^{\mathbb{Q}} \left[\frac{1}{1+\tilde{r}} \right]$$

But $x \mapsto \frac{1}{1+x}$ is a strictly convex function, so by Jensen's inequality, such an equality does not hold.

$\implies$ We have an arbitrage opportunity, showcasing that we cannot specify the model and the risk-neutral probabilities completely arbitrarily.

We will use the stochastic processes introduced earlier for modelling short-term rates.

The binomial model

Let us assume that the spot rate is governed by a binomial lattice model. I.e., we have

- A time horizon: $[0, T]$.
- N number of partitions of $[0, T]$.
- $\Delta t = \frac{T}{N}$. The time-step size.
- $s_{t,t+1}(s)$: The spot rate between time t and $t + 1$ in state s .

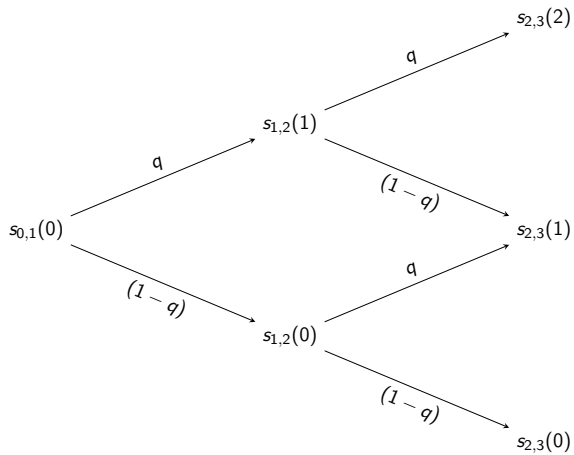
Note that $s_{t,t+1}$ is the spot rate that we use for discounting from period t to $t + 1$.

Thus there are $N + 2$ periods in a N -period model of the stochastic interest rate: $[0, 1, \dots, N, N + 1]$.

The states s are the nodes in the binomial lattice.

We will often assume that $q = \frac{1}{2}$, since we are allowed to choose it freely...

The model, visually



The binomial model: Zero-coupon bond prices

Price financial instruments exactly as we did options: work backwards in the binomial lattice and apply risk-neutral pricing.

Definition (Zero-coupon bond prices)

In the binomial model of stochastic interest rates, we denote by $B_{t,T}^s$ the zero coupon bond price at time t in state s of a zero coupon bond with maturity T .

A zero coupon bond with maturity $T_i \in (0, T]$ always has

$$B_{T_i, T_i}^s = 1$$

Working backwards we then have:

$$B_{t-1, T_i}^s = \frac{1}{1 + s_{t-1, t}} \mathbb{E}^{\mathbb{Q}} [B_{t, T_i}] = \frac{1}{1 + s_{t-1, t}} \left[q B_{t, T_i}^{s+1} + (1 - q) B_{t, T_i}^s \right]$$

Repeat all the way back to $t = 1, \dots, T_i$ and $s = 0, \dots, t - 1$.

The binomial model: General pricing

Having the zero coupon bond prices, we can back out the term structure of interest rates.

$$s_{0,t} = B_{0,t}^{-\frac{1}{t}} - 1$$

Just like earlier, we may calibrate the model to match the observed term structure in the market.

For a general financial instrument (maybe a derivative), simply find the value today V_0 by going through

$$V_{t-1}^s = \frac{1}{1 + s_{t-1,t}(s)} [qV_t^{s+1} + (1 - q)V_t^s]$$

where $t = 1, \dots, T$, $s = 0, \dots, t - 1$, and V_t^s is the time- t value of the instrument in state s .

The binomial model: Example

Suppose the current one-year short rate is $s_{0,1} = 5\%$ with continuous compounding, and this either increases to $s_{1,2} = 6\%$ or decreases to $s_{1,2} = 4\%$ with identical risk-neutral probabilities.

- What is the current price of a one-year ZCB?
- What will be the price of a one-year ZCB one year from now?
- What is the current price of a 2-year ZCB?
- Calculate the current term structure of yields!
- What is the current price of a call option written on a 2-year ZCB with strike price 95 and maturity 1 year?

The binomial model

Is the binomial model then free of arbitrage? Yes!

In order to have arbitrage, we need

$$V_{t-1}^s \leq 0, \quad V_t^{s+1} \geq 0, \quad V_t^s \geq 0$$

with at least one strict inequality. But

$$V_t^{s+1} \geq 0, \quad V_t^s \geq 0 \implies V_{t-1}^s \geq 0$$

as V_{t-1}^s is a convex combination of the two non-negative quantities.

The same logic applies if $V_t^s > 0$ or $V_t^{s+1} > 0$.

The binomial model: The forward equation

Working backwards through the model is tedious. For zero coupon bonds we are able to obtain a forward equation where we calculate prices iteratively, starting from time 0.

We can do this by means of *state prices*. Recall that $\psi_{t,s}$ is the price of receiving a payoff of 1 at time t in state s .

Obviously at time $t = 0$ we have

$$\psi_{1,0} = \frac{1 - q}{1 + s_{0,1}}, \quad \psi_{1,1} = \frac{q}{1 + s_{0,1}}$$

Standing at time $t = 1$ we can then simply deduce the state price $\psi_{t,s}$ by considering the probability of ending in that state and discount appropriately.

Illustration on blackboard

The binomial model: The forward equation (cont'd)

Theorem (Forward equation in binomial model)

In a binomial model, the state price $\psi_{t,s}$ can be calculated according to

$$\psi_{t,s} = \begin{cases} \frac{(1-q)}{1+s_{t-1,t}(s)} \psi_{t-1,s}, & \text{if } s = 0 \\ \frac{(1-q)}{1+s_{t-1,t}(s)} \psi_{t-1,s} + \frac{q}{1+s_{t-1,t}(s-1)} \psi_{t-1,s-1}, & \text{if } 0 < s < t \\ \frac{q}{1+s_{t-1,t}(s-1)} \psi_{t-1,s-1}, & \text{if } s = t \end{cases}$$

Proof on board

Zero-coupon bond prices can then be found as

$$B_{0,T} = \sum_{s=0}^T \psi_{T,s}$$

Callable/redeemable bonds

In Denmark it is very common to have mortgage loans with an option to redeem the loan at any point in time.

That means the loan-taker can always choose to pay out the remainder of the nominal loan amount prior to maturity.

This may be a good strategy if the interest rate has just fallen.

From investor's perspective, a callable bond is the sum of a non-callable bond and a call option with strike 100:

$$\text{Price}_{\text{callable}} = \text{Price}_{\text{non-callable}} - \text{Price}_{\text{option}}$$

The price of a callable bond is always lower than that of a non-callable bond. The added option to the bond adds value to the loan-taker.

Callable bond pricing

Example: Consider a 3-year callable bond in the form of an annuity. Assume the coupon rate is 3%. The cash flow from the perspective of the loan-taker is then

$$(1, -0.3535, -0.3535, -0.3535)$$

for each unit of currency loaned.

The remaining debt to be paid is

$$R = (1, 0.6765, 0.3432, 0)$$

The optimal strategy is then to call the bond if the remaining debt becomes less than the present value of the future payments.

So in each node we substitute the value V_t^s of the non-callable bond by $\min\{R_t, V_t^s\}$.

The price of the non-callable bond is calculated as usual.

Immunization

Interest rate derivatives are primarily used to hedge risk related to changes in interest rates.

When the interest rate is described by a binomial lattice, the process of immunization is analogous to how we hedged (replicated) using the underlying when pricing options.

I.e., create a dynamic portfolio with value process that replicates the payment stream we want to immunize.

To do this in practice, we would have to calibrate the binomial model to data, such that it matches the observed term structure of interest rates.

Such a calibration is usually not as easy as when calibrating to asset prices. The model lacks structure.

More flexibility: The Ho-Lee model

Ho and Lee (1986): the short rate as a one-factor model:

$$s_{t,t+1}(s) = a_t + b_t s$$

where b_t is the time-dependent factor loading. The factor itself is the state, s .

Spot rate = fixed component a_t plus a linear contribution from the previous spot rate with factor b_t governing the impact of the state.

Standard Ho-Lee: b_t is not time dependent, i.e., $b_t = b$.

Choose b from the variation in the observed spot rate, since $\mathbb{V}[s_{t,t+1}] = \left(\frac{b}{2}\right)^2$.

Given b , we then solve T equations in T unknowns to find the a_t 's.

Black-Derman-Toy

Black, Derman and Toy (1990): multiplicative model of the form

$$s_{t,t+1}(s) = a_t e^{b_t s}$$

Corresponding to the Ho-Lee model of $\log(s_{t,t+1}(s))$.

Again we may choose $b_t = b$ and calibrate the model to observed term structure.

Can also let b_t be time-varying, but then more equations to perform the calibration.

In that case match observed standard deviations to model-implied standard deviation as well.

Continuous-time models

Again, interested in modelling the interest rate continuously through time, such that we have a model for the spot rate at all points $t \in [0, T]$.

Model the instantaneous short rate r_t using an Itô process:

$$dr_t = a(r_t, t)dt + b(r_t, t)dW_t^{\mathbb{Q}}$$

where $(W_t^{\mathbb{Q}})_{t \in [0, \infty)}$ is a Brownian motion under the risk-neutral measure $\mathbb{Q}$.

Note: model formulated directly under the risk-neutral measure.

The most popular processes used for this are the affine diffusions, where

$$a(r_t, t) = \zeta_0 - \zeta_1 r_t, \quad b(r_t, t) = \sqrt{\gamma_0 + \gamma_1 r_t}$$

Continuous-time Ho-Lee

Taking the limit $\Delta t \rightarrow 0$ in the Ho-Lee model yields a process of the form

$$dr_t = \theta(t)dt + \sigma dW_t^{\mathbb{Q}}$$

where $\sigma > 0$ is the volatility, which plays the role of b from before.

Under this model the future short rate is normally distributed with

$$\mathbb{E}^{\mathbb{Q}}[r_T] = r_0 + \int_0^T \theta(s) ds, \quad \mathbb{V}^{\mathbb{Q}}[r_T] = \sigma^2(T - t)$$

$\theta(t)$ is the time-dependent trend and σ the standard deviation per unit time.

Note: negative interest rates will happen with probability 1 due to normal distribution.

The Vasicek model

Vasicek (1977): short rate follows a so-called Gaussian Ornstein-Uhlenbeck process:

$$dr_t = \kappa(\theta - r_t)dt + \sigma dW_t^{\mathbb{Q}}$$

where $\kappa, \theta > 0$.

The drift $\kappa(\theta - r_t)dt$ depends on the current level of r_t .

- If $r_t > \theta$, the drift is negative.
- If $r_t < \theta$, the drift is positive.

The process is pushed up or down around θ , depending on the current interest rate
 $\implies$ *mean-reversion*.

The Vasicek model (cont'd)

The stochastic differential equation in the Vasicek model can be solved by applying Itô's lemma to the process

$$Y_t = f(r_t, t) = e^{\kappa t} r_t$$

Doing some manipulation (on blackboard) we get

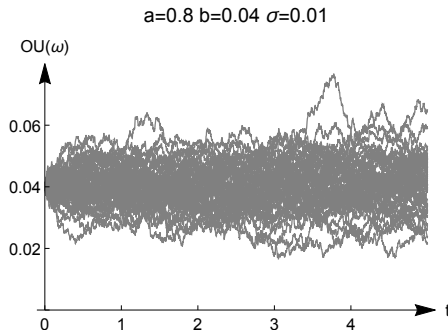
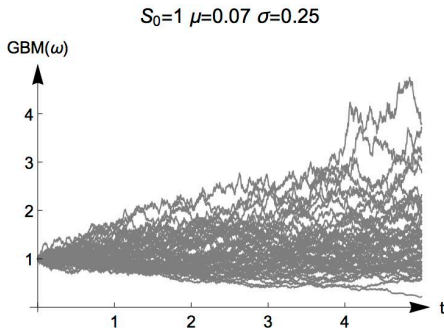
$$r_T = \theta + (r_t - \theta)e^{-\kappa(T-t)} + \sigma \int_t^T e^{-\kappa(T-s)} dW_s^{\mathbb{Q}}$$

Once again we end up with a normally distributed short rate, but with mean and variance that depends on the current level of r_t :

$$\mathbb{E}[r_T | r_t] = \theta + (r_t - \theta)e^{-\kappa(T-t)}, \quad \mathbb{V}^{\mathbb{Q}}[r_T | r_t] = \frac{\sigma^2}{2\kappa} \left(1 - e^{-2\kappa(T-t)}\right)$$

What happens when $T \rightarrow \infty$?

Vasicek model vs BSM model



Note: $\lim_{t \rightarrow \infty} \sqrt{\text{Var}[S_t]} = \infty$ vs $\lim_{t \rightarrow \infty} \sqrt{\text{Var}[r_t]} = \frac{0.01}{\sqrt{2 \times 0.8}} \approx 0.008$, constant.

The Cox-Ingersoll-Ross model (CIR)

The Vasicek model still allows for negative interest rates, although only for brief periods of time due to mean reversion.

Cox, Ingersoll and Ross (1985): to avoid negative rates, model of the short rate

$$dr_t = \kappa(\theta - r_t)dt + \sigma\sqrt{r_t}dW_t^{\mathbb{Q}}$$

where $\kappa, \theta, \sigma > 0$. Also has mean-reversion, but we see that the noise coming from the Brownian motion is scaled according to the level of r_t :

If the interest rate is low, volatility is low and vice versa.

When $r_t = 0$, the volatility term completely disappears, and the process is pushed away from 0 and towards θ .

The Cox-Ingersoll-Ross model (CIR) (cont'd)

One can show that if $r_t > 0$ and $2\kappa\theta > \sigma^2$, then the process is strictly positive for all $T > t$. (The *Feller condition*)

When calibrating a CIR model to data, the Feller condition is, however, often not satisfied.

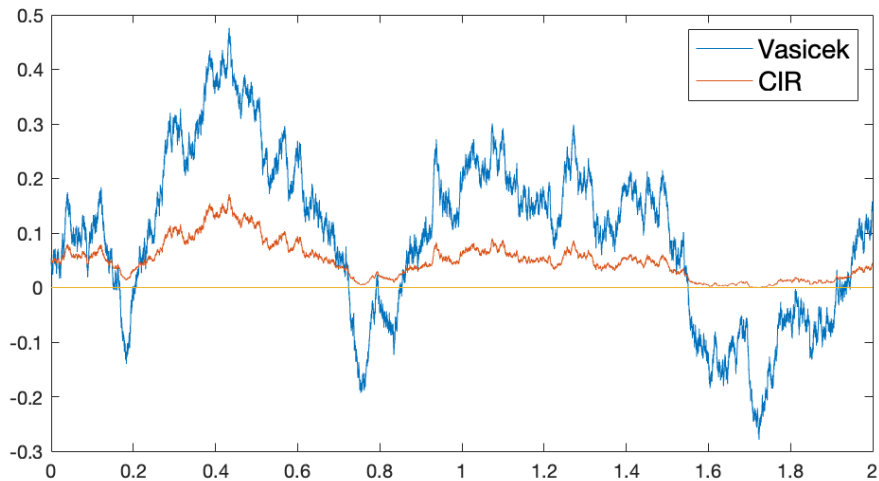
The distribution of the short rate r_T is no longer normal, but a non-central χ^2 .

In the long-run ($T \rightarrow \infty$) we have

$$r_T \sim \text{Gamma}(\alpha, \beta), \quad \alpha = \frac{2\kappa\theta}{\sigma^2}, \quad \beta = \frac{2\kappa}{\sigma^2}$$

The CIR model is also used extensively in finance to model stochastic volatility. This model was proposed by Heston (1993).

Vasicek vs CIR process



Interest rate derivatives

When pricing derivatives, we know that in absence of arbitrage the Black-Scholes equation should be satisfied.

Assume that the short rate follows an Itô process

$$dr_t = a(r_t, t)dt + b(r_t, t)dW_t^{\mathbb{Q}}$$

Given an interest rate derivative on the short rate r_t with price process $f(r_t, t)$, the Black-Scholes equation reads

$$r_t f(r_t, t) = \frac{\partial f(r_t, t)}{\partial t} + a(r_t, t) \frac{\partial f(r_t, t)}{\partial r_t} + \frac{1}{2} b(r_t, t)^2 \frac{\partial^2 f(r_t, t)}{\partial r_t^2}$$

Given the boundary condition at the endpoint, $F(r_T, T)$, we solve the equation backwards in time.

Zero-coupon bond prices in continuous time

The most fundamental interest rate security is the zero-coupon bond. Let $B(r_t, t, T)$ denote the time t price of a zero-coupon bond with maturity T , given the short rate r_t .

If the term structure of interest rates is flat, i.e. $r_t = r$ for all $t \geq 0$, then

$$dr_t = 0$$

For ease of notation, put

$$f(r_t, t) = B(r_t, t, T) = B(t, T)$$

Then

$$\frac{dB(t, T)}{dt} = rB(t, T) \iff d \log(B(t, T)) = r dt$$

Zero-coupon bond prices in continuous time (cont'd)

This differential equation has the solution

$$\log(B(t, T)) = rt + C$$

where C is some constant.

The boundary condition $B(T, T) = 1$ gives us the C as

$$\log(B(T, T)) = rT + C \implies C = -rT$$

Therefore the price of the zero coupon bond is

$$B(t, T) = e^{-r(T-t)}$$

Zero-coupon bond prices in continuous time (cont'd)

If interest rate dynamics are more complex, solving the BSM equation becomes more complicated.

Assume that interest rates follow the special case of the Ho-Lee model:

$$dr_t = \mu dt + \sigma dW_t^{\mathbb{Q}}$$

We can guess that the pricing formula is

$$B(r_t, t, T) = A(t, T)e^{-r_t(T-t)}$$

We may easily check when this satisfies the Black-Scholes equation and obtain an expression for $A(t, T)$:

$$A(t, T) = e^{-\frac{1}{2}\mu(T-t)^2 + \frac{1}{6}\sigma^2(T-t)^3}$$

Affine models

Affine diffusion models are quite general models of the interest rate, popular due to obtainable ZCB prices. We have

$$B(r_t, t, T) = e^{-a(T-t) - b(T-t)r_t}$$

where $a(\tau), b(\tau)$ are functions such that $a(0) = b(0) = 0$ and satisfying the following ODEs

$$\begin{aligned}\frac{1}{2}\gamma_1 b(\tau)^2 + \zeta_1 b(\tau) + b'(\tau) &= 1 \\ \frac{1}{2}\gamma_0 b(\tau)^2 - \zeta_0 b(\tau) + a'(\tau) &= 0\end{aligned}$$

where $\tau = T - t > 0$.

In some cases, we can solve these equations analytically, other times only numerically.

These type of affine processes pop up everywhere in finance.

Vasicek model: Pricing securities

Either use risk-neutral pricing,

$$f_t = E_t^{\mathbb{Q}} \left[e^{-\int_t^T r_u du} f_T \right]$$

Alternatively, a hedging argument (combined with Itô's lemma) shows how the equation

$$r f(t, r) = f_t(t, r) + f_r(t, r)\kappa(\theta - r) + \frac{1}{2} f_{rr}(t, r)\sigma^2$$

must be solved by the price of any security.

- Analogous to solving the BSM equation for derivatives on a stock.

Vasicek model: Pricing securities (cont'd)

Any of the two approaches yields:

Proposition

When $\kappa \neq 0$, bond prices in the Vasicek model satisfy

$$P(t, T) = e^{-a(t, T) - b(t, T)r_t}$$

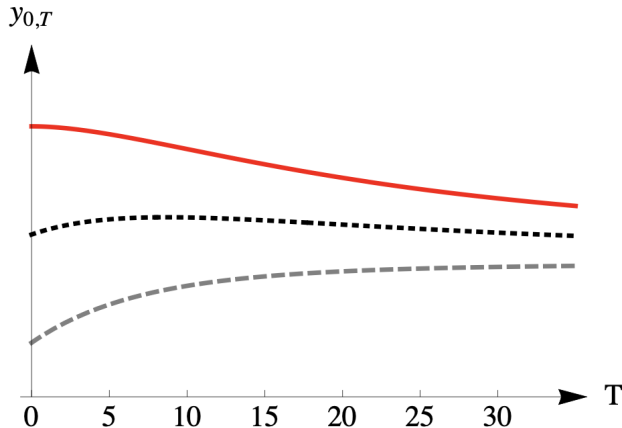
with

$$b(t, T) = \frac{1 - e^{-\kappa(T-t)}}{\kappa} \text{ and}$$

$$a(t, T) = - \left(\theta - \frac{\sigma^2}{2\kappa^2} \right) (b(t, T) - (T - t)) + \frac{\sigma^2}{4\kappa} b(t, T)^2$$

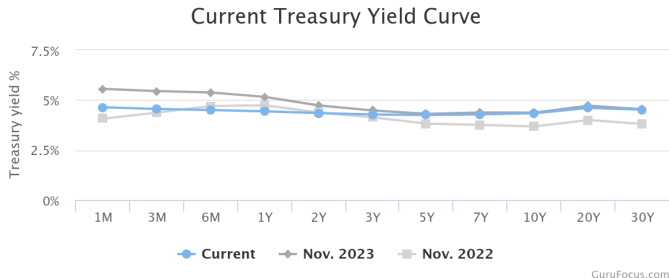
Vasicek model: Pricing securities (cont'd)

Yield curve can be increasing (when $\theta - \frac{3}{4} \frac{\sigma^2}{\kappa} > r_0$), decreasing ($r_0 > \theta$), or hump-shaped (in between):



Vasicek model: Pricing securities (cont'd)

But not every yield curve is possible:



An interest rate model should match the observed bond prices.

→ e.g., the Hull-White model allows for drift $\kappa(\theta(t) - r_t)$ to match data.

Summary

First steps towards modelling interest rates as stochastic processes, using the binomial model or diffusions based on Brownian motion.

We have focused on computing ZCB prices, the most fundamental interest rate-related financial instrument.

In the binomial model this is straightforward to do either recursively or iteratively.

In the continuous time models, things are slightly more complicated, but “nice” formulas are available.

We have also introduced callable bonds and discussed their pricing—one example are Danish mortgage loans that are callable at par value.

Investment and Finance

Review

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Review of the 2nd half of the course

- SDF, state prices, and risk-neutral pricing.
- Derivatives: forwards, futures, and swaps; pricing.
- Derivatives: options.
- Stochastic processes and Itô's Lemma.
- The binomial lattice model for derivative pricing.
- The Black Scholes model, formula and the BS SDE.
- Interest rate models and prices of interest rate derivatives and bonds.

General NA pricing

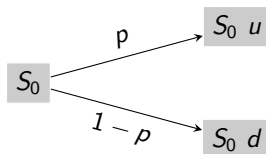
- Recall: arbitrage, law of one price.
- Utility functions:
 - Risk aversion
 - investors maximize their expected utility $\max \mathbb{E}[u(X)]$
 - Euler equation: $P = \mathbb{E}[\delta \frac{u'(c_1)}{u'(c_0)} X] = \mathbb{E}[\Lambda X]$, SDF.
- State pricing:
 - FTAP: no arbitrage $\equiv$ exists a $\psi \in \mathbb{R}_+$ that solves $P = D\psi$
 - Market completeness
 - Swaptions
- Risk-neutral pricing:
 - Changing measure from $\mathbb{P}$ to $\mathbb{Q}$
 - FTAP rewrite: no arbitrage $\equiv$ exists probability measure $\mathbb{Q} = (q_1, \dots, q_S)$ such that

$$P_i = \frac{1}{1 + r_f} \mathbb{E}^{\mathbb{Q}} [d^i]$$

Basic derivatives: Model-free discussion

- Forwards.
 - Setup, OTC, payoff, pricing.
 - $F_0 = \frac{S_0}{d_{0,T}}$, proof by NA—replication or hedging.
 - Cost of carry, convenience yield.
- Futures.
 - Pricing and relation to forwards.
 - $G_{0,T} = \mathbb{E}^{\mathbb{Q}}[S(T)]$ while $F_{0,T} = \mathbb{E}^{\mathbb{Q}}[S(T)] + \frac{\text{Cov}^{\mathbb{Q}}(d_{0,T}, S(T))}{B_{0,T}}$
 - Hedging with futures: minimum variance.
- Swaps.
 - Commodity swaps.
 - Interest rate swaps.
- Options.
 - Payoffs, pricing.
 - Put-Call parity
 - * $c_t - p_t = S_t - Ke^{-r(T-t)}$
 - * $S_t - K \leq C_t - P_t \leq S_t - Ke^{-r(T-t)}$

Binomial Trees



- Hedging/Replication arguments yield

$$f_t = \frac{1}{1 + r_f} \mathbb{E}^{\mathbb{Q}}[f_T],$$

with $q = \mathbb{Q}[\text{"up"}] = \frac{1+r_f-d}{u-d}$.

- The expected return under $\mathbb{Q}$ on the risky asset is r_f ; $\mathbb{Q}$ is a *risk-neutral measure*.
- Can price and hedge American options in the binomial setting.

Black-Scholes-Merton Model: Hedging approach

- The price of the risky asset evolves like

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t \Leftrightarrow S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$$

for two constants $\mu > 0$ and $\sigma > 0$, and a Brownian motion W under the real-world probability measure $\mathbb{P}$.

- A short derivative is *perfectly hedged* by investing $\phi_{S,t}$ in the stock:
 $\Pi_t = -f_t + \phi_{S,t}$, hence

$$d\Pi_t = -df_t + \phi_{S,t} \frac{dS_t}{S_t} = r\Pi_t dt.$$

- Together with Itô's Lemma, we find that the price $f(t, S_t)$ of the derivative satisfies

$$r f(t, S) = f_t(t, S) + r S f_S(t, S) + \frac{1}{2} \sigma^2 S^2 f_{SS}(t, S)$$

Black-Scholes-Merton Model: RN approach

- The price of the risky asset evolves like

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t \Leftrightarrow S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$$

for two constants $\mu > 0$ and $\sigma > 0$, and a Brownian motion W under the real-world probability measure $\mathbb{P}$.

- The price f_t of any European derivative written on the risky asset satisfies

$$f_t = \mathbb{E}_t^{\mathbb{Q}} \left[e^{-r(T-t)} f_T \right],$$

for the *risk-neutral* probability measure $\mathbb{Q}$.

- The process $dW_t^{\mathbb{Q}} \equiv dW_t + \frac{\mu-r}{\sigma}dt$ is a Brownian motion under $\mathbb{Q}$. In particular,

$$\frac{dS_t}{S_t} = rdt + \sigma dW_t^{\mathbb{Q}}$$

under $\mathbb{Q}$.

Black-Scholes-Merton Formulas

- The arbitrage-free price of a European call is

$$c_t = S_t N(d_1) - K e^{-r(T-t)} N(d_2),$$
$$\text{with } d_1 = \frac{\ln\left(\frac{S_t}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t},$$

and N being the cdf of a $\mathcal{N}(0, 1)$.

- The arbitrage-free price of a European put is

$$p_t = K e^{-r(T-t)} N(-d_2) - S_t N(-d_1)$$

- The Greeks: $\Delta_c = N(d_1)$, Γ , ...

Models of the Short Rate

- Interest rate trees
 - Same pricing technique as before.
 - Forward equation / state pricing.
 - Ho-Lee, Black-Derman-Toy to match yield curve.
- Affine models, e.g. Vasicek: $dr_t = \kappa(\theta - r_t)dt + \sigma dW_t^{\mathbb{Q}}$
 - Mean-reversion
 - Close form

$$r_T = \theta + (r_t - \theta)e^{-\kappa(T-t)} + \sigma \int_t^T e^{-\kappa(T-s)} dW_s^{\mathbb{Q}}$$

- Hedging argument (and Itô's lemma) shows

$$r f(t, r) = f_t(t, r) + f_r(t, r)\kappa(\theta - r) + \frac{1}{2}f_{rr}(t, r)\sigma^2$$

- Pricing the ZCB: $B(t, T) = e^{-a(t, T) - b(t, T)r_t}$.
- Cox-Ingersoll-Ross (CIR) keeps interest rates positive